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(54) **THERAPEUTIC INHIBITORS OF THE
REVERSE MODE OF ATP SYNTHASE**

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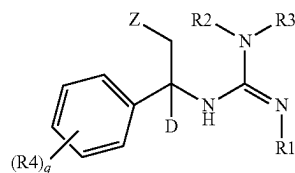
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(57)

ABSTRACT

Compounds of the following formula, and pharmaceuti-
cally-acceptable salts, solvates, hydrates and prodrugs
thereof, formula (A) are useful to preferentially inhibit the
ATP-hydrolysing mode of ATP synthase, and are thereby
useful for treating various diseases and orders including
cancer, particularly cancers that utilise the Warburg effect.



(A)

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FIGURE 1

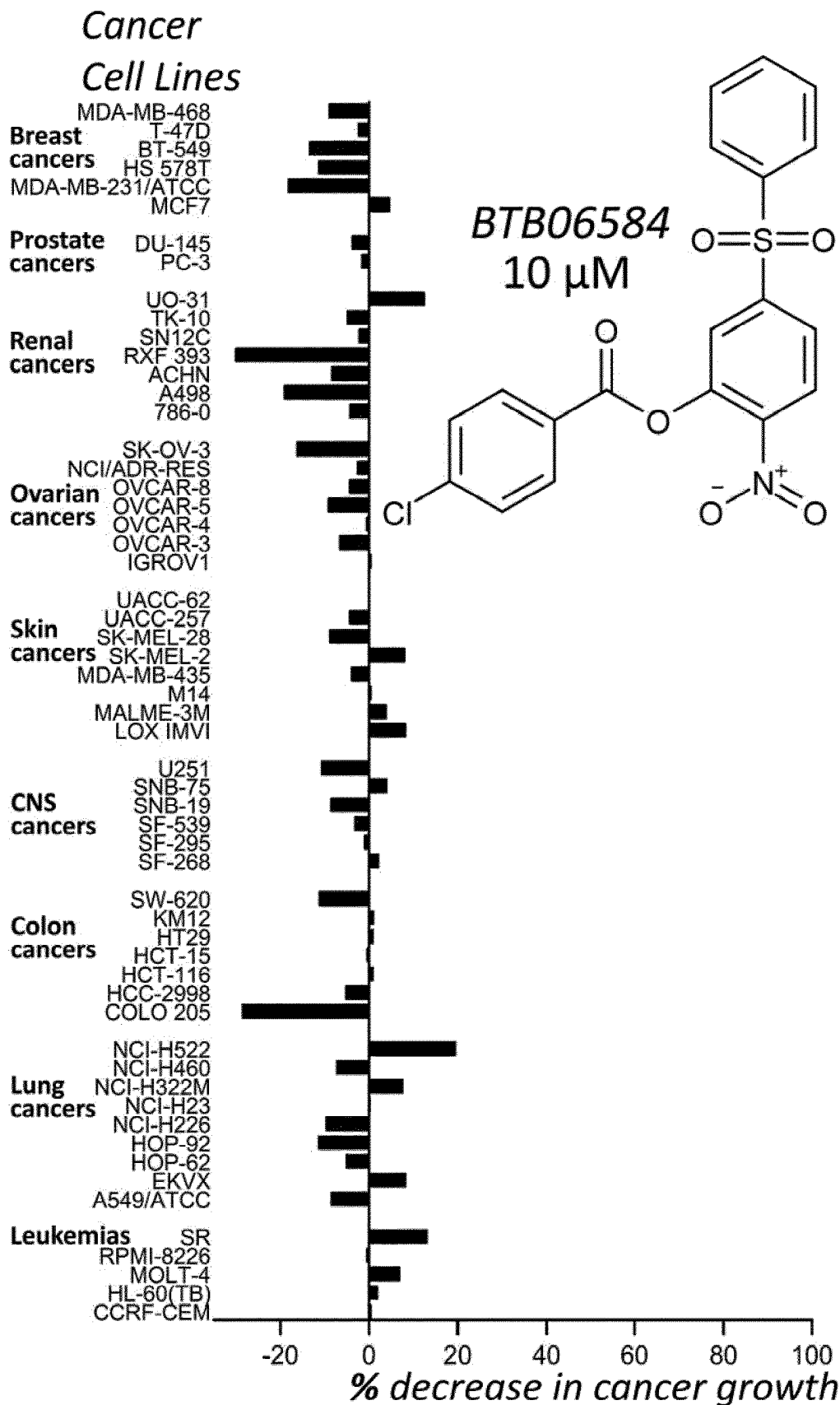


FIGURE 2

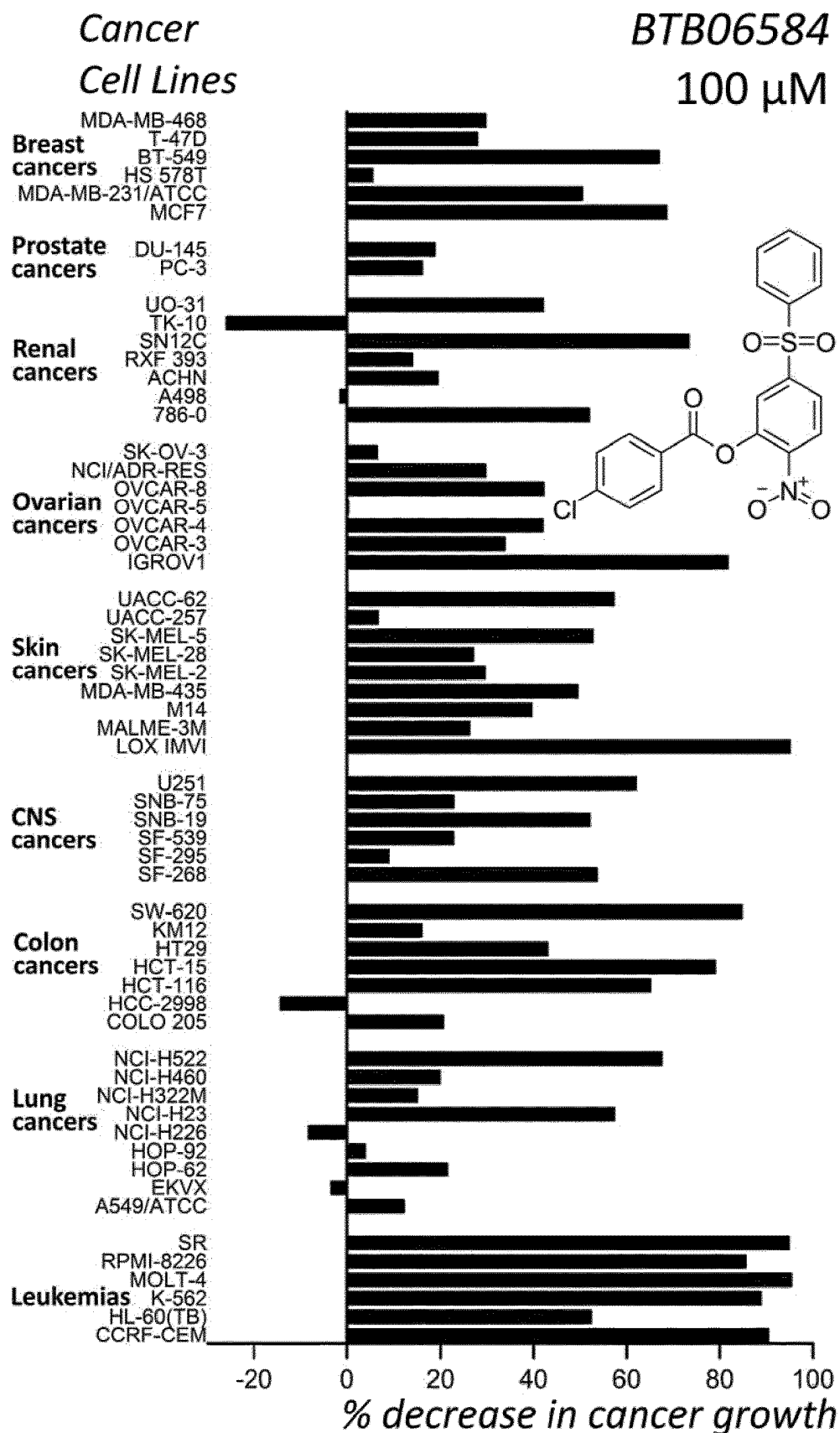


FIGURE 3

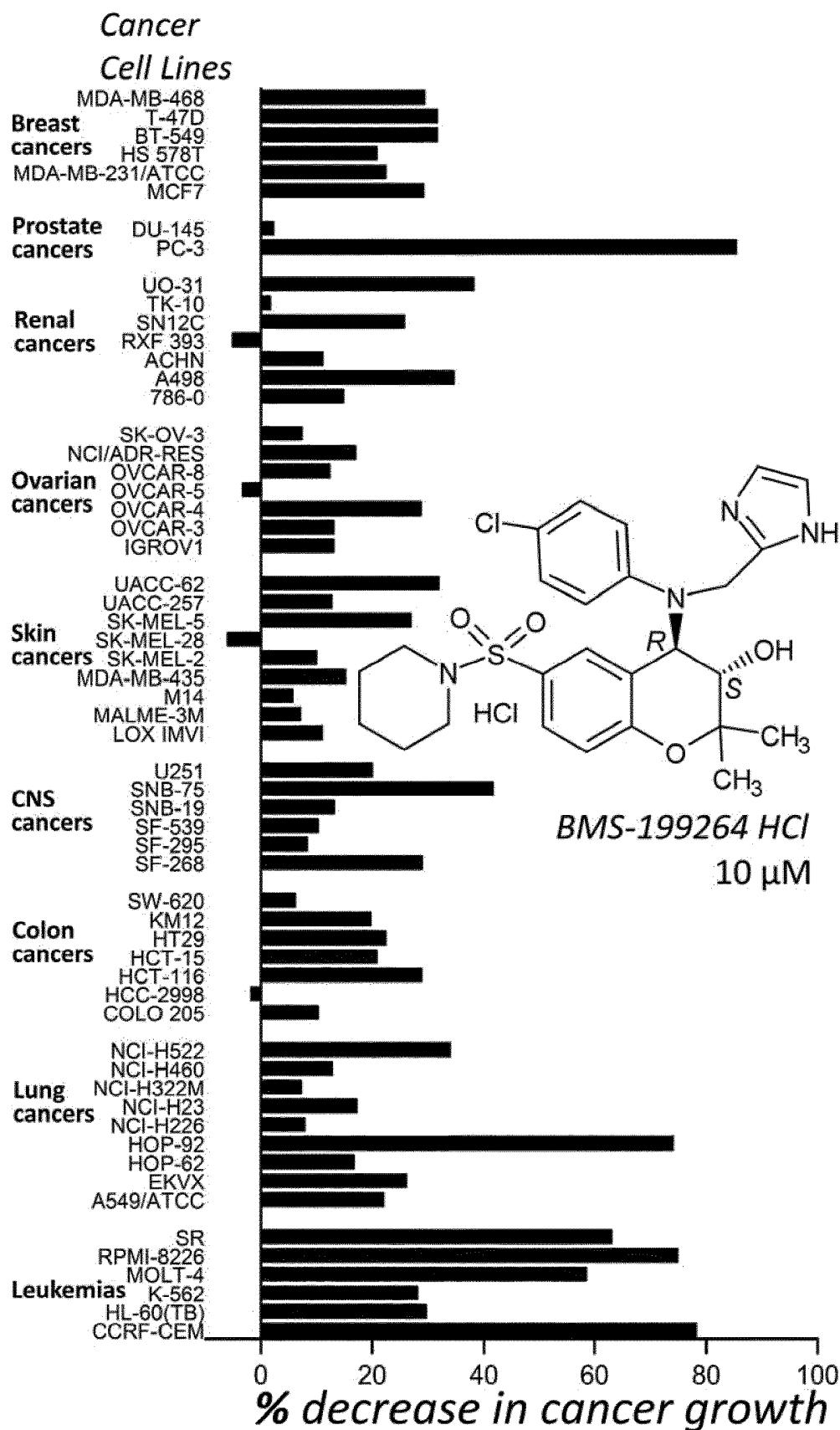


FIGURE 4

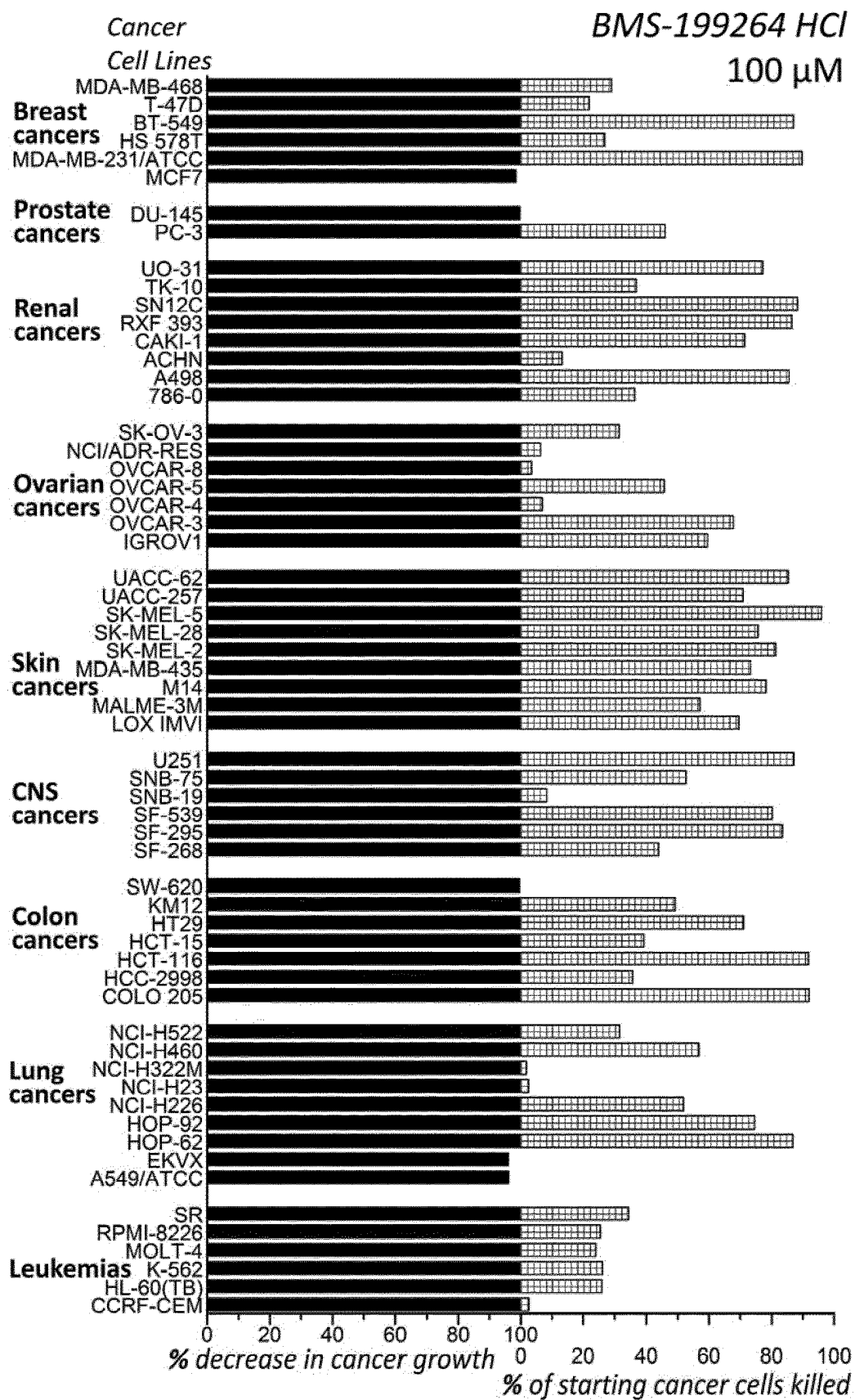


FIGURE 5

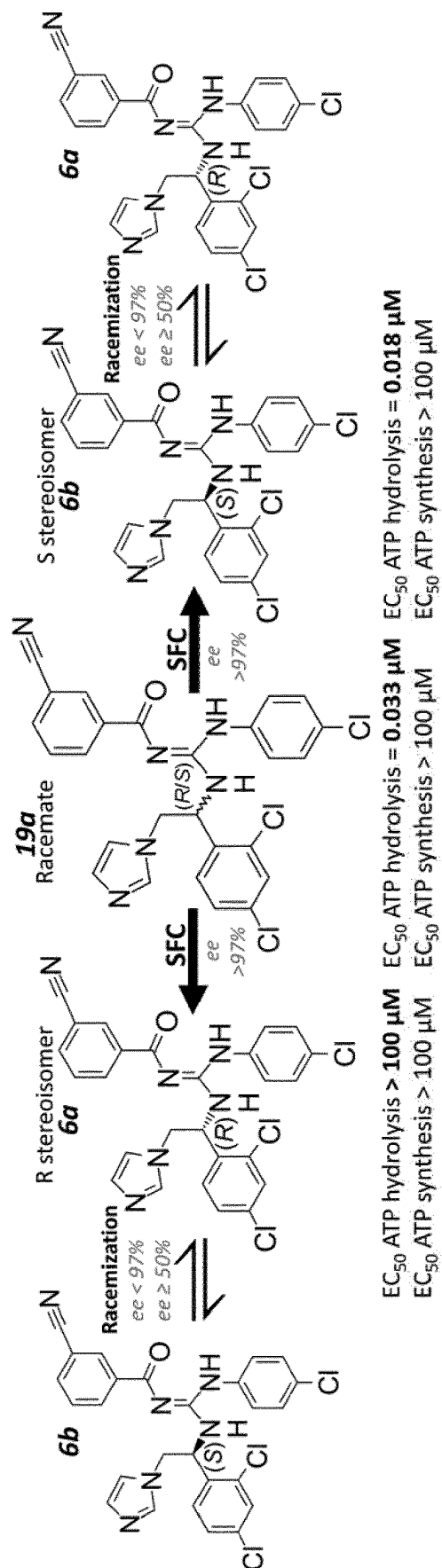


FIGURE 5 cont.

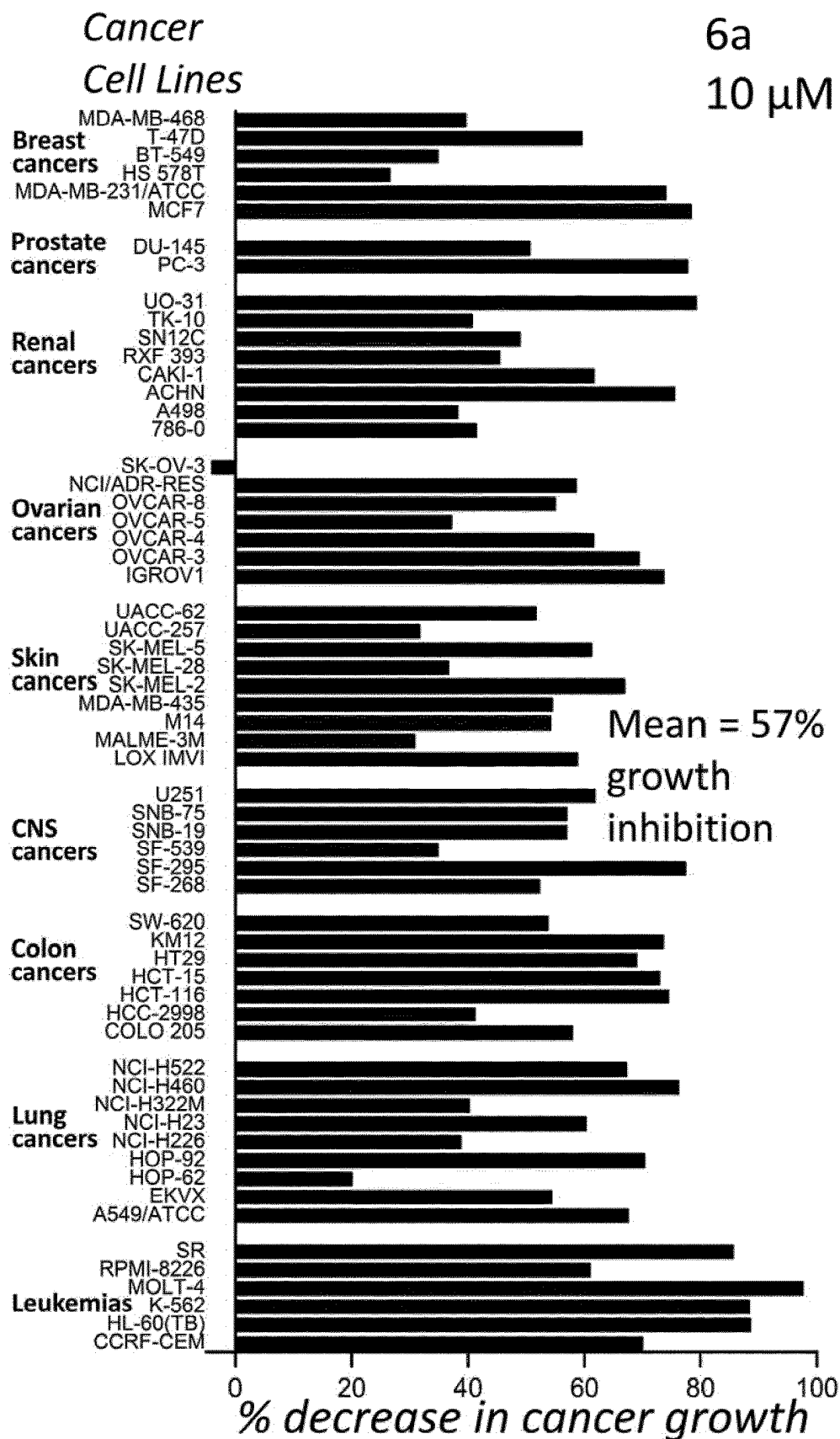
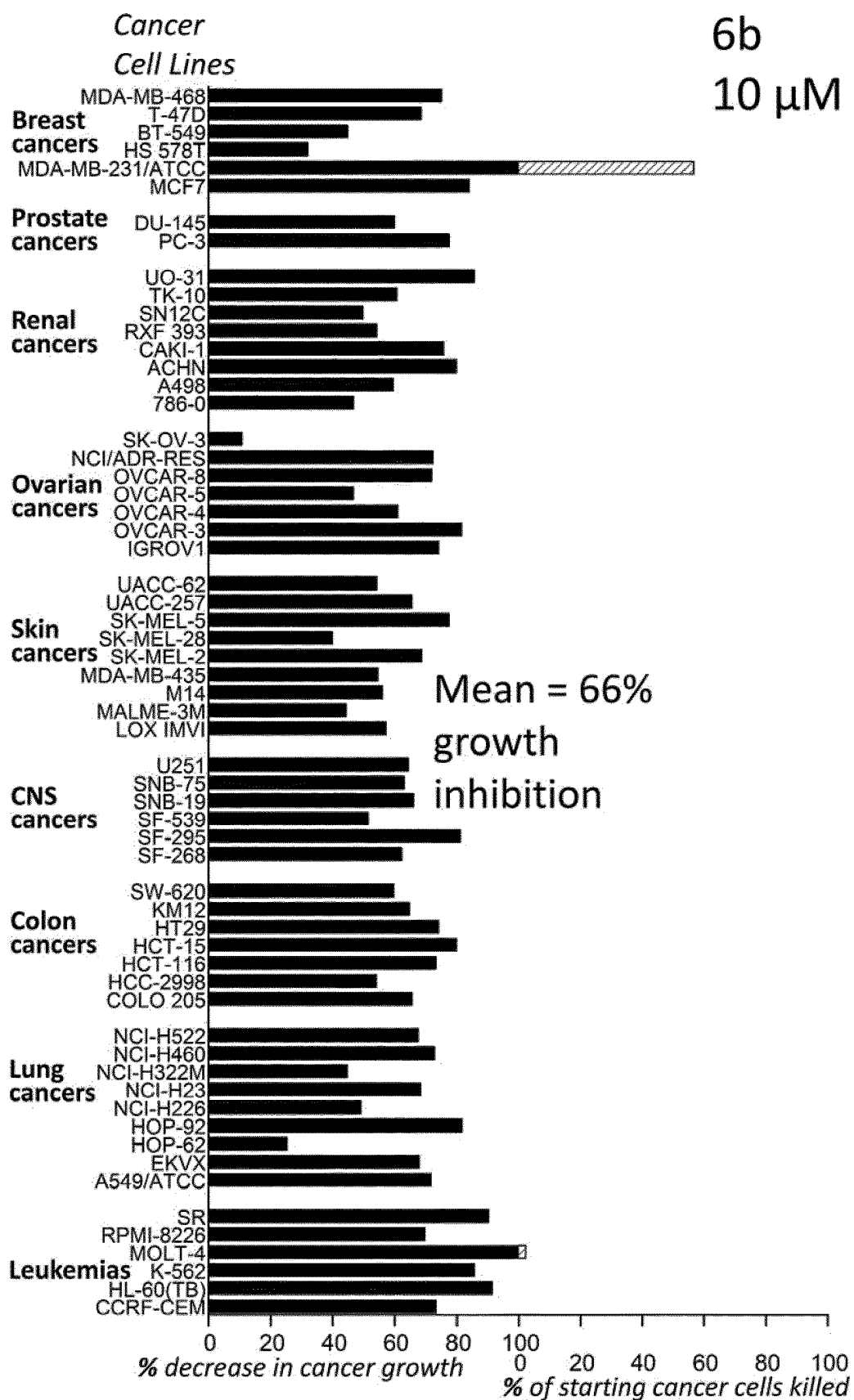
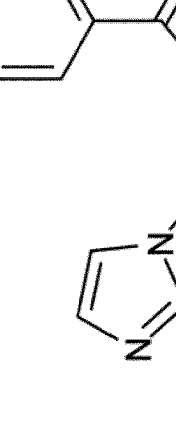


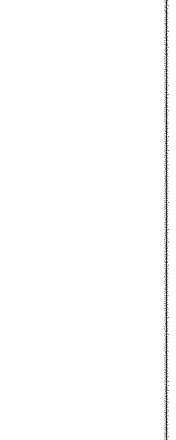
FIGURE 5 cont.





BTB06584

Anti-cancer activity scales
 with EC₅₀ F₁F₀ ATP hydrolysis



19a

BMS-199264 hydrochloride

FIGURE 6 cont

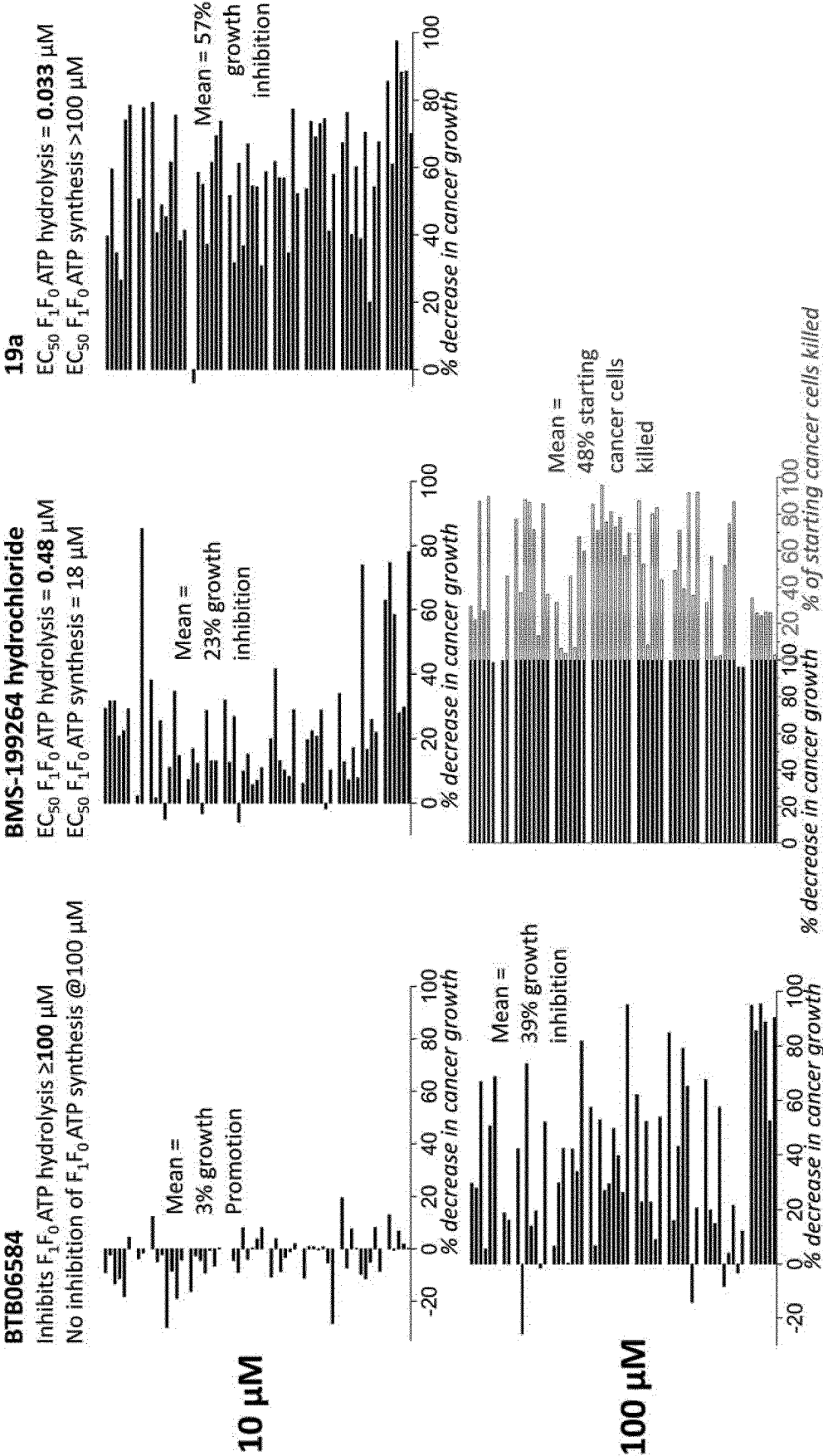
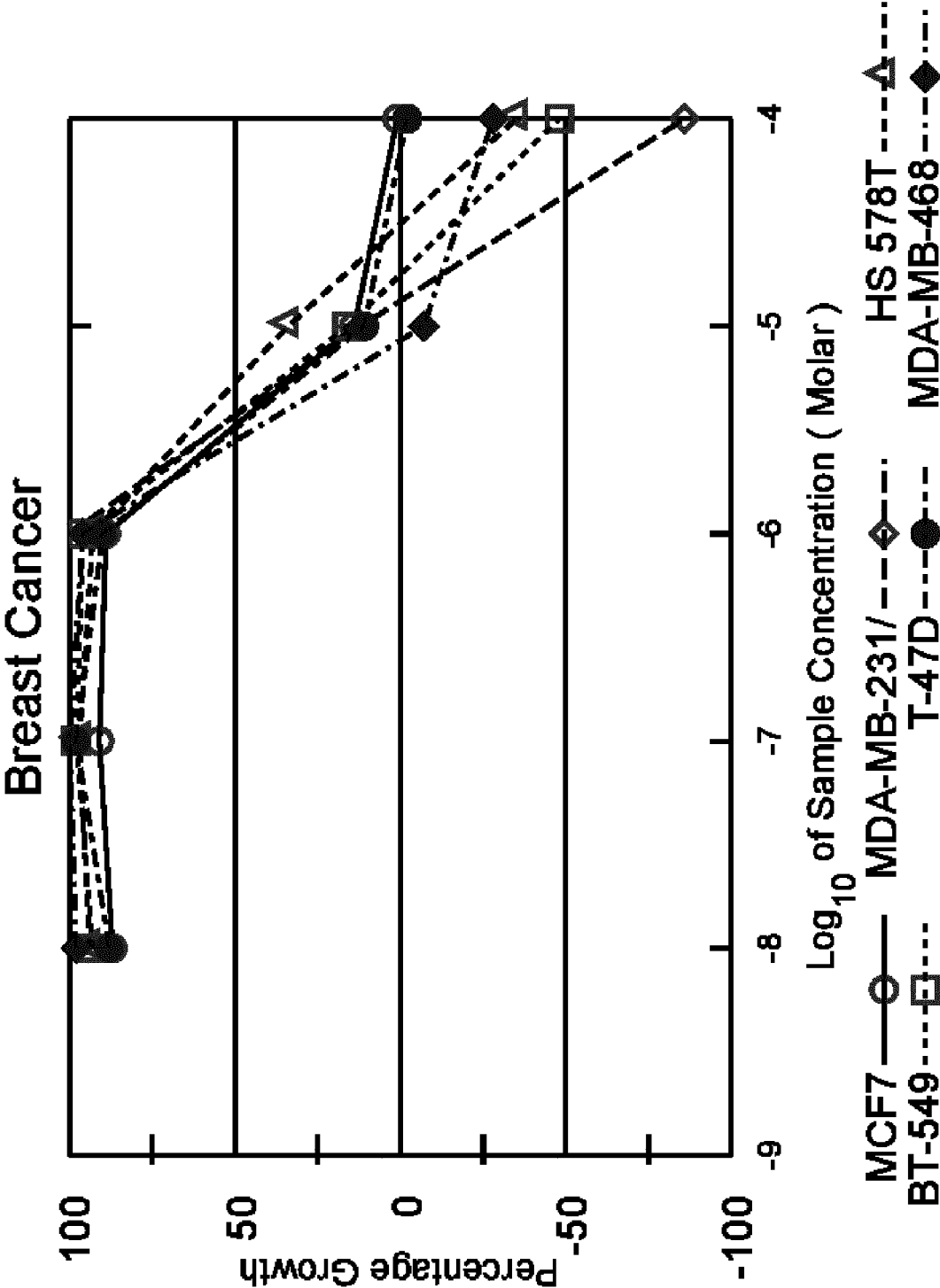


FIGURE 7A



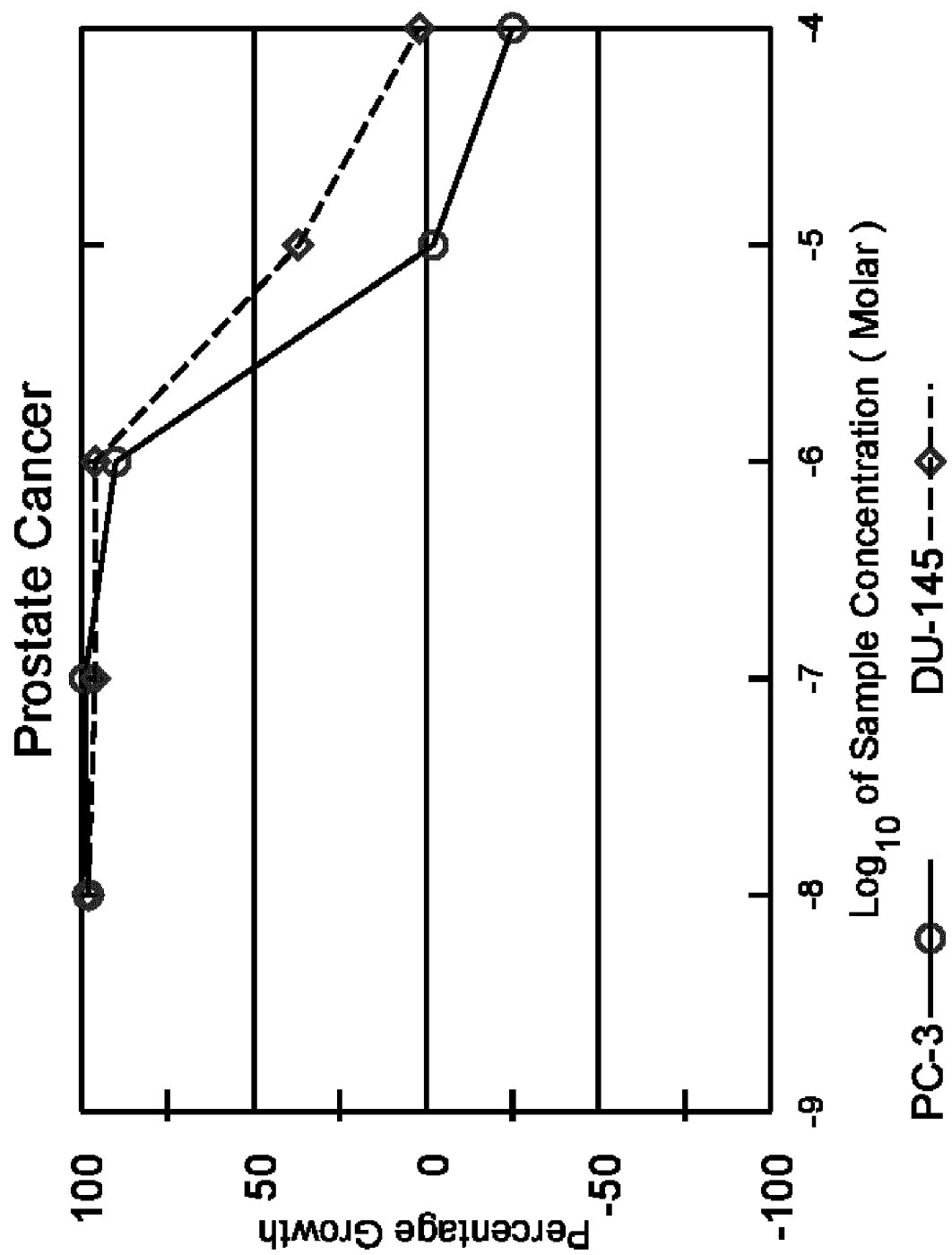


FIGURE 7B

FIGURE 7C

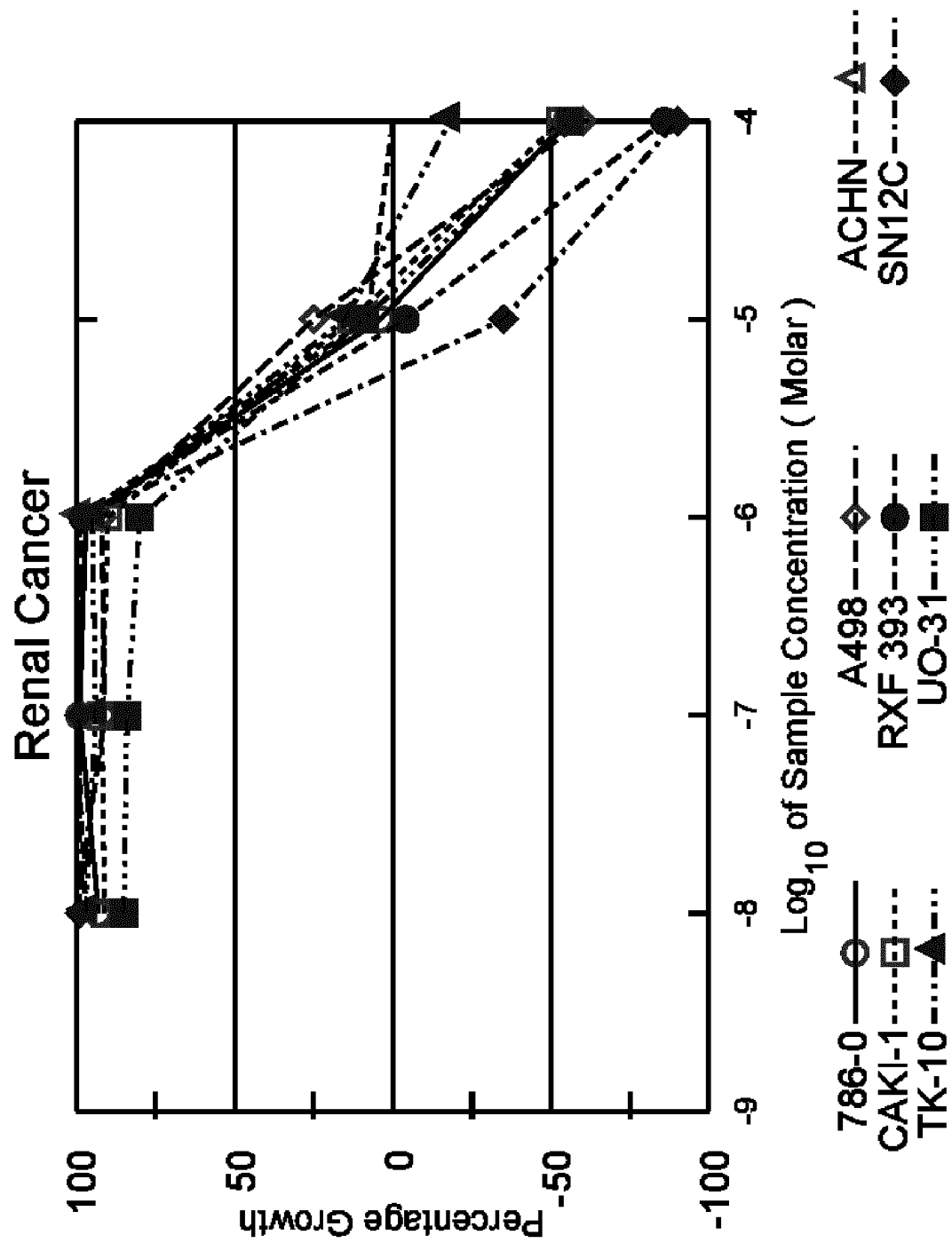


FIGURE 7D

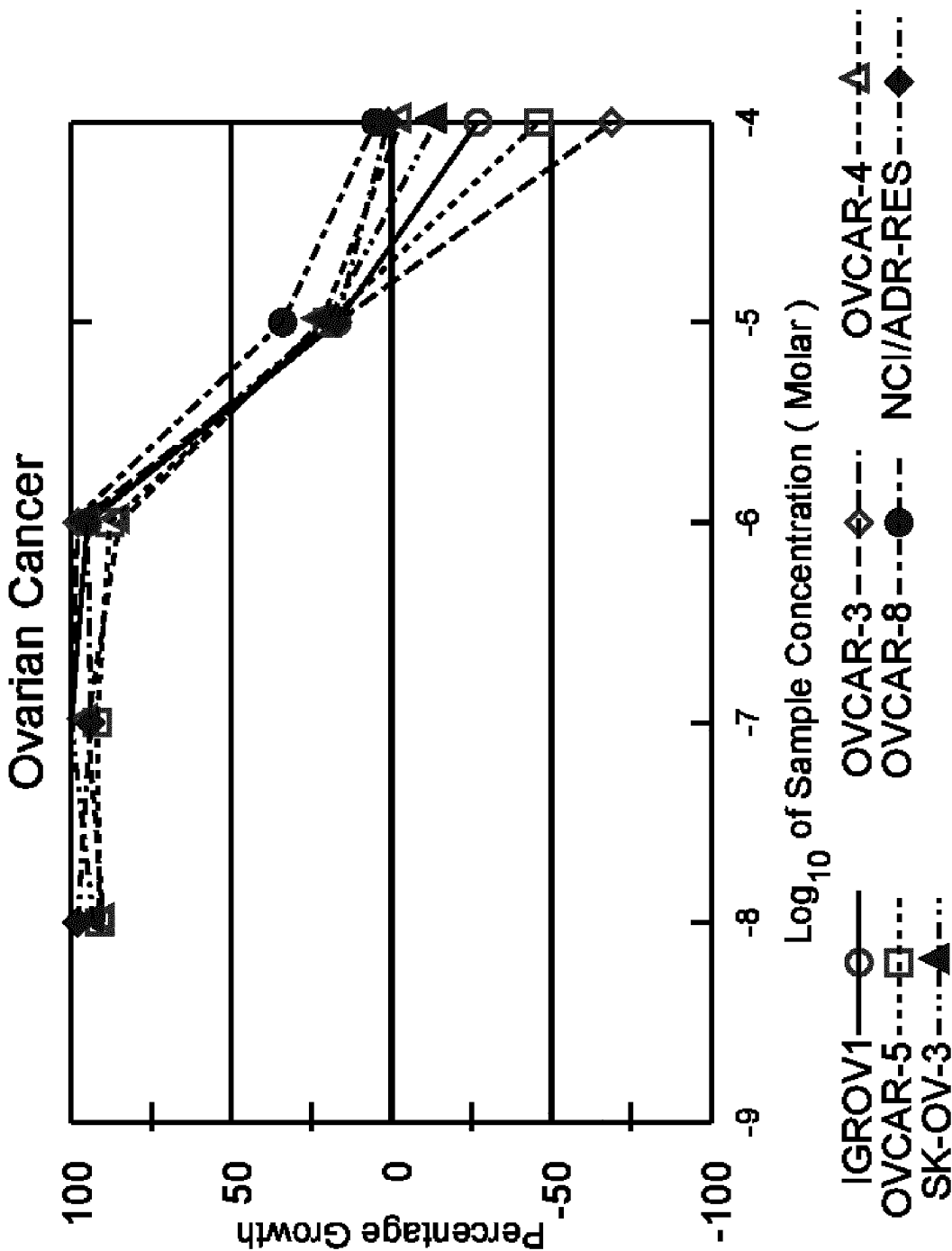


FIGURE 7E

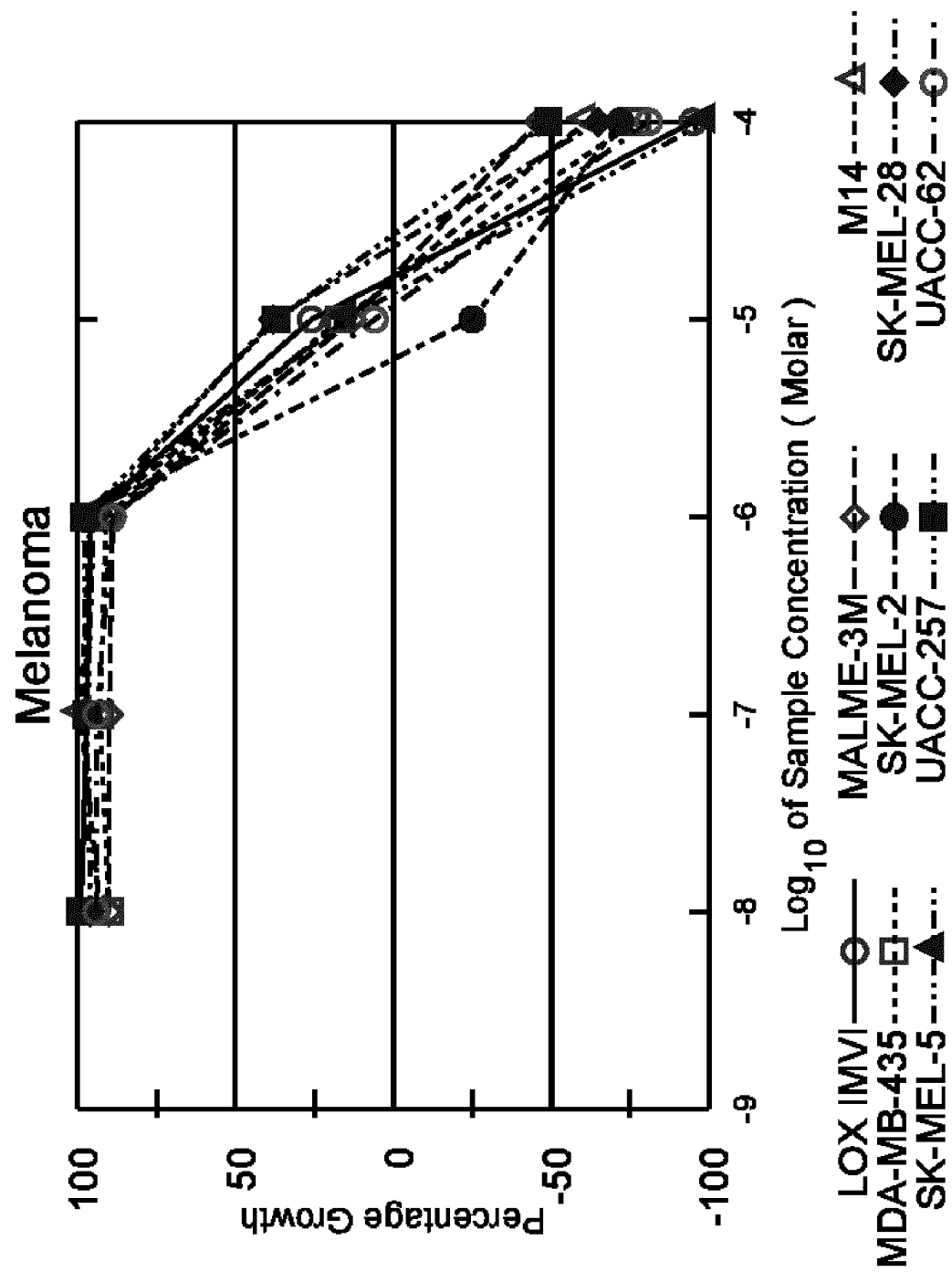


FIGURE 7F

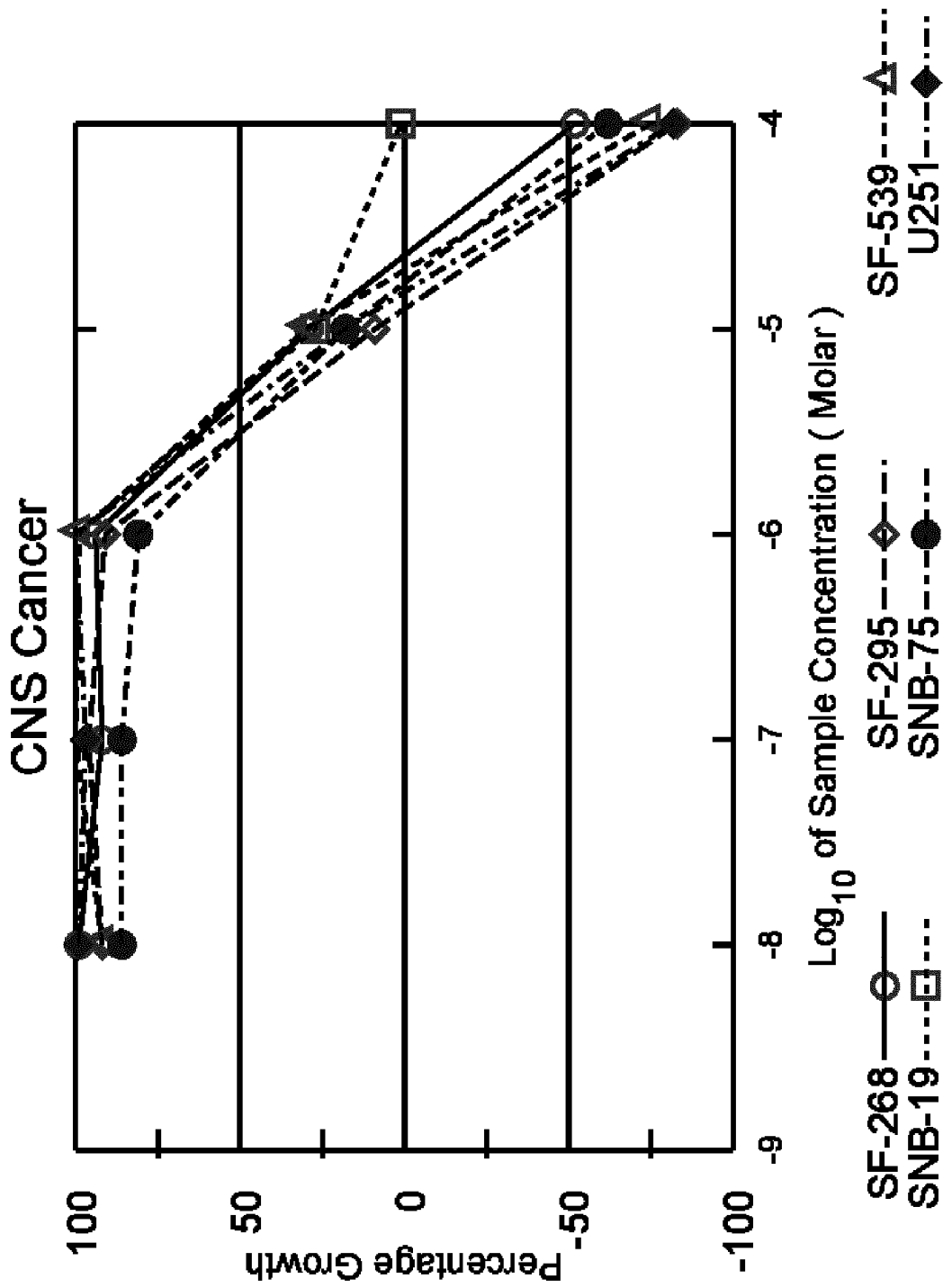


FIGURE 7G

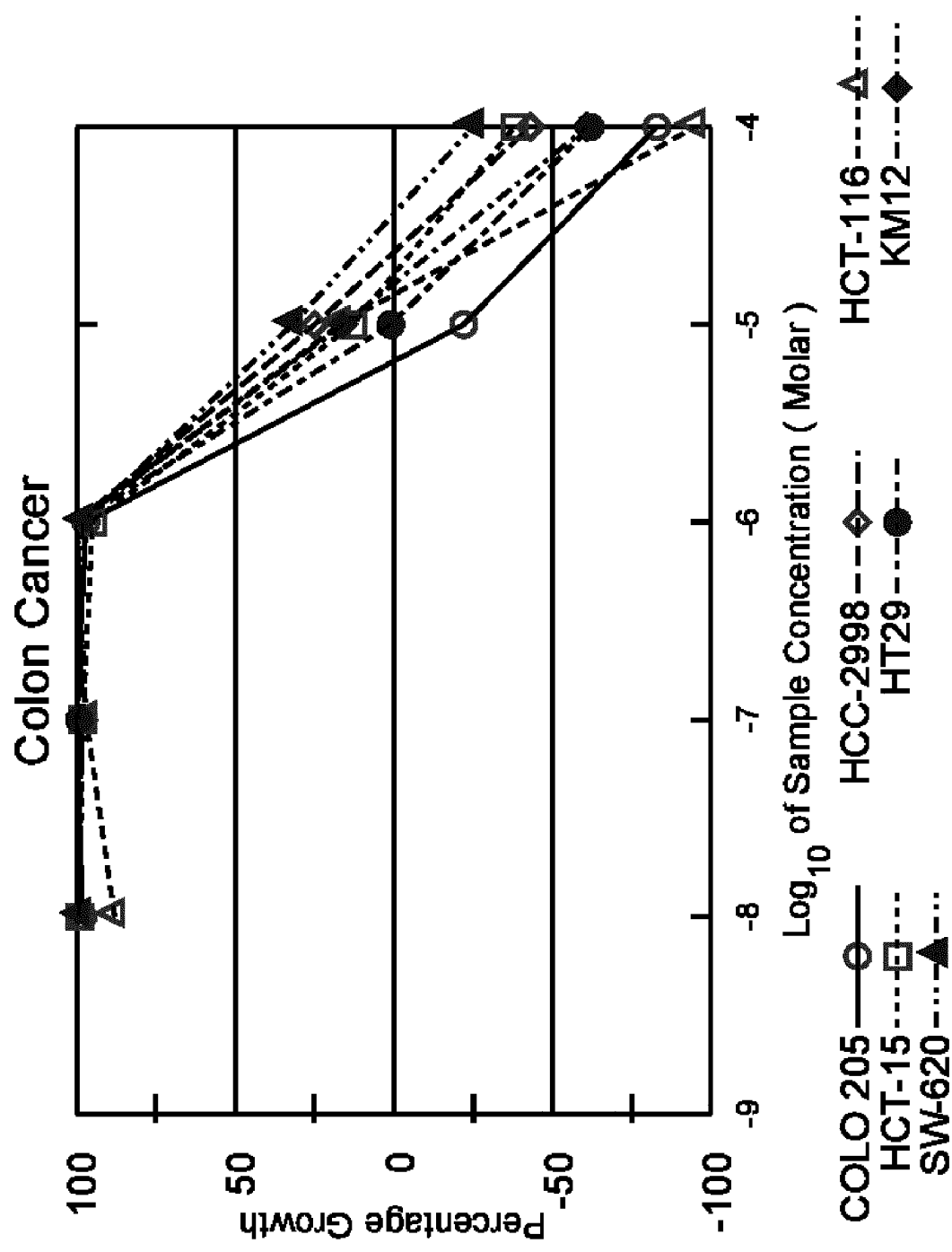


FIGURE 7H

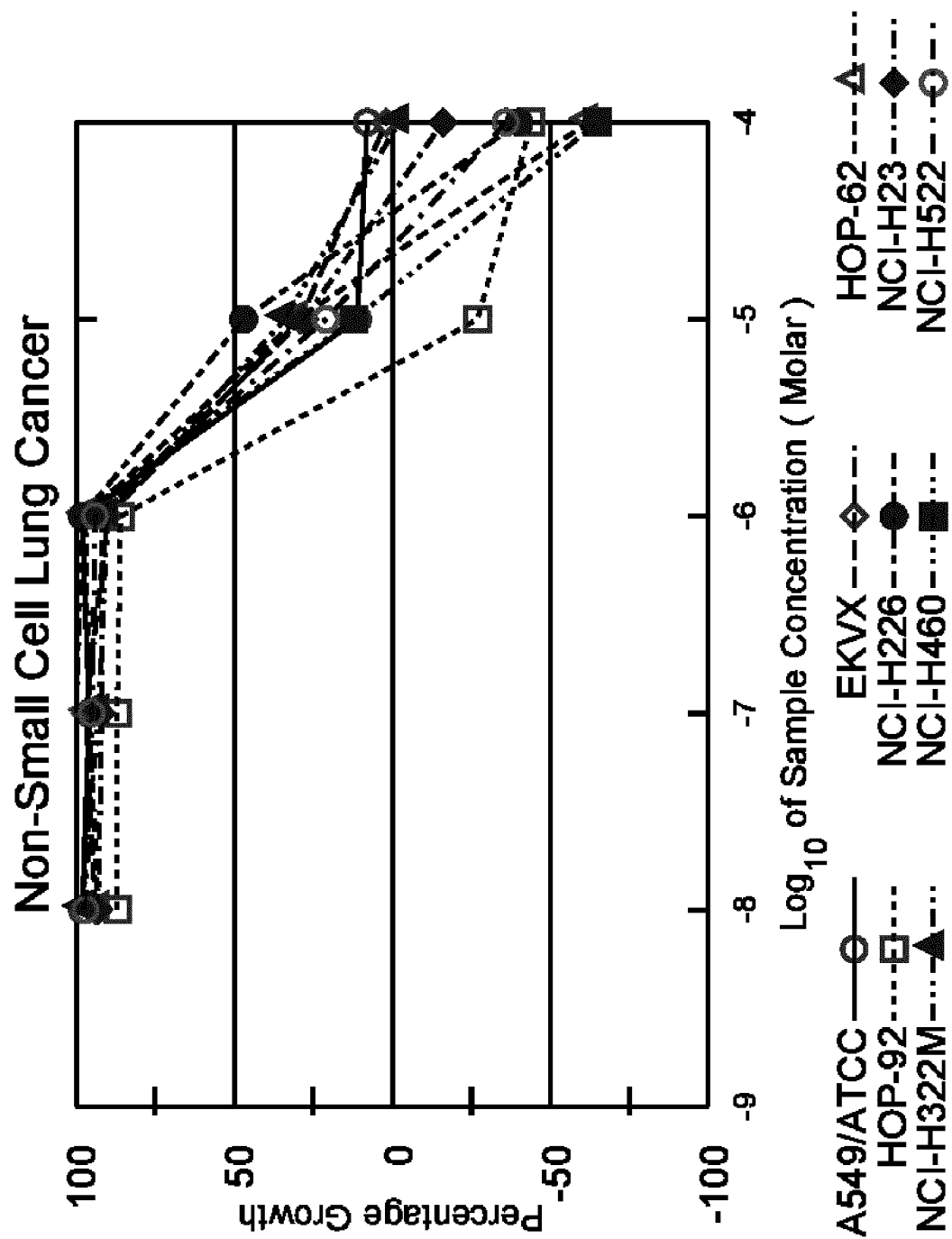


FIGURE 71

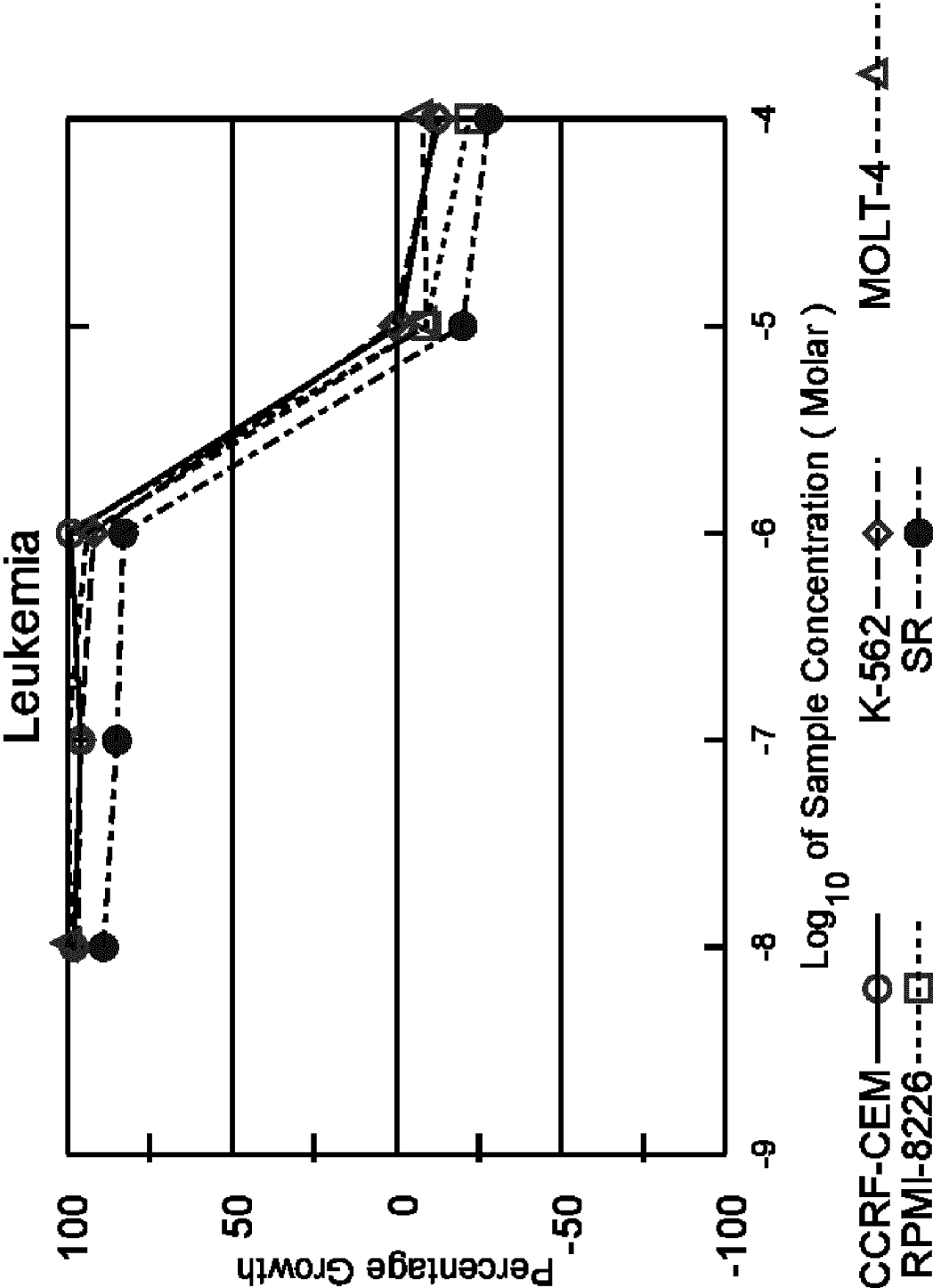


FIGURE 8

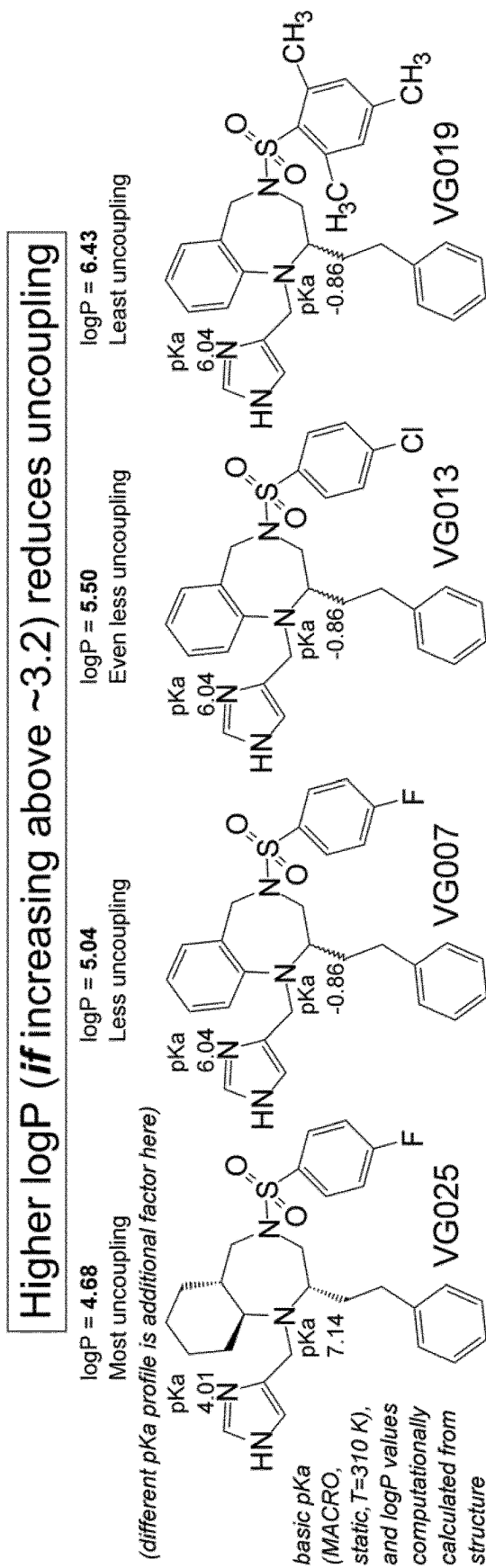


FIGURE 8 cont.

<i>Uncoupling = increased O₂ consumption</i>					
	VG025	VG007	VG013	VG019	BENCHMARK DRUGS
ΔO_2 consumption (without respiratory chain inhibitor)	-1.19% (10 μ M) +75% (100 μ M)	+25% (100 μ M)	+3.32% (10 μ M) +20% (100 μ M)	-0.34% (100 μ M)	Oligomycin (3 μ M) -40% FCCP (500 nM) +50%
$\Delta \psi_{IM}$ (without respiratory chain inhibitor)	-7.3% (10 μ M) -47.5% (100 μ M)	0%	0%	0%	Oligomycin (3 μ M) none μ FCCP (500 nM) -50%
$\Delta \psi_{IM}$ (with respiratory chain inhibitor)	-46% (10 μ M) -64% (100 μ M)	-42.61% (100 μ M)	-30.15% (10 μ M) -40.79% (100 μ M)	-29.74% (100 μ M)	Oligomycin (3 μ M) -49.1% FCCP (500 nM) -52.5%

μ . should hyperpolarize the membrane slightly: presumably the assay used isn't sensitive enough to detect small hyperpolarizations

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THERAPEUTIC INHIBITORS OF THE REVERSE MODE OF ATP SYNTHASE

RELATED APPLICATIONS

This application claims the priority benefit of GB application numbers GB1700772.5 (filed 17 Jan. 2017), GB1706046.8 (filed 14 Apr. 2017), GB1707945.0 (filed 17 May 2017), GB1710198.1 (filed 27 Jun. 2017), GB1711250.9 (filed 13 Jul. 2017), GB1715756.1 (filed 28 Sep. 2017), GB1715758.7 (filed 28 Sep. 2017), GB1715938.5 (filed 1 Oct. 2017), GB1716492.2 (filed 9 Oct. 2017), GB1800092.7 (filed 4 Jan. 2018), GB1800291.5 (filed 8 Jan. 2018) and GB1800581.9 (filed 15 Jan. 2018). The entire teachings of these applications are incorporated herein by reference.

FIELD OF THE INVENTION

This invention discloses compounds that preferentially inhibit the ATP-hydrolysing mode of ATP synthase, pharmaceutical compositions of these compounds, and methods of use for treating subjects known to have various diseases or disorders including cancer (e.g. diagnosed with), subjects suspected of having various diseases or disorders including cancer or subjects at risk of developing various diseases or disorders including cancer. In a particular embodiment, the subject is a human.

BACKGROUND OF THE INVENTION

ATP Synthase

ATP synthase (also known as F_1F_0 ATP synthase, F_0F_1 ATP synthase, F_1F_0 -ATPase, F_0F_1 -ATPase) is located at the inner mitochondrial membrane (IM). It can use the proton motive force (pmf) to generate ATP from ADP and P_i [1-3]. ATP synthase is reversible and—depending on its substrate/product concentrations, the pmf and the voltage across inner mitochondrial membrane $\{\psi_{IM}\}$ —it can work “forwards” (passaging protons, making ATP) or “backwards” (pumping protons, consuming ATP): its “forward” and “reverse” modes respectively, which may also be termed F_1F_0 ATP synthesis and F_1F_0 ATP hydrolysis respectively.

Inhibitors of ATP Synthase

There are drug inhibitors of ATP synthase, reviewed in [4] (herein incorporated in its entirety). Some inhibitors disproportionately/selectively inhibit the reverse mode, as compared to the forward mode, of ATP synthase [4-13]. Macrolides are a class of polyketide. So macrolide F_1F_0 ATP synthase inhibitors are polyketide F_1F_0 ATP synthase inhibitors, and these terms are used interchangeably herein. Polyketide F_1F_0 ATP synthase inhibitors (e.g. oligomycin) inhibit the forward mode, more than the reverse mode, of ATP synthase [11]. Oligomycin is well known in the art as an inhibitor of F_1F_0 ATP synthase, and thence oxidative phosphorylation and aerobic respiration [3]. Human life relies upon aerobic respiration. Indeed, the importance of breathing (O_2 in, CO_2 out) is widely appreciated. Thence the danger of oligomycin is easily apparent.

IF_1 is an endogenous protein, encoded by the $ATPIF1$ gene, which can selectively block the reverse mode of ATP synthase [4]. Its activity is pH sensitive and low, but non-zero, at normal matrix pH, and significant upon matrix acidification, caused by collapse of the proton motive force across the mitochondrial inner membrane.

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Prior Art Teaches That Molecules of This Disclosure are Not Anti-Cancer Therapeutics

Polyketide F_1F_0 ATP synthase inhibitors (e.g. oligomycin) are poisonous to cancer [14] and normal [15] cells. Indeed, just 1 mg/kg oligomycin kills healthy rats ($n=10$) within 48 hours; $LD_{50}=0.5$ mg/kg [15]. Normal cells typically need to use F_1F_0 ATP synthase in its forward mode and so blocking this mode is typically lethal. Thus, polyketide F_1F_0 ATP synthase inhibitors are not suitable as anti-cancer therapeutics: indeed, cytarabine, ossamycin and peliomycin don't work in xenograft mouse models of cancer (data in [16], oligomycin untested) because a therapeutic window is absent because, to repeat, polyketide F_1F_0 ATP synthase inhibitors are highly poisonous to normal cells, whilst not even being poisonous to all cancer cells: e.g. ineffective against glycolytic cancers exhibiting the Warburg effect [14]. [17] used oligomycin in a xenograft cancer mouse model but only by applying oligomycin to the cancer cells before they were inoculated into mice, and washing the excess oligomycin off before inoculation into the mice (by culture for 2 days in drug free medium). They did the study like this (atypical, as clear to someone of the art) because oligomycin toxicity is not discriminate for cancer in a mammal. Obviously this experiment has no clinical parallel or utility. The synthesis/structure of some molecules of this disclosure has been disclosed in prior disclosures [P1, P2, P3], wherein these structures are speculated to be anti-cancer medicines merely by analogy to the anti-cancer activity of polyketide F_1F_0 ATP synthase inhibitors in [14]. Indeed, to mirror and use the restriction of [14], these disclosures restrict their suggestion to “cancers having tumor cells that do not exhibit the Warburg effect” [P3] i.e. they restrict to cancers using oxidative phosphorylation (OXPHOS) and ATP synthase, in its forward mode, to generate ATP. But what undermines this (postulated) approach is that this aerobic profile is what normal cells typically use also, especially on aggregate across an organism: well known to those of the art (evidence: importance of breathing to mammalian life). By this analogy to polyketide F_1F_0 ATP synthase inhibitors, these disclosures speculate these molecules are safe anti-cancer therapeutics. When in fact, by this analogy, they actually teach the opposite. This is clear when [14] isn't considered in isolation, as it shouldn't be, but alongside the rest of the literature e.g. [15], [16] and the knowledge of someone of the art (well known that normal cells need to use F_1F_0 ATP synthase in its forward mode, to generate ATP, and that oligomycin blocks this, and is potentially dangerous). So, these prior disclosures [P1, P2, P3] teach someone of the art, that these structures are, by their chosen analogy to polyketide F_1F_0 ATP synthase inhibitors, not suitable for anti-cancer therapy. It isn't sufficient to kill cancer to be an anti-cancer therapeutic. This killing must be selective, leaving normal cells alive. Metabolic poisons such as cyanide or oligomycin do not fit this criterion. By distinction, the present invention discloses selective killing of cancer cells, at compound concentrations harmless to normal cells. This couldn't have been anticipated from the prior art. Furthermore, this selective anti-cancer activity is pronounced for cancers that do exhibit the Warburg effect. Distinctly, the present disclosure discloses experimental data. Its inventive step is to show that its compounds are safe anti-cancer therapeutics, exactly because of their distinction from polyketide F_1F_0 ATP synthase inhibitors. There is a broad therapeutic margin for the compounds of this disclosure as a virtue of the distinctive (from oligomycin) way they work, leveraging differences between normal and cancer cells, discovered and disclosed as part of this invention.

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Indeed, the compounds of this disclosure can kill highly glycolytic cancers exhibiting the Warburg effect. These cancers tend to be the most dangerous, with the worst prognosis (numerous studies find this: representatives: [18-20]).

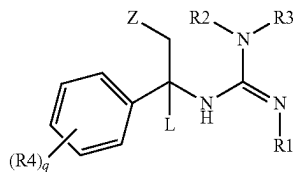
Molecules of this disclosure don't just exert anti-cancer activity. They can also affect normal cells, making their metabolism more efficient, which can cause weight gain/reduce weight loss/maintain body weight, all of which combats cachexia. For example, cancer driven cachexia, which is the leading cause of death in cancer patients. By contrast, polyketide F_1F_0 ATP synthase inhibitors are toxic to normal cells, denying them energy, rather than enabling them more energy, by efficiency gain, as molecules of this disclosure can do.

In this disclosure, molecules are shown to exert anti-cancer activity by inhibiting F_1F_0 ATP hydrolysis. This directly opposes the prior art, wherein experimental "results indicate that [even] under severe hypoxic conditions, ATP synthase does not hydrolyze ATP in cancer cells" and that "in cancer cells IF_1 overexpression fully prevents ATP synthase hydrolytic activity" [21]. Indeed, the prior art finds that inhibiting F_1F_0 ATP hydrolysis conveys advantage to cancer [21]. Thence the prior art teaches that the compounds of this disclosure, which inhibit F_1F_0 ATP hydrolysis, will assist rather than compromise cancer, and thence this invention, which discloses discovery of the opposite, is novel over the prior art.

All publications, patents and patent applications mentioned or cited in this disclosure are herein incorporated, in entirety, by reference. This disclosure uses IC_{50} and EC_{50} interchangeably, for a process being inhibited or reduced. Chemical structures were drawn using the chemical drawing feature in [31], and if a drawing feature is unknown to the reader they are referred to its documentation, or to explore the software themselves: all clear to those of the art. Hydrogen on structures is typically not shown, present implicitly, but it is shown for some presented structures "On Hetero and Terminal" [31] groups.

SUMMARY OF THE INVENTION

Disclosed is the discovery of a cancer-specific drug target: the reverse mode of ATP synthase. Indeed, new experimental data, disclosed herein, demonstrates that molecules which specifically inhibit F_1F_0 ATP hydrolysis can exert specific anti-cancer activity, at concentrations that do not harm normal cells. Any anti-cancer drug that targets/inhibits F_1F_0 ATP hydrolysis is componentry to this invention. This disclosure discloses numerous anti-cancer drug working examples, some of which are also new compositions of matter, and discloses rationale and methods to find further working examples, which are, in turn, componentry to this invention and encompassed by this disclosure. The best mode for prevention or treatment of cancer in a subject, particularly with cancer exhibiting the Warburg effect, is to use a pharmaceutical composition with an effective amount of one or compounds of the following formula,



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or a pharmaceutically-acceptable salt, solvate, hydrate or prodrug thereof, wherein:

L is alkyl, substituted alkyl or any atom or isotope permitted by valence;

R_1 is cyano, $-SO_2R_8$, $-C(=O)R_9$, or heteroaryl;

R_2 is (i) independently hydrogen, alkyl, or substituted alkyl, or (ii) taken together with R_3 forms a heterocyclo;

R_3 is (i) independently alkyl, substituted alkyl, alkylthio, aminoalkyl, carbamyl, B_B -aryl, B_B -heterocyclo, B_B -heteroaryl, or B_B -cycloalkyl, or (ii) taken together with R_2 forms a heterocyclo;

Z is heteroaryl provided that when R_1 is cyano, Z is not 2-pyridinyl;

B_B is a bond, C_{1-4} alkylene, C_{2-4} alkenylene, substituted C_{1-4} alkylene, substituted C_{2-4} alkenylene, $-C(=O)NR_{19}-$, $-C_{1-4}$ alkylene- $C(=O)NR_{19}-$, or substituted C_{1-4} alkylene- $C(=O)NR_{19}-$;

R_4 at each occurrence is selected independently of each other R_4 from the group consisting of halogen, alkyl, haloalkyl, nitro, cyano, haloalkoxy, OR_{25} , SR_{25} , $NR_{25}R_{26}$, $NR_{25}SO_2R_{27}$, SO_2R_{27} , $SO_2NR_{25}R_{26}$, CO_2R_{26} , $C(=O)R_{26}$, $C(=O)NR_{25}R_{26}$, $OC(=O)R_{25}$, $-OC(=O)NR_{25}R_{26}$, $NR_{25}C(=O)R_{26}$, $NR_{25}CO_2R_{26}$, aryl, heteroaryl, heterocyclo and cycloalkyl;

R_8 is alkyl, substituted alkyl, aryl, or heteroaryl;

R_9 is $-NR_{10}R_{11}$, alkyl, substituted alkyl, alkoxy, alkylthio, cycloalkyl, aryl, heteroaryl, heterocycle, or $-CO_2R_{12}$;

R_{10} and R_{11} are (i) independently selected from hydrogen, alkyl, substituted alkyl, alkoxy, heterocyclo, cycloalkyl, aryl, and heteroaryl; or (ii) taken together form a hetero cyclo or heteroaryl;

R_{12} and R_{19} are hydrogen or alkyl;

R_{25} and R_{26} are independently selected from hydrogen, alkyl, or substituted alkyl, or taken together form a heterocyclo or heteroaryl ring;

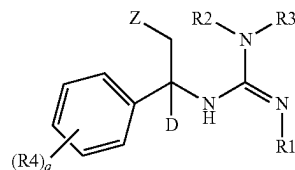
R_{27} is alkyl or substituted alkyl, and

q is 0, 1, 2, or 3.

In some embodiments, the S-enantiomer of the compound is in enantiomeric excess. In some embodiments, the enantiomeric excess of S-enantiomer exceeds 70%.

In some embodiments, L is hydrogen or deuterium.

In some embodiments, the compound is a compound according to the formula

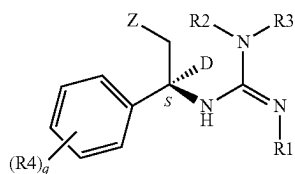


or pharmaceutically-acceptable salts, solvates, hydrates and prodrugs thereof, wherein

D is deuterium (enrichment, for example, exceeding 40% deuterium incorporation at shown position, and optionally at other positions also).

In some embodiments, the compound is a compound according to the formula

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or pharmaceutically-acceptable salts, solvates, hydrates and prodrugs thereof, wherein

D is deuterium (enrichment, for example, exceeding 40% deuterium incorporation at shown position, and optionally at other positions also);

S symbolises the S stereoisomer, for example, in enantiomeric excess (ee) exceeding 70%.

Herein, the terms “S-stereoisomer” and “S-enantiomer” refer to the arrangement of groups around the chiral centre shown in the structure above, regardless of the specific identities of the variables such as Z, L and R_4 within the structure.

DETAILED DESCRIPTION OF THE INVENTION

As used herein with reference to the utilities described, the terms “treating” or “treatment” encompass both responsive and prophylaxis measures designed to inhibit or delay the onset of the disease or disorder, or to alleviate, ameliorate, lessen, reduce, modulate or cure the disease or disorder and/or one or more of its symptoms. The terms “subject” and “patient” refer to organisms to be treated by the compounds/methods of the present invention and can refer to a human or animal.

The invention of this disclosure hinges on its discovery, disclosed herein, that some cancers rely on F_1F_0 ATP hydrolysis, even under normoxia (indeed under hyperoxia: ~21% O_2), during some or all of their cell cycle. Evidence herein: compounds of this disclosure, which specifically inhibit F_1F_0 ATP hydrolysis, slow cancer proliferation at concentrations that they do not harm normal cells.

In some of the most dangerous cancers, refractory to present [chemo/radio] therapies, during some or all of their cell cycle, reactive oxygen species (ROS) decrease [NADPH], because it is consumed in ROS mitigation processes, and this then pulls through increased pentose phosphate pathway (PPP) and glycolytic flux. But such a pivotal increase in glycolytic/PPP flux can only occur because of F_1F_0 ATP hydrolysis, a distinctive feature to these cancers, which stops ATP produced by glycolysis from accumulating and slowing glycolysis by negative feedback inhibition of key glycolytic enzymes. This increased PPP flux maintains [NADPH] and ROS mitigation. In this way, these cancers can maintain a very high ROS mitigation capability, maintain very low intracellular [ROS], and tend to be the most resistant to conventional [chemo/radio] therapies, which work, or often don't work (!), by increasing [ROS]. Compounds of this disclosure undermine this process/resistance. By inhibiting F_1F_0 ATP hydrolysis, they increase the anti-cancer efficacy of any chemical or treatment that increases reactive oxygen species (ROS) in cancer cells. An embodiment of this disclosure is any such co-treatment(s). Indeed, a compound(s) of this disclosure increases the success rate of standard of care [chemo/radio] therapies and permits their use at lower dosing, which reduces their horrendous side-effects. This disclosure encompasses a compound(s) of this invention in co-therapy with chemotherapy, or radiotherapy

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or any US Food and Drug Administration (FDA) approved drug(s) or treatment, for example, a drug approved for cancer treatment. Chemotherapies are well known to those of the art, including, but not limited to, cisplatin, carboplatin, taxol, oxaliplatin etc.

In other embodiments, a compound(s) of this disclosure is used as cancer therapy alone. Indeed, this is a much more cancer-targeted therapeutic approach. The most dangerous cancers use this distinctive metabolism, with ATP synthase distinctively in reverse, consuming glycolytic ATP, to yield high glycolytic rate, thence abundant glycolytic intermediates for biosynthesis and, crucially, to keep [ROS] low (as prior disclosed), which is necessary to cancer immortality (limitless replicative potential) and thence danger. This distinction is targeted, by compound(s) of this disclosure, without significant damage to normal cells. Normal adult cells normally use a different metabolism, with ATP synthase more in forward mode, and a higher ATP yield from glucose, but at the cost of higher [ROS] and mortality.

This reliance of normal cells upon the forward mode of ATP synthase makes them exquisitely susceptible to oligomycin. The compounds of this disclosure are useful for anti-cancer treatment, unlike oligomycin, because of their distinction from oligomycin, which couldn't have been foreseen without the inventive steps of this disclosure. In normal cells that are actively respiring (known as state 3 respiration [3]), inhibitors of the forward mode of ATP synthase (e.g. oligomycin) cause a state 3 to state 4 transition, hyperpolarize ψ_{IM} , decrease O_2 consumption and reduce [ATP] (so called “modulators” of the forward mode of ATP synthase, e.g. Bz-423, can also cause one or more of these effects) whilst a specific inhibitor of the reverse mode of ATP synthase does not exert these effects at a working concentration ([12-13], herein incorporated in their entirety). However, at this working concentration, after inhibition of the respiratory chain (e.g. blocked by rotenone, or some other respiratory chain inhibitor, or by a reduced O_2 concentration), a specific inhibitor of the reverse mode of ATP synthase will depolarise ψ_{IM} . This feature distinguishes a molecule that inhibits the reverse mode of ATP synthase significantly more than it inhibits/modifies the forward mode of ATP synthase, and/or inhibits/modifies ATP synthesis. Such a molecule, put into use as an anti-cancer therapeutic, is an embodiment of this invention. A further embodiment is the process/method of seeking new anti-cancer molecules by assaying whether a candidate molecule can depolarise ψ_{IM} when ψ_{IM} is maintained by F_1F_0 ATP hydrolysis (e.g. when OXPHOS is blocked by a respiratory chain inhibitor or insufficient O_2), but that can't hyperpolarize ψ_{IM} and/or decrease O_2 consumption, when ψ_{IM} is maintained by proton pumping by complexes of the respiratory chain. If a candidate molecule meets these requirements, it is an anti-cancer therapeutic, as determined by the invention of this disclosure.

Some cancers intrinsically rely upon ATP synthase in reverse, as revealed by experimental data of this disclosure, and further cancers can have this reliance imposed upon them, to maintain ψ_{IM} in the hypoxia of a solid tumour, which also makes them susceptible to drugs of this disclosure. Significant lactate release is correlated with the most dangerous cancers and poor patient outcomes (numerous studies find this: example: [22]). High lactate release indicates high glycolytic rate, which F_1F_0 ATP hydrolysis enables, and which drugs of this disclosure attack. This invention confronts the most deadly cancers by discovering/disclosing a cancer-specific weakness, and the means to selectively attack it.

All the following molecules are—in use as anti-cancer therapeutics—embodiments of this invention: (1) Molecules that inhibit the reverse, and not the forward, mode of ATP synthase, (2) Molecules that inhibit the reverse more than than forward mode of ATP synthase, (3) Molecules that inhibit the reverse mode of ATP synthase, and not its forward mode, but that shuttle protons across the mitochondrial inner membrane, dissipating the pmf as heat (uncoupling [3]), which reduces F_1F_0 ATP synthesis, and in a further embodiment: uncoupling molecules that reduce F_1F_0 ATP hydrolysis more than F_1F_0 ATP synthesis, (4) Molecules that inhibit ATP hydrolysis more than ATP synthesis at the mitochondrial inner membrane, (5) Molecules that have a lower IC_{50} or EC_{50} for F_1F_0 ATP hydrolysis than F_1F_0 ATP synthesis. This invention discloses the process/method of using one or more molecular species, each with one or more of the characteristics in the aforementioned numbered points, as an anti-cancer medicine or treatment. Some examples are presented in this disclosure. Any cancer therapy or treatment or drug that leverages, relies upon, utilises or targets that cancers employ ATP synthase in its reverse mode is an embodiment of this disclosure.

Mechanistic Distinction From Polyketide F_1F_0 ATP Synthase Inhibitors

The compounds of this invention act by a distinctly different mechanism, upon cancer cells, than oligomycin. Drugs that act against the same molecular target have a similar pattern of activity against the different cancer cell lines of the NCI-60 assay i.e. the smaller, and the larger, of their GI50 values are against the same cell lines (GI50 is compound concentration that causes 50% growth inhibition of a cell line relative to no-drug control). The degree of (dis)similarity can be measured using the COMPARE algorithm [23-24], which employs a Pearson correlation coefficient. For example, [25] found that the COMPARE algorithm can successfully group different FDA-approved anti-cancer drugs by their method of action using their NCI-60 GI50 data. Oligomycin A (NSC: 717694 [16]) inhibits F_1F_0 ATP synthase [4, 14] and so do other polyketides: cytovaricin (NSC: 349622 [16]), ossamycin (NSC: 76627 [16]) and peliomycin (NSC: 76455 [16]); indeed, their NCI-60 pattern responses (GI50 values) correlate with that of oligomycin A: 0.896, 0.809 and 0.865 respectively (COMPARE algorithm output, all significant at $p < 0.05$). However, the NCI-60 pattern response (GI50 values) of BMS-199264 is uncorrelated to that of oligomycin A (0.009). This mechanistic distinction is vital because polyketide F_1F_0 ATP synthase inhibitors are poisonous to normal cells [15], which means they fail in cancer xenograft mouse experiments [16] and are without clinical utility.

Higher HIF-1 α (and lower pyruvate kinase {liver isoenzyme}, lower aspartate aminotransferase 2 {mitochondrial} and lower ATP synthase) gene expressions are reported to be a marker of the Warburg effect [14] and correlate (at $p < 0.05$) with insensitivity to the polyketide F_1F_0 ATP synthase inhibitor, cytovaricin (Table 1 of [14]). By contrast, using the same cell lines and gene expression data set used to make Table 1 of [14], BMS-199264 sensitivity (GI50) does not correlate (at $p < 0.05$) with any of these gene expressions. And actually higher HIF-1 α expression, a marker of the Warburg effect, correlates (0.714, but statistically insignificant at $p < 0.05$) with higher sensitivity to BMS-199264. FIG. 5 of [14] presents apotolidin resistant NCI-60 cell lines, resistant because they utilise the Warburg effect [14], but the majority of these cell lines are more sensitive to BMS-19264 than the average, with a lower GI50 value than the average GI50 value (3.9 μ M) for BMS-199264.

The lower the bioenergetic cellular index (BEC) of a cancer cell [18], the more it demonstrates the Warburg effect and the more it relies on glycolytic rather than oxidative metabolism. BEC is, by one measure [19], the ratio amount of the β subunit of F_1 ATPase (β -F1-ATPase; gene: ATP5B) to that of Glyceraldehyde 3-phosphate dehydrogenase (GAPDH). I calculated BEC for the same cell lines analysed for Table 1 of [14], using the mRNA transcript amounts of ATP5B and GAPDH in each cell line, data sourced from [26-27], and then calculating the $([ATP5B]/[GAPDH])$ transcript ratio) for each of these cancer cell lines. Using transcript data rather than protein data is a limitation, but [28] report that a protein's cellular amount is generally well correlated (0.76) to its mRNA transcript amount, at least for cells in the NCI-60 assay, for the protein subset they studied. And furthermore, [14] relied on transcript data, so best comparison with [14] is made using such data. Polyketide F_1F_0 ATP synthase inhibitors don't work well against cancer cells exhibiting the Warburg effect [14] and, indeed, for the cell lines analysed (same ones used as for Table 1 in [14]) there is a significant (at $p < 0.05$) negative Pearson correlation between $\log_{10}(\text{GI50})$ and BEC for oligomycin A (-0.9411). So, this correlation shows that the more a cancer uses Warburg metabolism, the less its danger is mitigated by oligomycin A. This significantly reduces the utility of oligomycin A as a cancer medicine because a low BEC score (indicating Warburg metabolism) is characteristic to some of the most dangerous cancers, with the worst patient outcomes [18-20]. By contrast, there is no significant (at $p < 0.05$) Pearson correlation (+0.3639) for BMS-199264 and BEC. This means that, distinctly from the polyketide F_1F_0 ATP synthase inhibitors, its anti-cancer action is not restricted to those, often less dangerous, cancers that don't utilise Warburg metabolism.

Molecules of this disclosure undermine cancer by inhibiting the reverse mode of ATP synthase. It is true that polyketide F_1F_0 ATP synthase inhibitors also inhibit this mode, but distinctly, in addition, they also inhibit the forward mode of ATP synthase, indeed more potently [11], and whilst they can exert anti-cancer activity, because this forward mode is vital to many cancers, it is also vital to many normal cells. This makes polyketide F_1F_0 ATP synthase inhibitors unsuitable as clinical molecules. Molecules of this disclosure are therapeutic because of their distinction from, not their similarity to, polyketide F_1F_0 ATP synthase inhibitors.

[14] sum up with "Many cancer cells maintain a high level of anaerobic carbon metabolism even in the presence of oxygen, a phenomenon that is historically known as the Warburg effect. From our results, we conclude that macrolide inhibitors of the mitochondrial F_0F_1 -ATP synthase selectively kill metabolically active tumor cells that do not exhibit the Warburg effect". So, [14] find that these macrolides only kill cancers reliant upon OXPHOS, so using F_1F_0 -ATP synthase in its forward mode to generate ATP (which unfortunately is also the metabolic profile of many key types of normal cell) and thus macrolide inhibition of the forward mode of F_1F_0 -ATP synthase is key to this (unspecific) anti-cancer activity. By contrast, the molecules of this disclosure exert anti-cancer activity by inhibition of the reverse mode of ATP synthase. BMS-199264 [4, 7, 9, 10, 11], BTB06584 [13] and 19a [5] have been described previously, as molecules that can inhibit this mode, and this invention discloses their utility as anti-cancer therapeutics, with supporting experimental data, thence identifying new cancer drugs and, more importantly, a new cancer specific drug target: F_1F_0 -ATP hydrolysis, which is the most funda-

mental invention of this disclosure. BTB06584 (100 μM) exerts anti-cancer activity (FIG. 2), despite not inhibiting F_1F_0 -ATP synthesis, as a function of inhibiting F_1F_0 -ATP hydrolysis (at ≥ 100 μM), and critically it isn't harmful to normal cells (mouse cortical neurons) at this concentration [13]. Its anti-cancer potency (none at 10 μM , observed at 100 μM) matches its inhibitory potency for F_1F_0 -ATP hydrolysis (none at 10 μM , requires ≥ 100 μM [13]). BMS-199264 (10 μM) exerts anti-cancer activity (FIG. 3). It doesn't harm normal cells (ex vivo rat heart) at this concentration [11]. In NCI five-dose testing [29-30], the mean GI50 for BMS-199264 is 3.9 μM (FIG. 7), which is lower/better than 62% of the 102 FDA approved cancer drugs in [24], their mean GI50 values sourced from Table 1 of [24]: all are directly comparable because they too are sourced from the NCI-60 five-dose assay. At 10 μM , 19a exerts more anti-cancer activity than BMS-199264 (FIG. 6), despite it having less effect on F_1F_0 -ATP synthesis, because it inhibits the reverse mode of ATP synthase more potently than BMS-199264. Again, a vindication that the molecules of this disclosure exert anti-cancer activity by inhibiting the reverse mode of ATP synthase, which distinguishes them from the macrolides and, distinctly, makes them usable therapeutically. Indeed, molecules of this disclosure don't appreciably inhibit the forward mode of ATP synthase, in sharp distinction to the macrolides. The compounds that contain a protonable nitrogen atom in their imidazole reduce F_1F_0 -ATP synthesis in SMPs because they shuttle protons across the mitochondrial inner membrane, dissipating the proton motive force (uncoupling). FIG. 8 presents structure-activity data for such uncoupling in whole cells, using compounds that are also componentry to this invention as anti-cancer drugs. BMS-199264 ($\log P=3.79$, calculated [31]) uncouples more than 19a ($\log P=5.97$, calculated [31]) because its $\log P$ is closer to the $\log P=3.2$ (calculated) optimum for uncoupling [32].

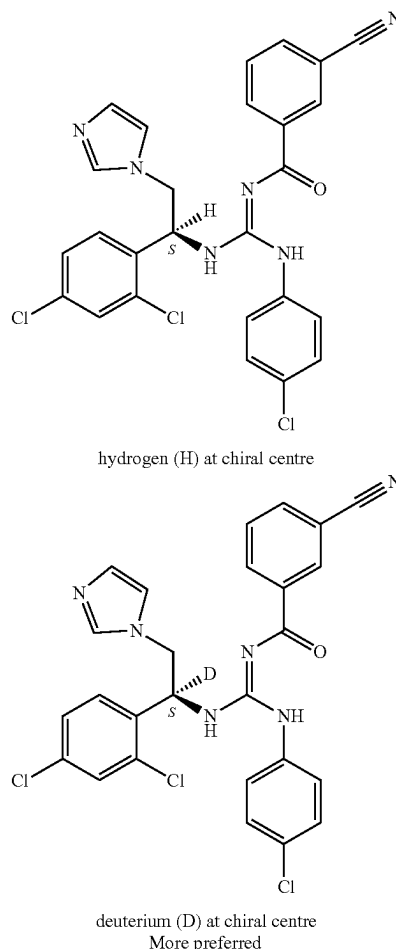
19a is a racemate, wherein the S stereoisomer, and not the R stereoisomer, inhibits F_1F_0 -ATP hydrolysis [5-6]. I tried to test the anti-cancer activity of the separated stereoisomers. They were successfully separated by superfluid chromatography (SFC). But subsequently underwent racemization during the NCI-60 tests. One stereoisomer sample conveyed slightly better anti-cancer activity than the other, revealing a slight enduring enantiomeric excess (ee) of S stereoisomer. Both samples ultimately contained a significant proportion of S stereoisomer and both had strong anti-cancer activity (FIG. 5). The Pearson correlation coefficient ($R=0.8$) for their patterns of anti-cancer activity is significant (at $p<0.05$). Racemization of the S stereoisomer is slowed by replacing the hydrogen atom on its chiral carbon with a deuterium atom and this is a new composition of matter, which is componentry to this invention, as is the method/process of using it for anti-cancer therapy. With this modification, the enantiomeric excess (ee) of the eutomer endures for longer and so per-unit anti-cancer activity is better, for longer. Analogy by the macrolide inhibitors of [14] would suggest that the S and R stereoisomers have equal anti-cancer activity, and that this would be weak, because they are both comparably weak reducers of F_1F_0 -ATP synthesis ($\text{EC}_{50}>100$ μM in SMP assays). By contrast, by the invention of this disclosure, the S stereoisomer specifically is revealed to be a potent anti-cancer therapeutic.

Stereoisomerism

For some molecules of this disclosure, one of its stereoisomers has much lower IC_{50} than the other for inhibiting F_1F_0 -ATP hydrolysis, and so, by the invention of this disclosure, this is the preferred stereoisomer for anti-cancer

use. Indeed, a form with high enantiomeric excess (ee) for this preferred stereoisomer is the preferred embodiment for anti-cancer therapy, e.g. $\text{ee}\Rightarrow 70\%$, $\text{ee}\Rightarrow 95\%$, $>99\%$ more preferred, $=100\%$ most preferred. However, ee can be eroded by racemization. This invention discloses an improvement. Embodied by this disclosure are permutations of each of its chiral molecules, wherein the hydrogen attached to each chiral carbon is replaced with a deuterium, wherein the natural abundance of deuterium (0.015%) at this position is enhanced (non-limiting example: >3000 times greater than the natural abundance of deuterium, i.e. a $>40\%$ incorporation of deuterium). The deuterium Kinetic Isotope Effect (KIE) [33] slows racemization.

Best Mode



The structure on the left has a low EC_{50} against F_1F_0 -ATP hydrolysis (0.018 μM), its [EC_{50} F_1F_0 ATP synthesis/ EC_{50} F_1F_0 ATP hydrolysis] ratio $>5,556$. In rats, this drug (administered in polyethyleneglycol:water:ethanol, 1:1:1) is orally bioavailable (47%) with good pharmacokinetics (intravenously applied drug half-life in blood=2.1 hours, $C_{\text{max}}=21$ μM , volume of distribution=2.37 L/kg). The deuterated analogue on the right, wherein the hydrogen atom on the chiral carbon is replaced with deuterium, conferring greater stereoisomeric stability because of the kinetic isotope effect (KIE, [33]), is more preferred and is the best mode anti-cancer therapy. The greater the % deuterium enrichment at the chiral carbon (carbon atom number 21) and the greater the enantiomeric excess, the more preferred the embodiment.

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The most valuable innovation of this invention is not a presented structure but a discovered, disclosed principle: the best anti-cancer compound of this invention is a molecule that inhibits F_1F_0 ATP hydrolysis as potently and specifically as possible, whilst it inhibits, by direct binding, the forward mode of the ATP synthase molecule as little as possible: most preferably not at all.

Cachexia

Inhibiting F_1F_0 -ATP hydrolysis, as compounds of this invention do, treats/ameliorates/prevents/combat the cachexia driven by some highly glycolytic (Warburg) cancers. These cancers rely upon F_1F_0 ATP hydrolysis to burn glycolytic ATP, to release glycolysis from ATP feedback inhibition and permit a high glycolytic rate, with high lactate efflux, which the liver converts back to glucose at an energetic cost (Cori cycle, 6 ATP per lactate to glucose conversion [1]), which the cancer then recycles to lactate de novo: iteratively this contributes to cachexia. Compounds of this invention treat/ameliorate/prevent/combat this cachexia drive. In the next section, a 2nd mechanism that treats/ameliorates/prevents/combat cachexia is disclosed, whereby compounds of this invention inhibit F_1F_0 -ATP hydrolysis in normal, rather than cancer, cells.

Temperature

An effective amount of a compound of this invention, when applied to a subject, inhibits F_1F_0 -ATP hydrolysis, which conserves ATP, so less ATP needs to be synthesized, therefore respiration rate slows, thence metabolic heat production falls and body temperature can fall towards ambient temperature (if ambient < body temperature). So, when the ambient temperature isn't arduous (not requiring significant energy consuming physiological/behavioural adaptations to maintain body temperature) and dietary intake stays constant, weight gain/maintenance can occur, which can assist cachexia, for example cancer driven cachexia. This is clinically valuable because cachexia is the leading cause of death in cancer patients. If the ambient temperature is sufficiently close to the required body temperature, then the aforementioned decrease in heat generation is safe, because the body temperature can't fall below the ambient temperature. So, for example, if the ambient temperature is 37° C., inhibiting F_1F_0 -ATP hydrolysis could make body temperature fall to this ambient temperature, but not below it, and this is safe because ~37° C. body temperature is safe. Inhibiting F_1F_0 -ATP hydrolysis will reduce, but not abolish, metabolic heat production. So, body metabolism will still contribute to heating the body, just less so, which will shift the thermoneutral and thermal comfort zones (terms well known to those of the art [34], temperatures vary by species, as is well known to those of the art) to higher temperature(s). If the subject is located at a higher temperature to account for this shift, for example at their updated, higher thermoneutral temperature, or make behavioural adaptations (e.g. wearing more clothes), then this shift is harmless. An embodiment of this invention is setting the dosage of a compound(s) that inhibits F_1F_0 ATP hydrolysis with consideration of the ambient temperature, wherein higher dosages are permissible at higher ambient temperatures. The preferred ambient temperature for a dosage permits the subject to be thermoneutral, and/or thermal comfortable, without the metabolic heat (respiration) fraction driven by the F_1F_0 ATP hydrolysis that is lost because of this dosage. This temperature management issue is more important for smaller than larger animals, because surface area scales to volume by a fractional power (e.g. refer Kleiber's law) and so larger animals retain their generated heat better, and so a given percentage drop in (per unit mass) metabolism will cause a

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smaller drop in body temperature in a bigger animal. The aforementioned weight gain can be of great clinical/health/nutritional value, or aesthetic value (by non-limiting example: bodybuilders), or commercial value when applied to livestock/farm animals or any animal with a commercial value e.g. racing animals, such as horses. This invention encompasses the method/process of using molecules of this disclosure for these applications, or any others wherein weight, nutritional or energetic gain is wanted in an animal or human. An embodiment of this invention is a method in which a subject takes or is administered an effective amount of a compound(s) of this invention, for example a compound of Formula (I), (II), (III), (IV) or (V) or another compound that selectively inhibits F_1F_0 ATP hydrolysis, or a pharmaceutically-acceptable salt, solvate, hydrate or prodrug thereof, to treat/ameliorate/prevent/combat cachexia, cancer-associated/driven cachexia, weight loss or a disease or disorder that causes a higher than normal body temperature which can include, but isn't limited to, fever, pyrexia, hyperpyrexia, hyperthermia, malignant hyperthermia, neuroleptic malignant syndrome, serotonin syndrome, thyroid storm, or to cause greater metabolic/bioenergetic efficiency in the subject, enhancing their physical or mental performance or causing body weight gain.

This temperature aspect to the molecules of this disclosure isn't relevant to the NCI-60 studies. Because in these studies, the ambient temperature is controlled at 37° C. [30], which is optimal for cells, and so if these drugs make cellular temperature fall to ambient temperature, this is not detrimental. It can be an issue for laboratory animal studies though. Laboratory mice, for example, are typically kept at room temperature (e.g. 20 to 23° C.) which renders them very reliant upon additional metabolic/physiological/behavioural heat production because their thermoneutral zone is much higher, at 30 to 32° C. (can vary depending on strain, size, age, gender etc. [34]). An administered compound(s) of this disclosure, which inhibits F_1F_0 ATP hydrolysis, can add to the cold stress that laboratory mice endure when kept at room temperature. An embodiment of this invention is the process/method of keeping laboratory animals at, or close to, their thermoneutral zone when performing animal studies with a compound(s) of this disclosure. For example, keeping mice at 30 to 32° C. And in a further embodiment, at even higher temperature to compensate for the amount that an administered compound(s) of this disclosure, by inhibiting F_1F_0 ATP hydrolysis, shifts the animal's thermoneutral zone to a zone of higher temperature. The amount shifted will depend on the administered dosage, so in a further embodiment, the ambient temperature is set according to the dosage used. Wherein, for a compound of this disclosure, a higher ambient temperature, within safe limits, can make a greater compound dosage safer.

An embodiment of this invention is a method in which a subject takes or is administered an effective amount of a compound(s) of this invention, for example a compound of Formula (I), (II), (III), (IV) or (V) or another compound that selectively inhibits F_1F_0 ATP hydrolysis, or a pharmaceutically-acceptable salt, solvate, hydrate or prodrug thereof, to treat/ameliorate/prevent/combat a medical disease/disorder, wherein the subject is monitored, for example by a healthcare/research professional/worker (doctor, nurse, vet, pharmacist, laboratory technician, scientist) or machine/artificial intelligence substitute, for any adverse signs/symptoms/non-normality after compound administration (in an embodiment for 5 minutes, in a further embodiment for 10 minutes, and in a further embodiment for longer) and in a particular embodiment for signs of reduction in body temperature

(methods well known to those of the art, in a particular embodiment the subject's body temperature is monitored) and/or the dosage administered is set, and/or modified (e.g. increased in graduations), by information from this subject wellness/normality/temperature monitoring and/or the subject is located at an ambient temperature (e.g. in a temperature controlled room/enclosure/confine or climate) that maintains their body temperature within safe limits whilst they have an effective amount of compound in their system. An embodiment of this invention is the process/method of considering the ambient temperature in the decision of whether to take or administer a compound(s) of this disclosure, and at what dosage. In an embodiment, a period of medical observation, by a clinical or healthcare professional (e.g. pharmacist), occurs after the subject takes or is administered a compound(s) of this disclosure for the first time, and in a further embodiment when the compound dosage is increased or decreased. In a further embodiment, during this period of medical observation, the subject stays in a location that has medical facilities and/or expertise to treat/combate hypothermia (well known to those of the art), in non-limiting example embodiments this is a hospital or clinic or pharmacy or workplace of healthcare professionals. In an embodiment, during this period of medical observation, the patient stays in a temperature controlled room or area, or at a location where one is available nearby, and if the patient displays signs or symptoms of hypothermia, feels uncomfortable, or their body temperature falls, they can be located in a higher ambient temperature. In an embodiment, while the subject takes or is administered a compound(s) of this invention, or in a monitoring period after it, they stay in a room at a safe ambient temperature for having a compound(s) of this disclosure (non-limiting examples: wherein the ambient temperature is close to the desired body temperature, $\sim 37^{\circ}\text{C}$., or exceeding it within safe limits) and are monitored by observation, and in a further embodiment their body temperature is monitored (methods well known to those of the art), as the controlled room temperature is reduced to a different temperature, in a further embodiment to, at or near, the ambient climatic temperature of that geography at that time, or colder. In a further embodiment, this process/method is iterated until the greatest dosage is found at which the subject has a safe body temperature at, or near, the ambient climatic temperature of that geography at that time or at the ambient temperature(s) at which the subject will spend their time at over their course of compound administration, or that their ambient temperature might fall to at some time over their course of compound administration, wherein the course of compound administration is the period during which the subject has an effective amount of compound in their body.

Uncoupling is Virtuous

The imidazole containing compounds of this disclosure inhibit F_1F_0 ATP hydrolysis and uncouple (shuttle protons across the mitochondrial inner membrane (IM), eroding the proton motive force, pmf). The former can exert a specific anti-cancer activity, because it undermines the means some cancers maintain ψ_{IM} in normoxia (experimentally shown by data of this disclosure) or in hypoxic tumours, and the compound's uncoupling can also exert specific anti-cancer activity, explained now. The imidazole containing compounds of this disclosure bind ATP synthase at the IF_1 binding site. In normal cells they bind ATP synthase at this site and are sequestered from uncoupling, and the ATP they "save" by binding and inhibiting F_1F_0 ATP hydrolysis can (over)compensate for the ATP "lost" to their uncoupling. But some cancers have very high IF_1 expression (numerous

studies show this, e.g. refer [21]). And for some cancers, this is to inhibit F_1F_0 ATP hydrolysis, to make their OXPHOS more efficient, which allows them to maintain [ATP] at low $[O_2]$, and thence survive using OXPHOS in hypoxia (their heat generation is less but their temperature is maintained by heat conduction from surrounding tissues). This high IF_1 expression blocks the binding of these compounds to their binding site on ATP synthase, so the compounds aren't sequestered from uncoupling, and this uncoupling increases the O_2 requirement of this cancer which can't be met in the hypoxic microenvironment of its tumour, thence the cancer's intracellular [ATP] can't be maintained and its proliferation is slowed and/or it dies. So, herein, this invention discloses that the uncoupling aspect to the imidazole containing compounds of this disclosure can deliver additional, specific, anti-cancer activity, for example, against those cancers that don't rely upon F_1F_0 ATP hydrolysis. This invention discloses the process/method of using a compound(s) that can inhibit F_1F_0 ATP hydrolysis, and that can shuttle protons across the IM to dissipate the pmf (uncouple), as an anti-cancer therapeutic. Wherein the compound inhibits F_1F_0 ATP hydrolysis by direct interaction with ATP synthase, and reduces F_1F_0 ATP synthesis (primarily) by uncoupling. So, a compound needn't necessarily have a much lower EC_{50} for F_1F_0 ATP hydrolysis than F_1F_0 ATP synthesis, in an SMP assay, to be componentry to this invention as an anti-cancer therapeutic. Indeed, even compounds with a lower EC_{50} for F_1F_0 ATP synthesis than F_1F_0 ATP hydrolysis in an SMP assay can be componentry to this invention, as anti-cancer therapeutics, provided they do inhibit F_1F_0 ATP hydrolysis and provided their inhibition of F_1F_0 ATP synthesis is (primarily) because of uncoupling rather than inhibiting the forward mode of ATP synthase. Oligomycin, for example, does not fit these requisites. So, this invention discloses the method of using compounds that inhibit F_1F_0 ATP hydrolase, that don't inhibit F_1F_0 ATP synthase, and that uncouple the proton motive force, as anti-cancer therapeutics.

Deuterated Compounds of the Invention

Deuterium (D or ^2H) is a stable, non-radioactive isotope of hydrogen and has an atomic weight of 2.0144. Hydrogen naturally occurs as a mixture of the isotopes ^1H (hydrogen or protium), D (^2H or deuterium), and T (^3H or tritium). The natural abundance of deuterium is 0.015%. One of ordinary skill in the art recognizes that in all chemical compounds with a H atom, the H atom actually represents a mixture of H and D, with about 0.015% being D. Thus, compounds with a level of deuterium that has been enriched to be greater than its natural abundance of 0.015%, should be considered unnatural and, as a result, distinct from their non-enriched counterparts. All percentages given for the amount of deuterium present are mole percentages. It can be quite difficult in the laboratory to achieve 100% deuteration at any one site of a lab scale amount of compound (e.g., milligram or greater). When 100% deuteration is recited or a deuterium atom is specifically shown in a structure, it is assumed that a small percentage of hydrogen may still be present. Deuterium-enriched can be achieved by either exchanging protons with deuterium or by synthesizing the compound with deuterium enriched starting materials, which are commercially available.

Deuterium enriched refers to the feature that the compound has a quantity of deuterium that is greater than in naturally occurring compounds or synthetic compounds prepared from substrates having the naturally occurring distribution of isotopes. Embodiments of this invention include compounds of Formula (I), (II), (III), (IV) and (V)

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with one or more of their hydrogen atoms replaced by deuterium, at a greater frequency than the natural abundance of deuterium (0.015%) e.g. a non-limiting example: >3000 times greater than the natural abundance of deuterium (i.e. a >40% incorporation of deuterium at a hydrogen replacement position). Additional examples of the abundance of deuterium at a position in, or positions of, a compound embodiment of this invention include 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99 to about 100%. In certain embodiments, the abundance of deuterium at a position in, or positions of, a compound embodiment of this invention is at least 40%. In certain other embodiments, the abundance of deuterium at a position in, or positions of, a compound embodiment of this invention is at least 60%. In further embodiments, the abundance of deuterium is at least 75%. In yet other embodiments, the abundance of deuterium is at least 90%. It is to be understood that the deuterium-enriched compounds described herein can be combined with a pharmaceutically acceptable carrier to form a pharmaceutical composition.

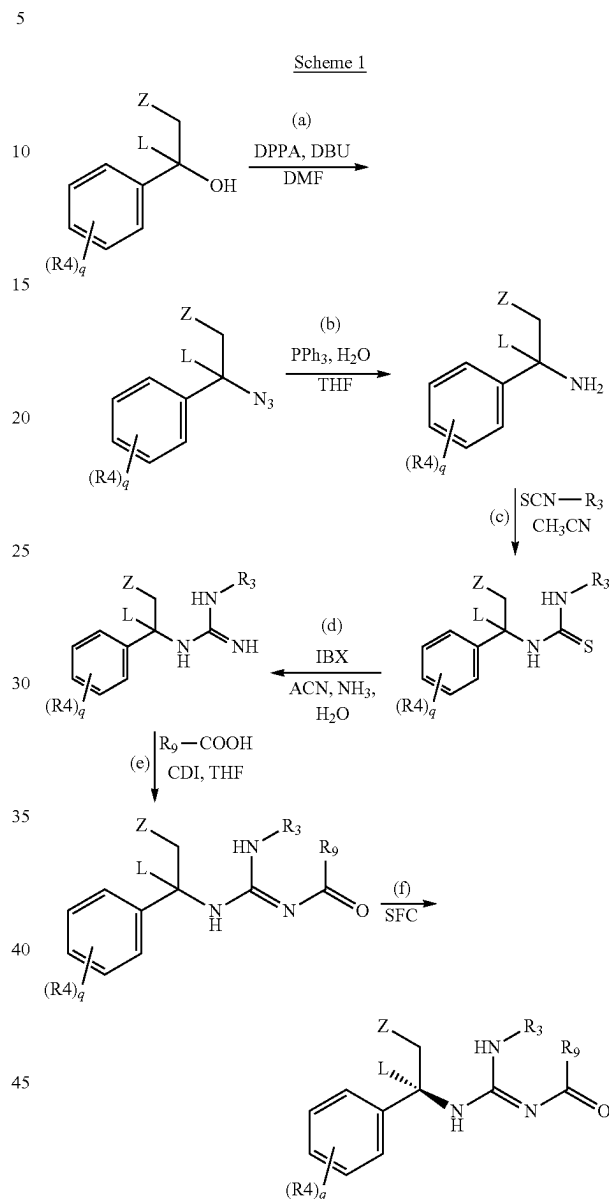
In the Description and Claims of this invention, when a position on a compound structure is designated deuterium (D), or said to have deuterium, or said to be enriched for deuterium, or even only said to be enriched, it is because the abundance of deuterium at that position is not at the natural value (0.015%) but instead, typically, in excess of 40%. The phrase 'enrichment at the chiral centre' herein, for example for a compound of Formula (I), means that the molar amount of deuterium at the chiral centre as a percentage of the total amount of all hydrogen isotopes at the chiral centre is greater than or equal to 40%, preferably greater than 40%, more preferably greater than 45%, and in ascending order of preference, $\geq 52.5\%$ deuterium enrichment at the chiral centre, $\geq 60\%$ deuterium enrichment at the chiral centre, $\geq 67.5\%$ deuterium enrichment at the chiral centre, $\geq 75\%$ deuterium enrichment at the chiral centre, $\geq 82.5\%$ deuterium enrichment at the chiral centre, $\geq 90\%$ deuterium enrichment at the chiral centre, $\geq 95\%$ deuterium enrichment at the chiral centre, $\geq 97\%$ deuterium enrichment at the chiral centre, $\geq 99\%$ deuterium enrichment at the chiral centre, $\geq 99.5\%$ deuterium enrichment at the chiral centre, 100% deuterium enrichment at the chiral centre. Greater % deuterium enrichment is preferred. Additional examples of the % deuterium enrichment at the chiral centre of a compound of this invention, for example a compound of Formula (I), are 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99 to about 100%.

Further possible isotopic variants of these structures are further embodiments of this disclosure. An embodiment of this disclosure is a molecule that inhibits the reverse mode, more than the forward mode, of ATP synthase, which has a deuterium in place of hydrogen (at a greater frequency than 0.015% e.g. >40%) at one or more places upon its structure, and/or any other isotopic substitution (at a greater than natural frequency).

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General Compound Synthesis

A general synthetic route applicable to some compounds of the invention is set out in Scheme 1 below.

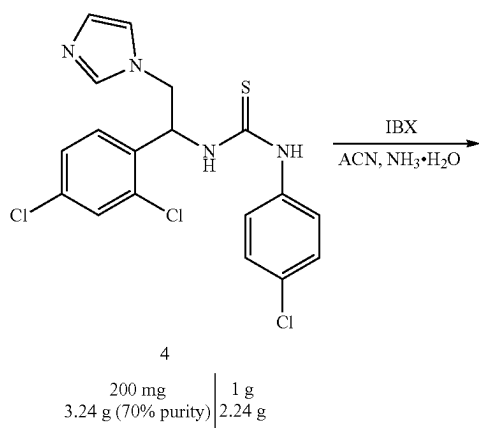
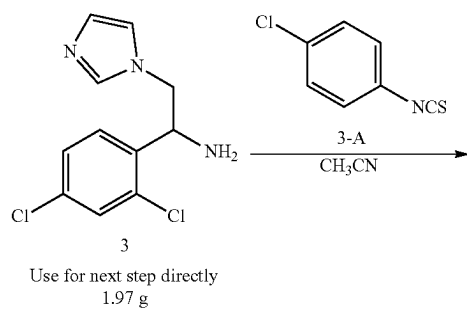
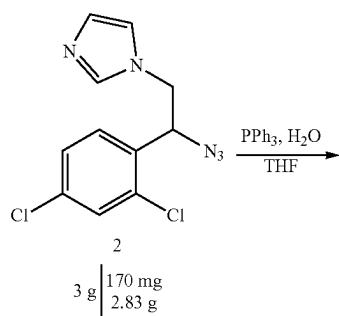
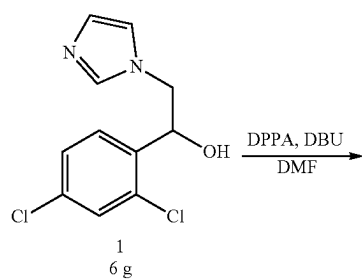


The person skilled in the art is able to make modifications to this general synthetic route, based on the common general knowledge and/or the content of prior art disclosures cited herein, in order to synthesise compounds of the invention where necessary.

Specific Compound Synthesis

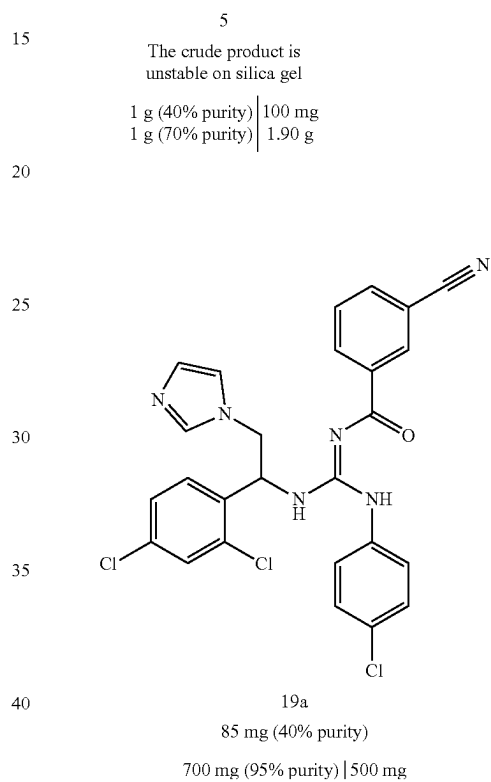
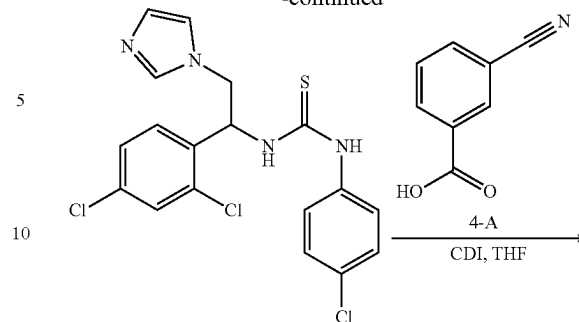
To perform the anti-cancer testing of this disclosure, BMS-199264 hydrochloride (pure BMS-199264 described in [4, 7, 9, 10, 11]) was purchased from Sigma-Aldrich. BTB06584 [13] was purchased from AdooQ Bioscience. Racemate 19a [5] was synthesised by the following synthesis route and separated into component stereoisomers using superfluid chromatography (SFC). Starting reagents were sourced commercially.

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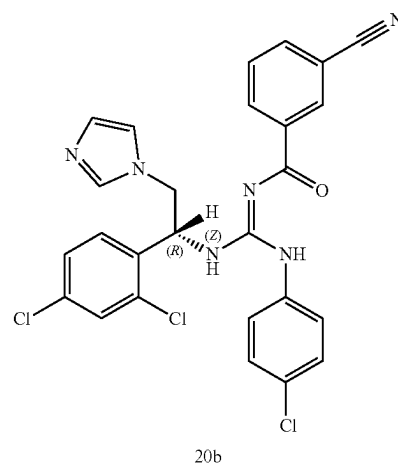
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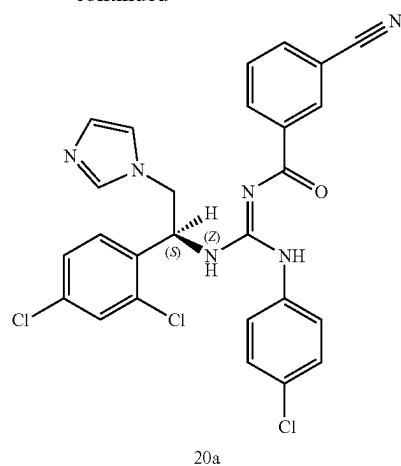
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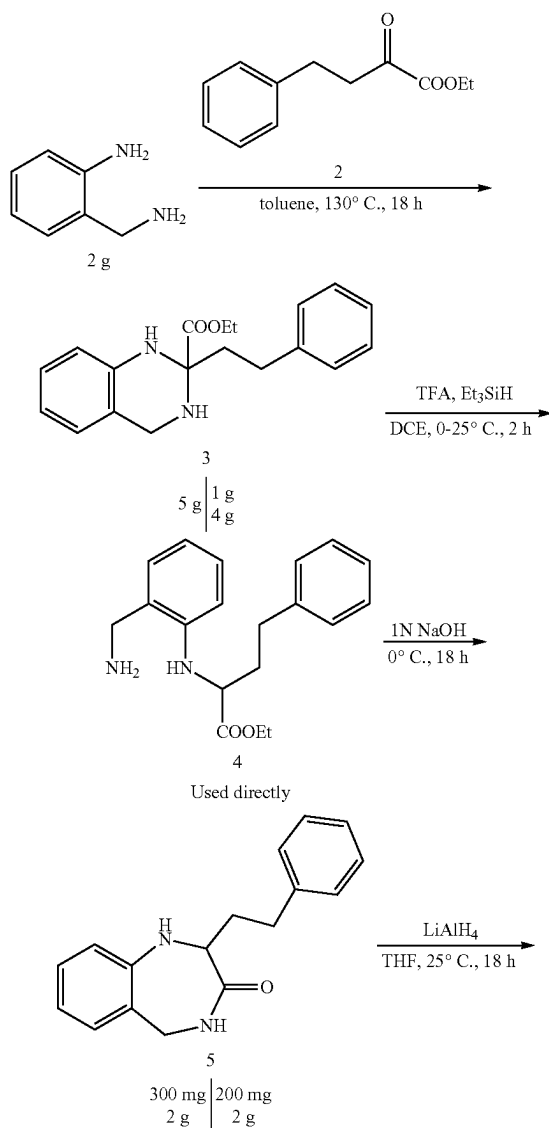


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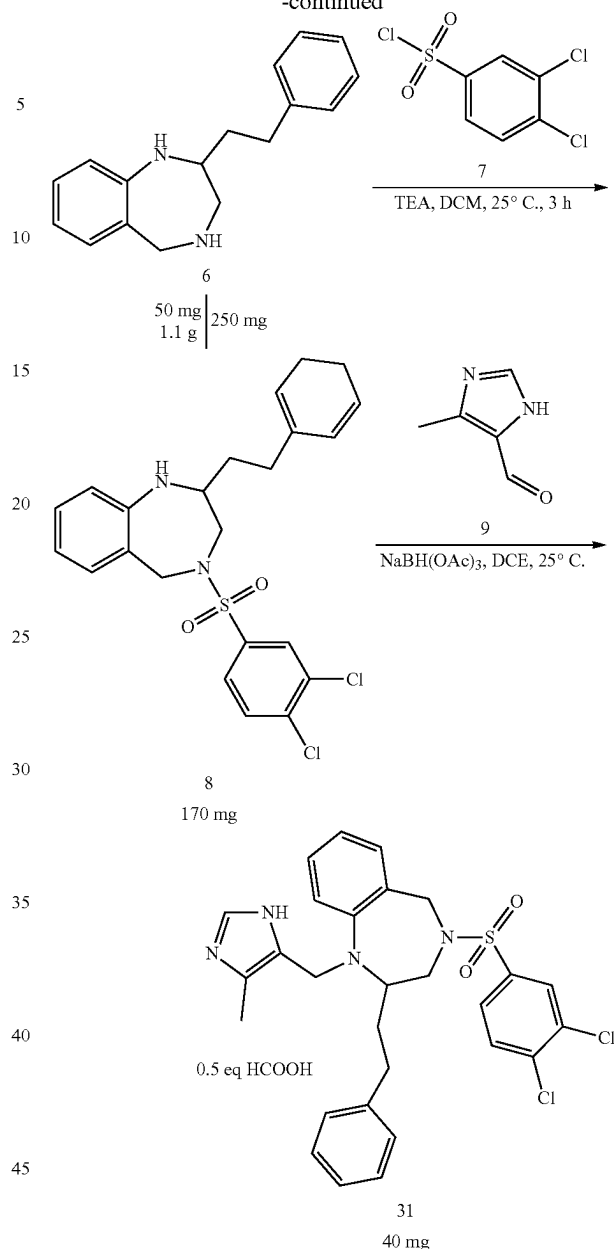


A synthesis route for structure 31 [8] as a formate.



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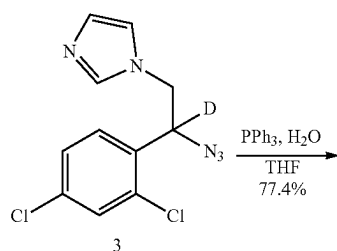
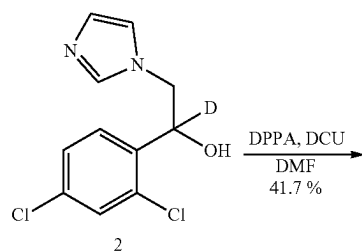
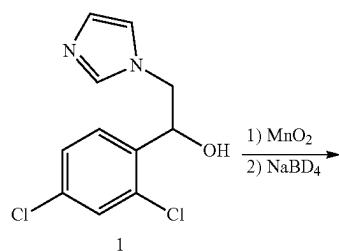
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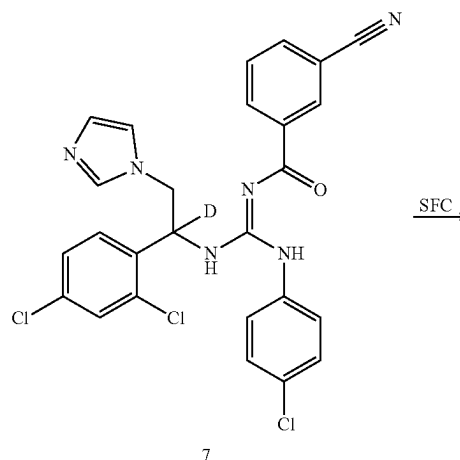
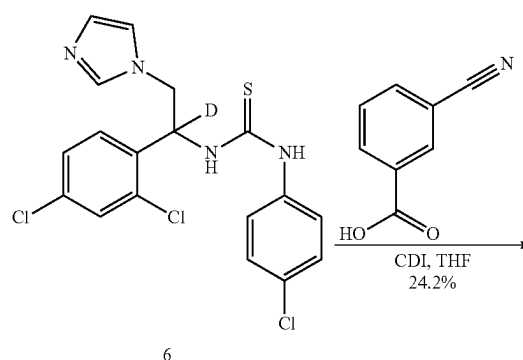
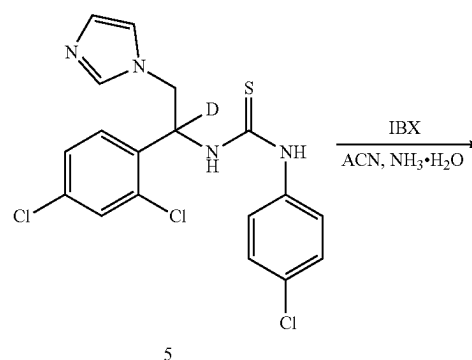
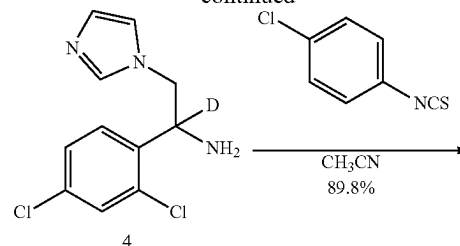
Molecule synthesis routes described in [5, 7, 8, P1, P2, P3, P4] (including references cited therein, where appropriate, and in their supplementary materials, all herein incorporated in their entirety)—for synthesizing molecules that inhibit F₁F₀ ATP hydrolysis more than F₁F₀ ATP synthesis—are componentry to this disclosure, as synthesis routes for synthesising anti-cancer molecules. In other embodiments of this invention, any given molecule synthesis route described in [5, 7, 8, P1, P2, P3, P4] is used with starting reagents, compounds, solvents and/or intermediates that have deuterium in place of hydrogen at some position(s). Such compounds are commercially available (e.g. refer C/D/N Isotopes Inc., Pointe-Claire). Or they can be created in house by invoking standard synthetic protocols known in the art for introducing isotopic atoms to a chemical structure. For example, a compound can be deprotonated by LiHMDS in tetrahydrofuran (THF) at -78 to -40° C. for 20 min,

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followed by quenching with deuterium oxide (D_2O , “heavy water”), to obtain a deuterated compound [33]. During these steps, a group upon which hydrogen is still desired over deuterium can be Boc protected and this Boc group removed subsequently using trifluoroacetic acid (TFA) treatment at room temperature. At the end, the level of deuterium can be checked by Proton NMR. The initial deprotonation step isn’t absolutely necessary as H/D exchange will occur when a molecule is quenched with D_2O , and this reaction can be catalysed, by acid, base or metal catalysts such as platinum. If, after D_2O quenching, the level of compound deuteration is insufficient (observed using Proton NMR) then the compound is quenched with D_2O , or some other deuterium containing solvent, for a longer period of time. Compounds of this disclosure can be synthesised in D_2O , during one or more chemical steps, or a starting compound, intermediate or final molecule of this disclosure can be incubated in D_2O to produce a deuterated version(s). So, deuterium-enriched compounds of this invention can be prepared by substituting a deuterium-enriched reagent or solvent for a non-isotopically labeled reagent or solvent in the synthetic schemes reported in [5, 7, 8, P1, P2, P3, P4]. An example is presented below, which shows a modified synthesis route from that presented previously, in order to produce a deuterated analogue, with deuterium in place of hydrogen on the chiral carbon. The scheme is provided for the purpose of illustrating the invention, and should not be regarded in any manner as limiting the scope or the spirit of the invention.

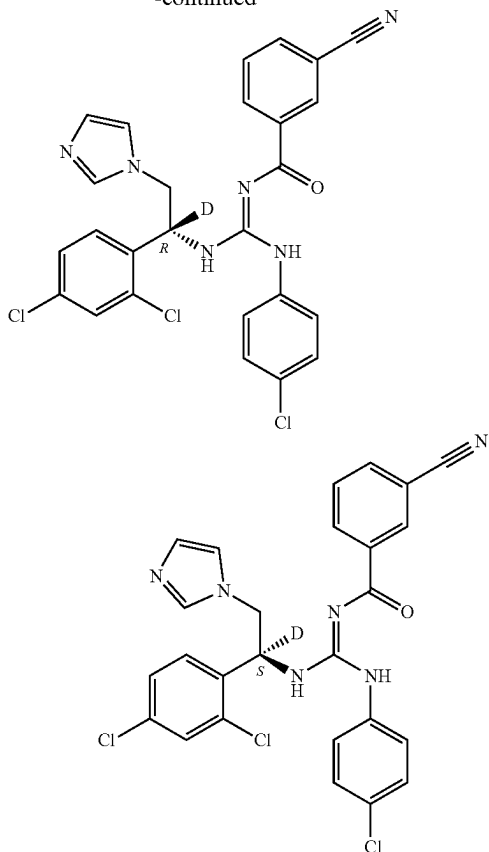
**22**

-continued



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-continued



Methods to Find Further Compounds Component to This Invention

A method to find compounds that slow the proliferation of and/or kill cancer cells is by screening/seeking compounds that preferentially inhibit the reverse mode of ATP synthase. For example, by separately assaying (in space and/or time) the compounds's effect upon ATP synthesis and ATP hydrolysis by ATP synthase (in its entirety or, less preferably, a component part of it). Then comparing these assay results. The greater the inhibition of reverse vs. forward mode, the more preferred this compound is for anti-cancer use. A further method is by screening/seeking compounds that inhibit ATP hydrolysis more than synthesis in submitochondrial particles (SMPs). ATP hydrolysis can be assayed by (non-limiting example) a spectroscopic assay for NADH fluorescence that incubates the SMPs with pyruvate kinase and lactate dehydrogenase enzymes (assay well-known to those of the art). ATP synthesis can be assayed by (non-limiting example) a spectroscopic assay for NADPH fluorescence that incubates the SMPs with hexokinase and glucose-6-phosphate dehydrogenase enzymes (assay well-known to those of the art). These assays are reported in in any one of [5, 7, 8, 11, 12, 13, 36], and/or as referenced therein, all of which are herein incorporated in their entirety. In these SMP assays, the criteria for a candidate anti-cancer compound is a low EC_{50} against ATP hydrolysis (thence anti-cancer activity) and a higher EC_{50} against ATP synthesis (thence safe for normal cells).

Cancer Types Particularly Targeted by This Invention

Particularly vulnerable to compounds of this invention: cancers that exhibit the Warburg effect (i.e. that produce ATP primarily by glycolysis, rather than oxidative phosphory-

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lation, even in abundant O_2), highly glycolytic cancers (which metabolize glucose and/or glutamine to lactate rather than metabolizing one or both fully with the use of oxidative phosphorylation) and cancers that reside in hypoxia, which forces them to produce ATP primarily by glycolysis. As explained in a preceding section, the imidazole containing molecules of this disclosure, with their uncoupling capability, can also attack cancers that reside in hypoxia, which use high IF_1 expression to enable oxidative phosphorylation at low $[O_2]$. Many cancers reside in hypoxia as tumours are often hypoxic.

So, if a cancer is highly glycolytic, either because of the Warburg effect (inherent glycolytic metabolism, regardless of $[O_2]$) or because of residing in hypoxia (imposed glycolytic metabolism, because of low $[O_2]$), or uses oxidative metabolism but resides in hypoxia (survival enabled by high IF_1 expression), it will be treated/ameliorated/prevented/combated by a compound of this invention. How to identify such cancers?

Cancers exhibiting the Warburg effect, or that have an imposed (by low $[O_2]$) glycolytic metabolism, are those that show up in positron emission tomography (PET) imaging using fluorine-18 (^{18}F) fluorodeoxyglucose (FDG), ^{18}F -FDG PET, optionally integrated with computed tomography (CT) [40]. FDG is a glucose analogue and glycolytic cancers take up more FDG than their surrounding tissue because glycolysis is an inefficient metabolism of glucose (yielding only ~ 2 ATP per glucose compared to ~ 31 ATP per glucose yielded by aerobic respiration [1-2]) and so they must uptake more glucose to obtain even the equivalent energy yield to nearby normal cells, which are using oxidative metabolism, as most normal cells do. So, if a cancer presents in this FDG-PET diagnostic (higher glucose uptake than surround), it is susceptible to a compound of this invention. Highly glycolytic cancers also release much lactate. So, if a patient has a high blood lactate level, noticeably above the normal non-pathological range, as clear to someone of the art, then their cancer is susceptible to a compound of this invention. Higher lactate levels in and around the cancer or tumour (than surrounding tissue) can also be detected using imaging technologies, for example 1H Magnetic Resonance Spectroscopy (1H -MRS) or chemical exchange saturation transfer magnetic resonance imaging (CEST MRI) [41], or other imaging modalities and methods of the art. So, if a cancer presents (higher [lactate] than surround) in a lactate imaging diagnostic it is susceptible to a compound of this invention. Cancer release of lactic acid acidifies its extracellular space and this acidification can be detected by imaging modalities, well known to those of the art e.g. [42-43], and if a cancer can be discriminated from its surrounding tissue by this method then it is susceptible to a compound of this invention. An oxygen-sensitive chemical probe can be used to obtain 3D maps of tissue pO_2 [44], and if a cancer is shown to reside in notable hypoxia then it is susceptible to a compound of this disclosure, because it is either glycolytic or using high IF_1 expression to enable oxidative metabolism, both of which make it susceptible to a compound of this invention. Imaging technologies can be integrated to improve signal to noise e.g. [44] integrate pO_2 and lactate imaging. Such integration can give added information: for example, a cancer producing much lactate in a high pO_2 environment is exhibiting the Warburg effect because it is heavily utilising glycolytic metabolism in abundant O_2 . Cancer gene expression markers and indicators of the Warburg effect, well known to those of the art e.g. [18-20], specify that a cancer is susceptible to a compound of this invention, wherein the cancer's genetic material can be

retrieved by biopsy, surgery, cancer cells or parts circulating in the bloodstream or some other method of the art.

If a cancer uses oxidative phosphorylation (OXPHOS) rather than glycolytic metabolism, and it does not already improve its OXPHOS efficiency by high IF_1 gene expression (which many cancers do e.g. refer [21]) then a compound of this invention, by preferentially inhibiting F_1F_0 ATP hydrolysis, will confer this efficiency gain and actually assist, rather than harm, this cancer. How to identify these cancers? A cancer's IF_1 gene expression, and particularly its gene expression ratio of IF_1 to a core ATP synthase sub-unit (e.g. ATP6), is informative. More so if compared to the corresponding gene expressions in a normal cell of its host tissue, so detecting difference from normal. If a cancer uses oxidative, rather than glycolytic, metabolism and does not have an appreciably higher IF_1 (or $IF_1/ATP6$ ratio) gene expression than its corresponding normal tissue then it isn't prudent to use a compound of this invention for cancer therapy. More simply, it is best to use a compound(s) of this invention against highly glycolytic cancers and some (non-limiting) imaging methods have been described herein to identify these.

This invention discloses a method of using a compound(s) that preferentially inhibits the ATP-hydrolysing mode of ATP synthase, for example a compound(s) of Formula (I), (II), (III), (IV) or (V), or a pharmaceutically-acceptable salt, solvate, hydrate or prodrug thereof, to treat/ameliorate/prevent/combat a cancer that preferentially uses glycolytic rather than oxidative metabolism, for example a cancer exhibiting the Warburg effect, and discloses methods to identify these cancers. Identification methods specified are to illustrate the invention and not to limit its scope: this invention encompasses all methods to identify highly glycolytic cancers, in order to identify cancers most amenable to treatment by a compound(s) of this invention.

So, innovatively and usefully, compounds of this disclosure are selected for anti-cancer therapy by metabolic feature of the cancer, which belie how the cancer survives and proliferates, and its weaknesses, weaknesses that compounds of this disclosure attack, rather than the typical, often too arbitrary, often unhelpful, allocation by tissue type, which is the present standard in the art. A diversity of cancers, from different tissues, will be susceptible to compounds of this invention, especially the most dangerous: glycolytic cancers, with high lactate efflux, often have the worst prognosis [18-20, 22]. Experimental data of this disclosure shows that compounds of this invention are effective against breast, prostate, renal, ovarian, skin, central nervous system, colon or lung cancer and leukaemia. Especially leukemia.

Compounds of the present invention treat tumour growth, treat metastasis, treat metastatic cancer, treat non-metastatic cancer, treat tumour implantation, are useful as an adjunct to chemo-/radio-therapy, treat cancers including, but not limited to, Chondrosarcoma, Ewing's sarcoma, Malignant fibrous histiocytoma of bone/osteosarcoma, Osteosarcoma, Rhabdomyosarcoma, Heart cancer, brain cancer, Astrocytoma, Brainstem glioma, Pilocytic astrocytoma, ependymoma, primitive neuroectodermal tumor, Cerebellar astrocytoma, Cerebral astrocytoma, malignant glioma, Medulloblastoma, Neuroblastoma, Oligodendroglioma, Pineal astrocytoma, Pituitary adenoma, Visual pathway and hypothalamic glioma, Breast cancer, Invasive lobular carcinoma, Tubular carcinoma, Invasive cribriform carcinoma, Medullary carcinoma, Male breast cancer, Phyllodes tumor, Inflammatory Breast Cancer, Adrenocortical carcinoma, Islet cell carcinoma, Multiple endocrine neoplasia syn-

drome, Parathyroid cancer, Pheochromocytoma, Thyroid cancer, Merkel cell carcinoma, intraocular melanoma, retinoblastoma, Anal cancer, Appendix cancer, cholangiocarcinoma, Carcinoid tumor, Colon cancer, Extrahepatic bile duct cancer, Gallbladder cancer, Gastric (stomach) cancer, Gastrointestinal carcinoid tumor, Gastrointestinal stromal tumor (GIST), Hepatocellular cancer, Pancreatic cancer, Rectal cancer, Bladder cancer, Cervical cancer, Endometrial cancer, Extragenital germ cell tumor, Ovarian cancer, Ovarian epithelial cancer (surface epithelial-stromal tumor), Ovarian germ cell tumor, Penile cancer, Renal cell carcinoma, Renal pelvis and ureter, transitional cell cancer, Prostate cancer, Testicular cancer, Gestational trophoblastic tumor, Ureter and renal pelvis, transitional cell cancer, Urethral cancer, Uterine sarcoma, Vaginal cancer, Vulvar cancer, Wilms tumor, Esophageal cancer, Head and neck cancer, Nasopharyngeal carcinoma, Oral cancer, Oropharyngeal cancer, Paranasal sinus and nasal cavity cancer, Pharyngeal cancer, Salivary gland cancer, Hypopharyngeal cancer, Acute biphenotypic leukemia, Acute eosinophilic leukemia, Acute lymphoblastic leukemia, Acute myeloid leukemia, Acute myeloid dendritic cell leukemia, AIDS-related lymphoma, Anaplastic large cell lymphoma, Angioimmunoblastic T-cell lymphoma, B-cell prolymphocytic leukemia, Burkitt's lymphoma, Chronic lymphocytic leukemia, Chronic myelogenous leukemia, Cutaneous T-cell lymphoma, Diffuse large B-cell lymphoma, Follicular lymphoma, Hairy cell leukemia, Hepatosplenic T-cell lymphoma, Hodgkin's lymphoma, Hairy cell leukemia, Intravascular large B-cell lymphoma, Large granular lymphocytic leukemia, Lymphoplasmacytic lymphoma, Lymphomatoid granulomatosis, Mantle cell lymphoma, Marginal zone B-cell lymphoma, Mast cell leukemia, Mediastinal large B cell lymphoma, Multiple myeloma/plasma cell neoplasm, Myelodysplastic syndromes, Mucosa-associated lymphoid tissue lymphoma, Mycosis fungoides, Nodal marginal zone B cell lymphoma, Non-Hodgkin lymphoma, Precursor B lymphoblastic leukemia, Primary central nervous system lymphoma, Primary cutaneous follicular lymphoma, Primary cutaneous immunocytoma, Primary effusion lymphoma, Plasmablastic lymphoma, Sézary syndrome, Splenic marginal zone lymphoma, T-cell prolymphocytic leukemia, Basal-cell carcinoma, Melanoma, Skin cancer (non-melanoma), Bronchial adenomas/carcinoids, Small cell lung cancer, Mesothelioma, Non-small cell lung cancer, Pleuropulmonary blastoma, Laryngeal cancer, Thymoma and thymic carcinoma, AIDS-related cancers, Kaposi sarcoma, Epithelioid hemangioendothelioma (EHE), Desmoplastic small round cell tumor, Liposarcoma. The compounds of the present invention treat cancers including, but not limited to, those that originate in the Testis, Cerebral cortex, Skin, Fallopian tube, Parathyroid gland, Small intestine, large intestine, Kidney, Skeletal muscle, Duodenum, Spleen, Epididymis, Bone marrow, Lymph node, Adrenal gland, Esophagus, Thyroid gland, Heart muscle, Tonsil, Lung, Prostate, Rectum, Anus, Adipose tissue, Colon, Stomach, Cervix, Gallbladder, Seminal vesicle, Breast, Ovary, Endometrium, Smooth muscle, Salivary gland, Pancreas, Urinary bladder, blood, brain, gum, head, liver, nasopharynx, neck, tongue, uterus.

Compounds of this Invention are Anti-Inflammatories

An embodiment of this invention is a method of using an effective amount of at least one compound of this disclosure, which inhibits F_1F_0 ATP hydrolysis, as an immunosuppressant and/or anti-inflammatory therapeutic. Because activated macrophages, unlike resting macrophages, use and

rely upon ATP synthase in its reverse mode, hydrolysing ATP [45]. Activated macrophages produce nitric oxide (NO), which switches down/off OXPHOS (NO increases the Km of Complex IV for O₂) and makes them reliant upon F₁F₀ ATP hydrolysis to maintain ψ_{IM} . If ψ_{IM} collapses, apoptosis ensues. Compounds of the present invention inhibit F₁F₀ ATP hydrolysis, thus attenuating the activated macrophage component to inflammation, and its pathologies, and treats/ameliorates/prevents/combats any disease or disorder associated with the undesirable activation or activity of macrophages, and/or any other NO producing cells of the innate immune system (e.g. monocyte-derived inflammatory dendritic cells), and/or immune or inflammation diseases/disorders/pathologies including, but not limited to, acute inflammation, chronic inflammation, systemic inflammation, inflammation because of infection or foreign bodies or injury or chemical or toxin or drug or stress or frostbite or burn or ionising radiation, inflammatory diseases/disorders/syndromes, Macrophage Activation Syndrome (MAS), autoinflammatory diseases/disorders/syndromes, age-related chronic inflammatory diseases ("inflammaging"), autoimmune diseases/disorders/syndromes, diseases/disorders of the innate immune system, sore throat, sore throat associated with cold or flu or fever, high-intensity exercise associated inflammation, ulcerative colitis, inflammatory bowel disease (IBD), irritable bowel syndrome (IBS), rheumatoid arthritis, osteoarthritis, psoriatic arthritis, atopic dermatitis, allergic airway inflammation, asthma, inflammation associated depression, exercise-induced acute inflammation, atherosclerosis, allergy, hay fever, anaphylaxis, inflammatory myopathies, drug-induced inflammation, systemic inflammatory response syndrome, sepsis-related multiple organ dysfunction/multiple organ failure, microbial infection, acute brain/lung/hepatic/renal injuries, acne vulgaris, celiac disease, celiac sprue, chronic prostatitis, colitis, diverticulitis, glomerulonephritis, hidradenitis suppurativa, hypersensitivities, interstitial cystitis, Mast Cell Activation Syndrome, mastocytosis, otitis, pelvic inflammatory disease (PID), reperfusion injury, rheumatic fever, rhinitis, sarcoidosis, transplant rejection, parasitosis, eosinophilia, type III hypersensitivity, ischaemia, chronic peptic ulcer, tuberculosis, Crohn's disease, hepatitis, chronic active hepatitis, immune hepatitis, ankylosing spondylitis, diverticulitis, fibromyalgia, systemic lupus erythematosus (SLE), Alzheimer's disease, Parkinson's disease, neurodegenerative disease, cardiovascular disease, chronic obstructive pulmonary disease, bronchitis, acute bronchitis, appendicitis, acute appendicitis, bursitis, colitis, cystitis, dermatitis, encephalitis, gingivitis, meningitis, infective meningitis, myelitis, nephritis, neuritis, periodontitis, chronic periodontitis, phlebitis, prostatitis, RSD/CRPS, rhinitis, sinusitis, chronic sinusitis, tendonitis, testiculitis, tonsillitis, urethritis, vasculitis, respiratory bronchiolitis-associated interstitial lung disease and desquamative interstitial pneumonia, interstitial lung disease, Löfgren syndrome, Heerfordt syndrome, monocytosis, liver fibrosis, steatohepatitis, nonalcoholic steatohepatitis, silicosis, histiocytoses, Langerhans' cell histiocytosis, haemophagocytic lymphohistiocytosis, pulmonary langerhans cell histiocytosis, obesity, type II diabetes, gout, pseudogout, organ transplant rejection, epidermal hyperplasia, chronic fatigue syndrome, graft versus host disease (GVHD), lymphadenopathy.

In clinical utility, the anti-inflammatory activity of compounds of this invention juxtaposes well with their aforementioned ability to reduce body temperature.

The anti-inflammatory action by compounds of this invention has an anti-cancer action. Because it reduces the

number of Tumour Associated Macrophages (TAMs) [46]. These can constitute a large component of tumour mass and their presence is often associated with poor patient prognosis because they can drive cancer pathology. Indeed, inflammation is now considered one of the hallmarks of cancer [47]. The anti-inflammatory action, and thence anti-cancer action, of these compounds synergises with their direct anti-cancer activities disclosed herein.

Macrophages can be subverted by pathogens, which hide inside them in safety from the immune system. Non-limiting examples of such pathogens are HIV (causes HIV/AIDS; HIV virus can lay latent in macrophages during antiretroviral therapy, wherein HIV virus becomes undetectable in blood, and then repopulate the virus in blood when antiretroviral therapy is interrupted or discontinued; HIV can replicate in macrophages [48]), *Mycobacterium tuberculosis* (causes tuberculosis), *Leishmania* parasite (causes Leishmaniasis), Chikungunya virus (causes Chikungunya), *Legionella pneumophila* (causes Legionnaires' disease), adenovirus (causes pink eye), *T. whipplei* (causes Whipple's Disease) and *Brucella* spp. (causes brucellosis). So, by exerting anti-macrophage activity, compounds of this disclosure can treat/ameliorate/prevent/combat such disorders and diseases. Because the compounds of this invention are selective for activated macrophages, an option is to activate macrophages before the compound administration, by administering to the patient an effective amount of a compound, protein, antibody or some other entity, e.g. pathogen, attenuated pathogen or pathogen component that activates macrophages. Some examples (non-limiting) of factors that can activate macrophages are cytokines such as interferon-gamma (IFN-gamma) and/or tumour necrosis factor (TNF), and/or IL-4, and/or IL-13, and/or IL-10, and/or IL-2, and/or IL-12, and/or IL-6, and/or IL-18 and/or chemokines (CCL3, CCL4, CCL5) and/or a bacterial endotoxin such as lipopolysaccharide (LPS), or a commercially available agent for macrophage activation in biological research (e.g. CAS 61512-20-7) or an antibody targeting a receptor on the macrophage cell surface or on the surface of a different cell type, which then activates a macrophage by mechanism. Macrophage activating antibodies are well known to those of the art.

An embodiment of this invention is the use of an effective amount of at least one compound of this invention, which inhibits F₁F₀ ATP hydrolysis, to treat/ameliorate/prevent/combat HIV infection, optionally with an effective amount of a compound, protein, antibody, pathogen or pathogen component that activates macrophages (isn't absolutely necessary because HIV activates macrophages [50-51], which drives the chronic inflammation pathology component to HIV infection) optionally in co-therapy with, or after, antiretroviral therapy (ART) or combination anti-retroviral therapy (cART). Even after prolonged cART, which drives plasma HIV down to undetectable levels, HIV-1 DNA and RNA is detectable in macrophages: they are an HIV reservoir that remains extant, even during cART, and that the virus can spread from during any interruption or termination of cART [49]: thence the vital importance of the methods and compounds herein. Furthermore, these compounds treat/ameliorate/prevent/combat HIV-associated chronic inflammation.

Macrophages mediate HIV virus neuroinvasion (and neuroinvasion by other viruses also e.g. SARS coronavirus) and compounds of this invention oppose this and treat/ameliorate/prevent/combat HIV-associated neurocognitive disorders (HAND) (and neurocognitive and neurodegenerative diseases/disorders caused by other viruses also e.g. SARS

coronavirus). The anti-HIV and anti-cancer activity of the compounds of this invention synergise to treat/ameliorate/prevent/combat HIV associated cancers: AIDS-defining cancers (Kaposi sarcoma, aggressive B-cell non-Hodgkin lymphoma, cervical cancer.) and non-AIDS defining cancers. This disclosure encompasses a compound(s) of this invention in co-therapy with any Food and Drug Administration (FDA) approved drug(s) or treatment for HIV or AIDS. Examples include, but aren't limited to, abacavir, didanosine, emtricitabine, lamivudine, stavudine, tenofovir, dioxiproxil fumarate (tenofovir DF, TDF), zidovudine (azidothymidine, AZT, ZDV), atazanavir, darunavir, fosamprenavir, indinavir, nelfinavir, ritonavir, saquinavir, tipranavir, enfuvirtide, maraviroc, dolutegravir, elvitegravir, raltegravir, cobicistat.

Non-limiting examples of autoinflammatory diseases/disorders/syndromes that the compounds of this invention treat/ameliorate/prevent/combat include, but aren't limited to, recurrent fever syndromes, which can be hereditary or acquired, characterized by recurrent fever associated with rash, serositis, lymphadenopathy and musculoskeletal involvement. Examples include familial mediterranean fever (FMF), TNF receptor —associated periodic syndrome (TRAPS), Hyperimmunoglobulinemia D with recurrent fever syndrome (HIDS), cryopyrin associated periodic syndrome (CAPS), Blau syndrome, Majeed syndrome, deficiency of interleukin-1 receptor antagonist (DIRA), mevalonate kinase deficiency, pyogenic-arthritis-pyoderma gangrenosum and acne syndrome (PAPA), periodic fever aphthous stomatitis pharyngitis adenitis (PFAPA) syndrome, Behcet's disease, Still's disease, Crohn's disease, Schnitzler's syndrome, Sweet's syndrome, NLRP12-associated autoinflammatory disorders, deficiency of interleukin-1 receptor antagonist (DIRA), pyoderma gangrenosum, cystic acne, aseptic arthritis, periodic Fever Associated with mevalonate kinase deficiency (hyperimmunoglobulin D Syndrome), Pyogenic Arthritis Pyoderma Gangrenosum Acne (PAPA) syndrome, Periodic Fever Aphthous Stomatitis, Pharyngitis and Adenopathy (PFAPA) syndrome, Adult-Onset Still's Disease (AOSD), Systemic Juvenile Idiopathic Arthritis (sJIA), Chronic Recurrent Multifocal Osteomyelitis (CRMO), Synovitis Acne Pustulosis Hyperostosis Osteitis (SAPHO) syndrome, Cryopyrin associated Periodic Syndrome (CAPS), Familial cold auto inflammatory syndrome (FCAS), Muckle-Wells syndrome (MWS), Familial cold urticarial, Neonatal onset multisystemic inflammatory disorder (NOMID), hereditary Periodic Fever Syndromes, Periodic Fever Syndromes, systemic autoinflammatory diseases.

Non-limiting examples of autoimmune diseases/disorders/syndromes that the compounds of this invention treat/ameliorate/prevent/combat include, but aren't limited to, Addison's disease, Agammaglobulinemia, Alopecia areata, Amyloidosis, Ankylosing spondylitis, Anti-GBM/Anti-TBM nephritis, Antiphospholipid syndrome, autoimmune angioedema, autoimmune dysautonomia, autoimmune encephalomyelitis, autoimmune hepatitis, autoimmune inner ear disease (AIED), autoimmune myocarditis, autoimmune pancreatitis, autoimmune retinopathy, autoimmune urticaria, axonal & neuronal neuropathy (AMAN), Baló disease, Behcet's disease, benign mucosal pemphigoid, Bullous pemphigoid, Castleman disease (CD), Celiac disease, Chagas disease, chronic inflammatory demyelinating polyneuropathy (CIDP), chronic recurrent multifocal osteomyelitis (CRMO), Churg-Strauss, Cicatricial pemphigoid, Cogan's syndrome, cold agglutinin disease, congenital heart block, coxsackie myocarditis, CREST syndrome, Berger's

disease, dermatitis herpetiformis, dermatomyositis, Devic's disease (neuromyelitis optica), discoid lupus, Dressler's syndrome, endometriosis, eosinophilic esophagitis (EoE), eosinophilic fasciitis, erythema nodosum, essential mixed cryoglobulinemia, Evans syndrome, fibromyalgia, fibrosing alveolitis, giant cell arteritis (temporal arteritis), giant cell myocarditis, glomerulonephritis, Goodpasture's syndrome, granulomatosis with polyangiitis, Graves' disease, Guillain-Barre syndrome, Hashimoto's thyroiditis, hemolytic anemia, immune hemolytic anemia, Henoch-Schönlein purpura (HSP), Herpes gestationis or pemphigoid gestationis (PG), hypogammaglobulinemia, IgA Nephropathy, IgG4-related sclerosing disease, Immune thrombocytopenic purpura (ITP), Inclusion body myositis (IBM), Interstitial cystitis (IC), juvenile arthritis, juvenile diabetes (Type 1 diabetes), juvenile myositis (JM), Kawasaki disease, Lambert-Eaton syndrome, Leukocytoclastic vasculitis, Lichen planus, Lichen sclerosus, Ligneous conjunctivitis, Linear IgA disease (LAD), Lupus, Lyme disease chronic, Meniere's disease, Microscopic polyangiitis (MPA), Mixed connective tissue disease (MCTD), Mooren's ulcer, Mucha-Habermann disease, Multiple sclerosis, Myasthenia gravis, Myositis, Narcolepsy, Neuromyelitis optica, Neutropenia, Ocular cicatricial pemphigoid, Optic neuritis, Palindromic rheumatism (PR) PANDAS, Paraneoplastic cerebellar degeneration (PCD), Paroxysmal nocturnal hemoglobinuria (PNH), Parry Romberg syndrome, Pars planitis (peripheral uveitis), Parsonage-Turner syndrome, Pemphigus, peripheral neuropathy, perivascular encephalomyelitis, Pernicious anemia (PA), POEMS syndrome, Polyarteritis nodosa, Polyglandular syndromes type I, II, III, Polymyalgia rheumatica, Polymyositis, Postmyocardial infarction syndrome, Postpericardiotomy syndrome, Primary biliary cirrhosis, Primary sclerosing cholangitis, Progesterone dermatitis, Psoriasis, Psoriatic arthritis, Pure red cell aplasia (PRCA), Pyoderma gangrenosum, Raynaud's phenomenon, Reactive Arthritis, Reflex sympathetic dystrophy, Relapsing polychondritis, Restless legs syndrome (RLS), Retroperitoneal fibrosis, Rheumatic fever, Rheumatoid arthritis, Sarcoidosis, Schmidt syndrome, Scleritis, Scleroderma, Sjogren's syndrome, Sperm & testicular autoimmunity, Stiff person syndrome (SPS), Subacute bacterial endocarditis (SBE), Susac's syndrome, Sympathetic ophthalmia (SO), Takayasu's arteritis, Temporal arteritis/Giant cell arteritis, Thrombocytopenic purpura (TTP), Tolosa-Hunt syndrome (THS), Transverse myelitis, Type 1 diabetes, Ulcerative colitis (UC), Undifferentiated connective tissue disease (UCTD), Uveitis, Vasculitis, Vitiligo, Wegener's granulomatosis (or Granulomatosis with Polyangiitis (GPA)), idiopathic thrombocytopenia purpura, splenomegaly.

DESCRIPTION OF THE DRAWINGS

For purposes of clarity, not every component is labelled in every figure, nor is every component of each embodiment of the invention shown where illustration is not necessary to allow those of ordinary skill in the art to understand the invention.

FIGS. 1, 2, 3, 4, 5, 6, 7: Experimental evidence: molecules that specifically inhibit the reverse mode of ATP synthase: specifically exert anti-cancer activity: representative, non-limiting examples.

FIGS. 1, 2, 3, 4, 5, 6 show results from the NCI-60 one-dose in vitro assay [29-30] at the Developmental Therapeutics Program (DTP), at the National Cancer Institute (NCI, Bethesda, MD, USA). Its protocol is well known to those of the art, and it tests the effect, if any, of a test

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compound on the growth/survivability of a cancer cell line as compared to the no compound control. When this protocol was first developed, a compound was tested against 60 cancer cell lines, hence the name NCI-60, but more recently this has been reduced to 59 cell lines, and there is some variation over time in the cancer cell lines making up this 59. However, a constant is that this 59 always has representative cell lines from leukemia, melanoma and cancers of the lung, colon, brain, ovary, breast, prostate and kidney. In a one-dose NCI compound test report, NCI report a number for each cell line, which they call "Growth Percent", which is its growth relative to the no-compound control, and relative to the time zero number of cells. This reported parameter allows detection of both growth inhibition (values between 0 and 100) and lethality (values less than 0). For example, an NCI "Growth Percent" value of 100 means no growth inhibition. Value of 40 means 60% growth inhibition. Value of 0 means no net growth over the course of the experiment. Value of -40 means 60% lethality. Value of -100 means all cells are dead. I don't present NCI one-dose data in this original format. Instead, if the NCI "Growth Percent" value for a cell is positive, it is manipulated: [100 minus this original NCI-60 "Growth Percent" data point], to yield the percentage "Growth Inhibition". If the original NCI "Growth Percent" value for a cell is negative, it is made positive to be the percentage of original cancer cells (at time zero) killed: "Percentage Killed" {and in these cases, of course, all growth has been inhibited, so percentage "Growth Inhibition" is then specified to be 100% for this cancer cell line}. In my one-dose figures, "Growth Inhibition" (0-100%) is presented on the x-axis and, if applicable, "Percentage Killed" (0-100%) further along on the x-axis. The latter is applicable when there is not just cancer growth inhibition but a reduction in the number of cancer cells from the start time i.e. when the compound is not merely slowing cancer growth, but is actively reducing the number of cancer cells from the starting number. In cases where there is only growth inhibition, only "Growth inhibition" is presented on the x-axis. In all cases, the greater the percentage number on the x-axis, for a given cancer cell line named on the y-axis, the greater the anti-cancer activity of this compound against this cancer cell line.

FIG. 1: No anti-cancer activity of BTB06584 at 10 μ M (NCI one-dose assay).

FIG. 2: Anti-cancer activity of BTB06584 at 100 μ M (NCI one-dose assay).

FIG. 3: Anti-cancer activity of BMS-199264 hydrochloride at 10 μ M (NCI one-dose assay).

FIG. 4: Anti-cancer activity of BMS-199264 hydrochloride at 100 μ M (NCI one-dose assay).

FIG. 5: 19a is a racemate. Superfluid chromatography (SFC) was used to separate 19a into its component R (6a) and S (6b) stereoisomers, which don't and do inhibit F_1F_0 ATP hydrolysis respectively, and two samples of opposite >97% enantiomeric excess (ee) were achieved. But during NCI one-dose (10 μ M) testing, these samples underwent racemization and their ee was eroded. Such that both samples contained S stereoisomer and both exerted anti-cancer activity. Racemization was advanced (Pearson correlation between results, $R=0.8$) but yet incomplete and the S sample still had more S stereoisomer, and greater anti-cancer activity, than the other sample. EC_{50} values are from SMP studies in [5-6].

FIG. 6: Anti-cancer potency of these molecules scales with their inhibition of (EC_{50}) F_1F_0 ATP hydrolysis. The experimental panels of this figure correspond to figures presented elsewhere in this disclosure and so, although cell

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lines aren't named on the y-axes (space didn't permit), any given horizontal bar in a panel can be linked to its cancer name by cross-correlating the panel to its corresponding figure herein, wherein cell line names are given. EC_{50} values for BMS-199264 and 19a are from SMP studies in [5-7], BTB06584 potency information (isn't an EC_{50}) is from a whole cell study [13].

FIG. 7: Anti-cancer activity of BMS-199264 hydrochloride. Results from the NCI-60 five-dose in vitro assay [29-30] at the Developmental Therapeutics Program (DTP), at the National Cancer Institute (NCI, Bethesda, MD, USA). In this assay, which is well known to those of the art, a compound is tested, in vitro, against 59 different cancer cell lines, sourced from 9 different tissue types, across 5 different concentrations. On the y-axis is the aforementioned "Growth Percentage" parameter used by NCI, which is growth relative to the no-compound control, and relative to the time zero number of cells. This parameter allows detection of both growth inhibition (values between 0 and 100) and lethality (values less than 0). GI50 is the compound concentration that causes 50% growth inhibition of a cell line relative to the no-drug control. Each cancer cell line has a GI50 value and the "mean GI50" of all 59 cell lines can be calculated. The mean GI50 for BMS-199264 hydrochloride is 3.9 μ M.

FIG. 8: Mechanistic distinction from oligomycin enables therapeutic utility. Drawn molecules of this figure have an imidazole group, with a protonable nitrogen atom that can shuttle protons across the mitochondrial inner membrane (IM), dissipating the proton motive force (pmf, uncoupling). This figure presents experimental data using the HL-1 cardiac muscle cell line (cancer derived, but now very cardiac differentiated e.g. spontaneously contracts and beats like heart cells). Refer to the "Benchmark Drugs" first, which produce cellular effects well known to those of the art [3]. Oligomycin here refers to Oligomycin B. Oligomycin binds ATP synthase and blocks its forward, proton passing, ATP synthesizing, mode. This means less protons pass through ATP synthase, less pmf is consumed per unit time, pmf increases, ψ_{IM} hyperpolarizes, electron flow along the respiratory chain slows, and O_2 consumption is decreased. Carbonilcyanide p-trifluoromethoxyphenylhydrazide (FCCP) is an uncoupler that shuttles protons across the IM, dissipates pmf (as heat), pmf decreases, ψ_{IM} depolarizes, electron flow along the respiratory chain speeds, and O_2 consumption is increased. Distinct from oligomycin, 3 molecules of this figure increase, rather than decrease, O_2 consumption, which signifies their mechanistic distinction from oligomycin: they reduce ATP synthesis more by uncoupling than any inhibition of forward mode ATP synthase. They all contain a protonable nitrogen atom, with a basic pKa value conducive to uncoupling i.e. a pKa value reasonably close to {pH of mitochondrial intermembrane space + pH of mitochondrial matrix}/2. Although for VG025, its most conducive pKa is on its main ring rather than its imidazole. In a NADPH-linked sub-mitochondrial (SMP) assay of ATP synthesis, these molecules would decrease ATP production because they dissipate pmf as heat, and so there is less pmf available for ATP production. In interpretation, this uncoupling could be incorrectly attributed to inhibition of the forward mode of ATP synthase and so the full mechanistic distinction of these molecules from oligomycin could be missed. This has been the case with other imidazole containing compounds of this disclosure, also with a protonable nitrogen in their imidazole, also with a pKa conducive to uncoupling, and wherein their inhibition of F_1F_0 ATP synthesis in the NADPH-linked SMP assay has been attrib-

uted to inhibiting the forward mode of ATP synthase [5-8], but wherein their uncoupling is likely to be the more predominant factor (extrapolated from data of this figure) and wherein they do not inhibit the forward mode of ATP synthase much, if at all, in stark distinction to oligomycin. Uncoupling capability, which decreases with increased logP (refer next paragraph), explains why different molecules of the present figure exert different effects on O₂ consumption. The high logP value of VG019 means its uncoupling is minimal and its effect on O₂ consumption is zero (rounded) at a concentration (100 μM) it inhibits the reverse mode of ATP synthase, in stark distinction to oligomycin (3 μM), which dramatically decreases O₂ consumption (~40%), because, distinctly, it potentially inhibits the forward mode of ATP synthase.

LogP=−3.2 is the optimal compromise for best passing a membrane: its hydrophobic core (selecting for high logP) and hydrophilic boundary layer (selecting for low logP) ([32], herein incorporated in its entirety). The imidazole containing molecules presented in this figure, and in this disclosure's drawings more generally, have logP>3.2 and present increased logP=decreased uncoupling. The uncoupling capability/liability of a molecule actually hinges on its intersection of pKa(s) and logP [32] but for the molecules in this disclosure's drawings, wherein the imidazole pKa values are, generally, all within a fairly narrow range, the more primary determinant to each molecule's uncoupling rate, relative to the others, is the molecule's logP value relative to the others.

The drawn molecules of this figure do inhibit the reverse mode of ATP synthase. When a respiratory chain inhibitor blocks electron flow, ψ_{IM} is maintained, not by proton pumping by the respiratory complexes, but by proton pumping by ATP synthase i.e. the reverse mode of ATP synthase. In the presented data, when the respiratory chain is blocked, the presented molecules depolarise ψ_{IM} because they block the reverse mode of ATP synthase. They don't affect ψ_{IM} by these means when the respiratory chain is operational, because ψ_{IM} isn't set/maintained by the reverse mode of ATP synthase in this case, but the molecules with stronger uncoupling capability, they can shuttle more protons across the IM (dissipate more pmf) than the respiratory chain can increase its rate to replace, and they do depolarise ψ_{IM} . When the respiratory chain is blocked, a stronger uncoupler in this figure depolarises ψ_{IM} more. Because not only does it inhibit the generator of ψ_{IM} (reverse mode ATP synthase), it simultaneously erodes ψ_{IM} (uncoupling).

Oligomycin does inhibit the reverse mode of ATP synthase. But distinctly it inhibits its forward mode more [11]. So, using oligomycin, there is no margin to inhibit the reverse mode (anti-cancer), without adversely affecting cells using OXPHOS i.e. most normal cells. Contrast this with molecule VG019 of this figure, for example, which can inhibit the reverse mode of ATP synthase, and yet—in observed distinction to oligomycin—does not affect cells using OXPHOS: it does not change their O₂ consumption or ψ_{IM} (at 100 μM). This grants it, in distinction to oligomycin, anti-cancer selectivity. Other molecules of this disclosure have even greater cancer selectivity. For example, the best mode (refer disclosure section of that name) inhibits F₁F₀ ATP hydrolysis>5,556 times more than F₁F₀ ATP synthesis, in NADH-linked and NADPH-linked SMP assays [5-6], whilst oligomycin—inversely—inhibits F₁F₀ ATP hydrolysis less than F₁F₀ ATP synthesis in such assays [11].

Computational calculations of logP and pKa were made using [31]. The data presented in this Figure is from [12] (herein incorporated in entirety), but the analysis/(re)inter-

pretation is novel. As is the process/method of using these molecules as anti-cancer therapeutics, which is componentry to this invention. The imidazole of the drawn molecules is 4-yl. Permutations, with 5-yl instead, are also disclosed by this invention as anti-cancer therapeutics.

EXAMPLE EMBODIMENTS OF THE INVENTION

The Drawings present embodiments of the invention. Further examples are enumerations of Markush Formulas (I), (II), (III), (IV) and (V), presented henceforth. Note: none of these share Markush symbols, which are symbols of the type: R_x, wherein x is an integer, well known to those of the art. They each have their own, as specified for each, in their own sections of this disclosure.

In this disclosure, the term "Formula [X]" is used when a statement is true for Formula (I), (II), (III), (IV) and (V), and all are being referred to independently. A compound of Formula [X] is a compound of Formula (I), or Formula (II), or Formula (III), or Formula (IV), or Formula (V), or any compound presented in this disclosure's Drawings.

This invention is described using these example embodiments but it isn't limited to these. These merely illustrate the invention. Compounds of other structures, which are identified as therapeutic inhibitors by the rationale and methods of the present invention, are also encompassed by the present invention.

Encompassed by this invention are methods of treating a subject suffering from a medical disease or disorder by administering an effective amount of at least one compound of Formula (I), (II), (III), (IV) or (V) or a pharmaceutically-acceptable salt, solvate, hydrate or prodrug thereof, or a pharmaceutical composition(s) comprising one or compounds of Formula (I), (II), (III), (IV) or (V). A very large number of diseases/disorders can be treated using compounds described herein. For example, but not limited to, the compounds described herein can be used to treat/ameliorate/prevent/combat a disease or disorder selected from cancer, cachexia, cancer driven cachexia, weight loss or a disease or disorder that causes a higher than normal body temperature which can include, but isn't limited to, fever, pyrexia, hyperpyrexia, hyperthermia, malignant hyperthermia, neuroleptic malignant syndrome, serotonin syndrome, thyroid storm, or to treat/ameliorate/prevent/combat Tumour Associated Macrophages (TAMs) or treat/ameliorate/prevent/combat any macrophage associated disease or disorder including, but not limited to, Macrophage Activation Syndrome (MAS), HIV, AIDS, HIV-associated neurocognitive disorders (HAND), HIV associated cancers, AIDS-defining cancers, non-AIDS defining cancers or treat/ameliorate/prevent/combat virus neuroinvasion via macrophages, as used for example by HIV and SARS coronavirus, or treat/ameliorate/prevent/combat neurocognitive and neurodegenerative diseases/disorders, for example those caused by, but not limited to, a virus or treat/ameliorate/prevent/combat acute/chronic/systemic inflammation or any inflammatory disease/disorder/syndrome or any autoimmune disease/disorder/syndrome or to treat/ameliorate/prevent/combat cardiovascular diseases and conditions associated with ischemia and associated conditions including, without limitation, ischemia-reperfusion injury, myocardial ischemia, ischemic heart disease, chronic stable angina pectoris, myocardial infarction, congestive heart failure, an acute coronary syndrome, muscle cell damage, necrosis, cardiac arrhythmias, non-Q wave MI, unstable angina, high blood pressure,

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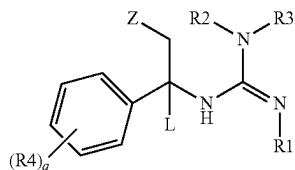
coronary artery disease, ischemic hypoxia, cyanosis, gangrene, acute limb ischemia, stroke, ischemic stroke, brain ischemia, vascular dementia, transient ischemic attack (TIA), ischemic colitis, mesenteric ischemia, angina pectoris, ischemic heart disease, ischemic neuropathy, hypoxic-ischemic encephalopathy, cerebral hypoxia, brain hypoxia, ischemia resulting from vascular occlusion, cerebral infarction, stroke and related cerebral vascular diseases (including cerebrovascular accident and transient ischemic attack), muscle cell damage and necrosis or to cause greater metabolic/bioenergetic efficiency in a subject, enhancing their physical or mental performance or causing body weight gain.

Example (I)

Summary of Formula (I)

This invention embodiment relates to compounds having the following formula:

Formula (I)



or a pharmaceutically-acceptable salt, solvate, hydrate or prodrug thereof, wherein:

L is alkyl, or substituted alkyl, or any atom or isotope permitted by valence;

R₁ is cyano, —SO₂R₈, —C(=O)R₉, or heteroaryl;

R₂ is (i) independently hydrogen, alkyl, or substituted alkyl, or (ii) taken together with R₃ forms a heterocyclo;

R₃ is (i) independently alkyl, substituted alkyl, alkylthio, aminoalkyl, carbamyl, B_B-aryl, B_B-heterocyclo, B_B-heteroaryl, or B_B-cycloalkyl, or (ii) taken together with R₂ forms a heterocyclo;

Z is heteroaryl provided that when R₁ is cyano, Z is not 2-pyridinyl;

B_B is a bond, C₁₋₄alkylene, C₂₋₄alkenylene, substituted C₁₋₄alkylene, substituted C₂₋₄alkenylene, —C(=O)NR₁₉—, —C₁₋₄alkylene-C(=O)NR₁₉—, or substituted C₁₋₄alkylene-C(=O)NR₁₉—;

R₄ at each occurrence is selected independently of each other R₄ from the group consisting of halogen, alkyl, haloalkyl, nitro, cyano, haloalkoxy, OR₂₅, SR₂₅, NR₂₅R₂₆, NR₂₅SO₂R₂₇, SO₂R₂₇, SO₂NR₂₅R₂₆, CO₂R₂₆, C(=O)R₂₆, C(=O)NR₂₅R₂₆, OC(=O)R₂₅, —OC(=O)NR₂₅R₂₆, NR₂₅C(=O)R₂₆, NR₂₅CO₂R₂₆, aryl, heteroaryl, heterocyclo and cycloalkyl;

R₈ is alkyl, substituted alkyl, aryl, or heteroaryl;

R₉ is —NR₁₀R₁₁, alkyl, substituted alkyl, alkoxy, alkylthio, cycloalkyl, aryl, heteroaryl, heterocycle, or —CO₂R₁₂;

R₁₀ and R₁₁, are (i) independently selected from hydrogen, alkyl, substituted alkyl, alkoxy, heterocyclo, cycloalkyl, aryl, and heteroaryl; or (ii) taken together form a hetero cyclo or heteroaryl;

R₁₂ and R₁₉ are hydrogen or alkyl;

R₂₅ and R₂₆ are independently selected from hydrogen, alkyl, or substituted alkyl, or taken together form a heterocyclo or heteroaryl ring;

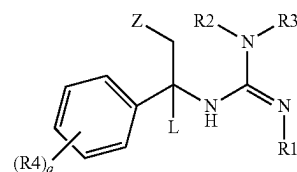
36

R₂₇ is alkyl or substituted alkyl, and

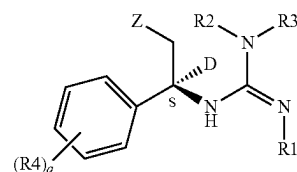
q is 0, 1, 2, or 3.

Preferred Compounds of Formula (I)

Preferred methods are to use, and preferred compounds are, compounds with the following formula, or pharmaceutically-acceptable salts, solvates, hydrates or prodrugs thereof,



and even more preferred methods are to use, and preferred compounds are, compounds with the following formula, or pharmaceutically-acceptable salts, solvates, hydrates or prodrugs thereof,



in which, in the preceding two structures shown:

L is hydrogen or deuterium;

D is deuterium (enrichment, for example, exceeding 40% deuterium incorporation at shown position, and optionally at other positions also);

S symbolises the S stereoisomer, for example, in enantiomeric excess (ee) exceeding 70%;

Z is triazolyl optionally substituted with one to two R₇ or imidazolyl optionally substituted with one to two R₇ and/or having fused thereto a benzene ring in turn optionally substituted with one to two R₇;

R₁ is cyano or —C(=O)R₉;

R₂ is hydrogen, alkyl, or benzyl;

R₃ is aryl or arylalkyl optionally substituted with alkyl, halogen, trifluoromethyl, OCF₃, cyano, nitro, amino, hydroxy, or methoxy;

R₄ is halogen, alkyl, trifluoromethyl, or OCF₃;

R₇ is alkyl, carbamyl or carbamylC₁₋₄alkyl;

R₉ is —NR₁₀R₁₁, alkyl, substituted alkyl, alkoxy, alkylthio, cycloalkyl, aryl, heteroaryl, heterocycle, or —CO₂R₁₂;

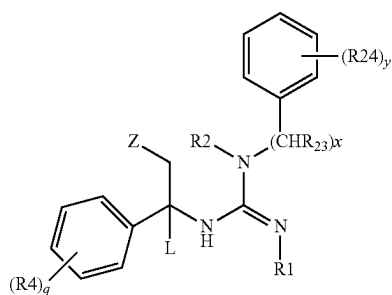
R₁₀ and R₁₁ are (i) independently selected from hydrogen, alkyl, substituted alkyl, alkoxy, heterocyclo, cycloalkyl, aryl, and heteroaryl; or (ii) taken together form a heterocyclo or heteroaryl;

R₁₂ is hydrogen or alkyl; and

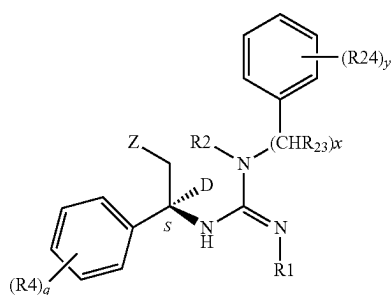
q is 0, 1, 2, or 3.

Further preferred are compounds having the following formula, or pharmaceutically-acceptable salts, solvates, hydrates or prodrugs thereof,

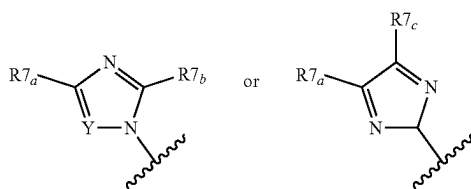
37



and even more preferred are compounds having the following formula, or pharmaceutically-acceptable salts, solvates, hydrates or prodrugs thereof,



in which, for the preceding two structures shown:
Z is



L is hydrogen or deuterium;
D is deuterium (enrichment, for example, exceeding 40% deuterium incorporation at shown position, and optionally at other positions also);

S symbolises the S stereoisomer, for example, in enantiomeric excess (ee) exceeding 70%;

Y is N, CH or CR_{7c};

R₁ is cyano or —C(=O)R₉;

R₂ is hydrogen or C₁₋₄alkyl;

R₄ is halogen, C₁₋₄alkyl, trifluoromethyl; or OCF₃;

R_{7a}, R_{7b}, and R_{7c} are hydrogen, alkyl, carbamyl or carbamylC₁₋₄alkyl, or R_{7a} and R_{7c} join to form an optionally substituted fused phenyl ring;

R₉ is —NR₁₀R₁₁, alkyl, substituted alkyl, alkoxy, alkylthio, cycloalkyl, aryl, heteroaryl, heterocycle, or —CO₂R₁₂;

R₁₀ and R₁₁ are (i) independently selected from hydrogen, alkyl, substituted alkyl, alkoxy, heterocyclo, cycloalkyl, aryl, and heteroaryl; or (ii) taken together form a heterocyclo or heteroaryl;

R₁₂ is hydrogen or alkyl;

R₂₃ is hydrogen, alkyl, hydroxyalkyl, or phenyl;

R₂₄ is alkyl, halogen, trifluoromethyl, cyano, halogen, hydroxy, OCF₃, methoxy, phenoxy, benzyloxy, cyano, or acyl, or two R₂₄ groups join to form a fused cycloalkyl or benzene ring;

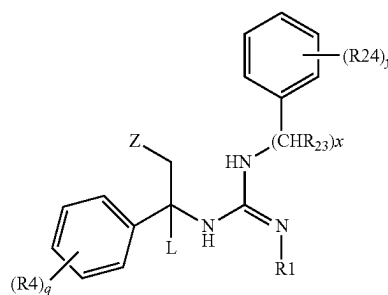
38

q is 1 or 2;

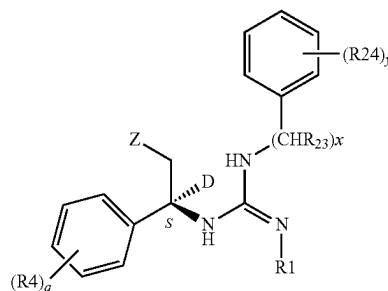
x is 0, 1, or 2; and

y is 0, 1, 2, or 3.

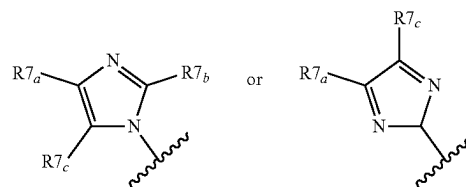
More preferred are compounds having the following formula, or pharmaceutically-acceptable salts, solvates, hydrates or prodrugs thereof,



and even more preferred are compounds having the following formula, or pharmaceutically-acceptable salts, solvates, hydrates or prodrugs thereof,



in which, for the preceding two structures shown:
Z is



L is hydrogen or deuterium;

D is deuterium (enrichment, for example, exceeding 40% deuterium incorporation at shown position, and optionally at other positions also);

S symbolises the S stereoisomer, for example, in enantiomeric excess (ee) exceeding 70%;

R₁ is cyano or —C(=O)R₉;

R₄ is halogen, C₁₋₄alkyl, trifluoromethyl, or OCF₃;

R_{7c} is hydrogen or R₇ and R_{7c} join to form a fused benzene ring optionally substituted with C₁₋₄alkyl or —(CH₂)₁₋₂—NHC(=O)C₁₋₄alkyl,

R_{7b} is hydrogen, C₁₋₄alkyl, or —(CH₂)₁₋₂—NHC(=O)C₁₋₄alkyl;

R₉ is a) —NR₁₀R₁₁

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b) C₁₋₈alkyl optionally substituted with one to two of:
i) SR₁₃, OR₁₃, NR_{13a}R_{13b}, halogen, trifluoromethyl, CO₂R_{13a}, and C(=O)NR_{13a}R_{13b};

ii) cycloalkyl optionally substituted with one to two of C(=O)H, C₁₋₄acyl, alkenyl, carbamyl, and/or phenyl in turn optionally substituted with halogen;

iii) phenyl or naphthyl optionally substituted with one to two of halogen, nitro, amino, alkyl, hydroxy, C₁₋₄alkoxy, or having fused thereto a five or six membered heterocyclo;

iv) pyridinyl, thiophenyl, furanyl, tetrahydrofuranyl, or azepinyl, optionally substituted with alkyl or having fused thereto a five to six membered carbocyclic ring optionally substituted with keto or C₁₋₄alkoxy;

c) C₁₋₄alkoxy;

d) C₁₋₄alkylthio;

e) CO₂alkyl;

f) 3 to 6 membered cycloalkyl optionally having up to four substituents selected from alkyl, halogen, cyano, alkenyl, acyl, alkylthio, carbamyl, and/or phenyl in turn optionally substituted with halogen; or having an aryl fused thereto;

g) phenyl optionally substituted with one to four of halogen, cyano, trifluoromethyl; nitro, hydroxy, C₁₋₄alkoxy, haloalkoxy, C₁₋₆alkyl, CO₂alkyl, SO₂alkyl, SO₂NH₂, amino, NH(C₁₋₄alkyl), N(C₁₋₄alkyl)₂, NHC(=O)alkyl, C(=O)alkyl, and/or C₁₋₄alkyl in turn optionally substituted with one to three of trifluoromethyl; hydroxy, cyano, phenyl, pyridinyl; and/or a five or six membered heteroaryl or heterocycle in turn optionally substituted with keto or having a benzene ring fused thereto;

h) pyridinyl, thiazolyl, furanyl, thiophenyl, and pyrrolyl optionally substituted with one to two of halogen, alkyl, and phenyl in turn optionally substituted with halogen or trifluoromethyl;

R₁₀ is hydrogen, alkyl, or alkoxy;

R₁₁ is alkyl, substituted alkyl, alkoxy, heterocyclo, cycloalkyl, aryl, or heteroaryl;

or R₁₀ and R₁₁, taken together form a heterocyclo or heteroaryl;

R₂₃ is hydrogen, alkyl, hydroxyalkyl, or phenyl;

R₂₄ is alkyl, halogen, trifluoromethyl, cyano, halogen, hydroxy, OCF₃, methoxy, phenyloxy, benzyloxy, cyano, or acyl, or two R₂₄ groups join to form a fused cycloalkyl or benzene ring;

q is 0, 1, or 2;

x is 0 or 1; and

y is 0, 1, or 2.

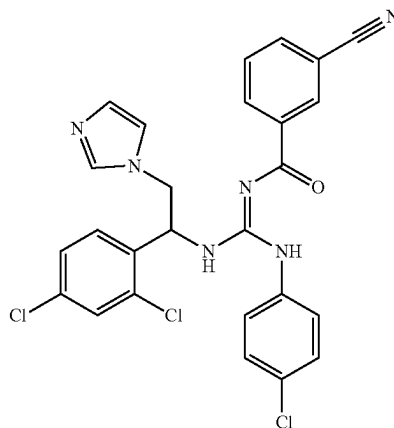
Most preferred are compounds as immediately defined above wherein, R₁ is cyano or —C(=O)R₉; R₉ is optionally substituted phenyl or phenyl C₁₋₄alkyl; x is 0 or 1; and q and y are 1 or 2. For this most preferred structure, its S stereoisomer is preferred. And further preferred is for its L group to be deuterium.

Example Embodiments of Formula (I)

Compounds from [5-6], selected as specific anti-cancer therapeutics by the invention of this disclosure, selected because they inhibit the reverse, more than the forward, mode of ATP synthase. EC₅₀ and IC₅₀ used interchangeably. EC₅₀ values for F₁F₀ ATP hydrolysis, and F₁F₀ ATP synthesis, in NADH-linked and NADPH-linked sub-mitochondrial (SMP) assays respectively, sourced from [5-6], are presented. [5-6] refer to these EC₅₀ values as IC₅₀ values for inhibiting F₁F₀ ATP hydrolase (reverse mode) and F₁F₀ ATP synthase (forward mode). However, this is incorrect.

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Because, as identified by the invention of this disclosure, explained herein, although these molecules inhibit F₁F₀ ATP hydrolase, their reducing of F₁F₀ ATP synthesis is not (predominantly) because of inhibiting F₁F₀ ATP synthase, but by uncoupling. More preferred molecules of this invention have a low EC₅₀ for F₁F₀ ATP hydrolysis, and a higher EC₅₀ for F₁F₀ ATP synthesis, and their ratio difference is large.

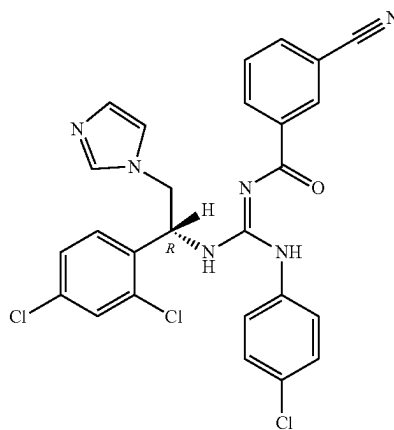


Racemate

EC₅₀ F₁F₀ ATP hydrolase = 0.033 ± 0.02 (μM)

EC₅₀ F₁F₀ ATP synthesis > 100 (μM)

EC₅₀ Ratio > 3,030



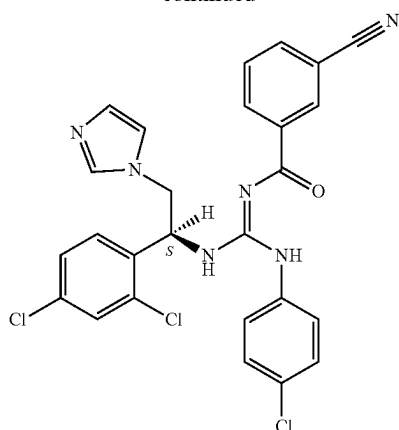
R stereoisomer

EC₅₀ F₁F₀ ATP hydrolase > 100 (μM)

EC₅₀ F₁F₀ ATP synthesis > 100 (μM)

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-continued



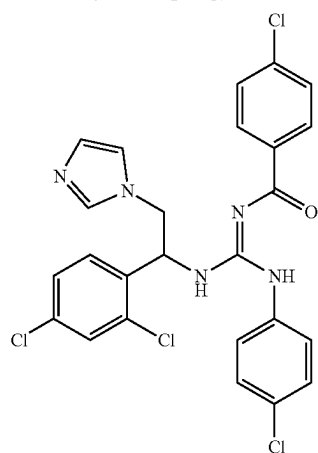
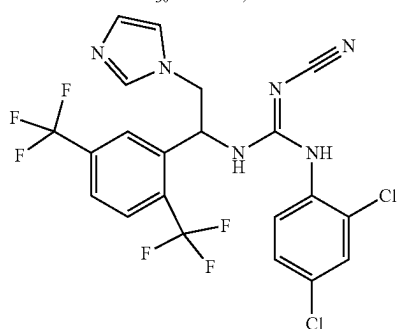
S stereoisomer Preferred

EC₅₀ F₁F₀ ATP hydrolase = 0.018 ± 0.016 (μM)EC₅₀ F₁F₀ ATP synthesis > 100 (μM)EC₅₀ Ratio > 5,556

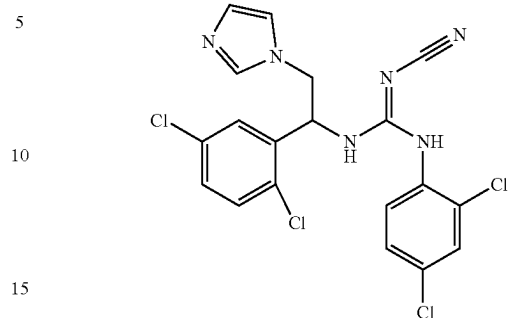
In rat: orally bioavailable (47%), i.v. half-life (2.1 hours),

C_{max} = 21 μM, volume of distribution V_{SS} = 2.37 L/kg

Further example embodiments of Formula (I), with SMP data, reinterpreted (as aforementioned, these molecules don't significantly inhibit F₁F₀ ATP synthase but do reduce F₁F₀ ATP synthesis by uncoupling), from [5],

EC₅₀F₁F₀ ATP hydrolase = 0.082 ± 0.03 (μM)EC₅₀F₁F₀ ATP synthesis > 100 (μM)EC₅₀ Ratio > 1,220EC₅₀F₁F₀ ATP hydrolase = 0.71 ± 0.34 (μM)EC₅₀F₁F₀ ATP synthesis > 100 (μM)EC₅₀ Ratio > 141**42**

-continued

EC₅₀F₁F₀ ATP hydrolase = 0.60 ± 0.16 (μM)EC₅₀F₁F₀ ATP synthesis > 100 (μM)EC₅₀ Ratio > 167

Further examples [5]:

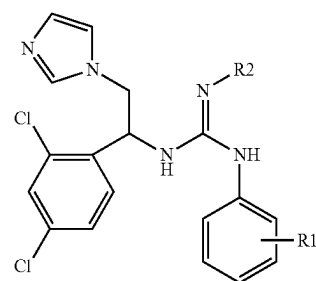
For all: EC₅₀ F₁F₀ATP synthesis > 100 μMEC₅₀ F₁F₀

ATP

hydro-

lase (μM)

R1	R2	
4-Cl	CN	8.8



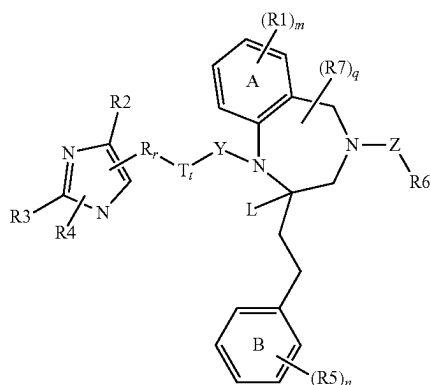
2-Cl	CN	2.23
2,3-Cl ₂	CN	2.49 ±
3-Cl	CN	0.72
4-Cl	C(=O)	0.28
4-CN—Ph		
4-Cl	C(=O)Et	2.27

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Example (II)

Summary of Formula (II)

This invention embodiment relates to compounds having the formula:



Formula (II)

or pharmaceutically-acceptable salts, solvates, hydrates or prodrugs thereof, wherein:

L is alkyl, or substituted alkyl, or any atom or isotope permitted by valence, for example hydrogen or deuterium;

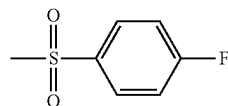
R₁ and R₅ are attached to any available carbon atom of phenyl rings A and B, respectively, and at each occurrence are independently selected from alkyl, substituted alkyl, halogen, cyano, nitro, OR₈, NR₈R₉, C(=O)R₈, CO₂R₈, C(=O)NR₈R₉, NR₈C(=O)R₉, NR₈C(=O)OR₉, S(O)_oR₉, NR₈SO₂R₉, SO₂NR₈R₉, cycloalkyl, heterocycle, aryl, and heteroaryl, and/or two of R₁ and/or two of R₅ join together to form a fused benzo ring;

R₂, R₃ and R₄ are independently selected from hydrogen, alkyl, and substituted alkyl, or one of R₂, R₃ and R₄ is a bond to R, T or Y and the other of R₂, R₃ and R₄ is selected from hydrogen, alkyl, and substituted alkyl;

Z and Y are independently selected from C(=O), —CO₂—, —SO₂—, —CH₂—, —CH₂C(=O)—, and —C(=O)C(=O)—, or Z may be absent;

R and T are selected from —CH₂—, —C(=O)—, and —CH[(CH₂)_p](Q)—, wherein Q is NR₁₀R₁₁, OR₁₀ or CN;

R₆ is selected from alkyl, alkenyl, substituted alkyl, substituted alkenyl, aryl, cycloalkyl, heterocycle, and heteroaryl; provided that where R₂ is hydrogen, Z—R₆ together are not —SO₂-Me or



R₇ is selected from hydrogen, alkyl, substituted alkyl, alkenyl, substituted alkenyl, aminoalkyl, halogen, cyano, nitro, keto (=O), hydroxy, alkoxy, alkylthio, C(=O)H, acyl, CO₂H, alkoxycarbonyl, carbamyl, sulfonyl, sulfonamide, cycloalkyl, heterocycle, aryl, and heteroaryl;

R₈ and R₉ are independently selected from hydrogen, alkyl, substituted alkyl, cycloalkyl, heterocycle, aryl, and heteroaryl, or R₈ and R₉ taken together to form a heterocycle or heteroaryl, except R₉ is not hydrogen when attached to a sulfonyl group as in SO₂R₉;

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R₁₀ and R₁₁ are independently selected from hydrogen, alkyl, and substituted alkyl;

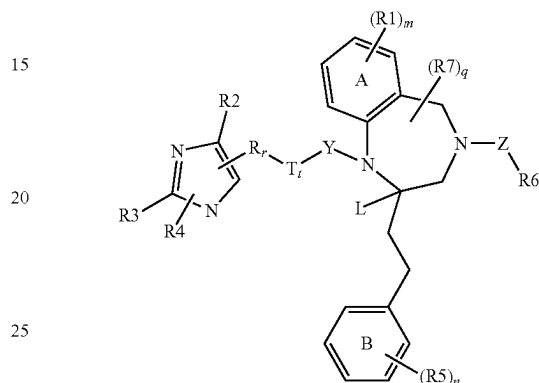
m and n are independently selected from 0, 1, 2 and 3

o, p and q are independently 0, 1 or 2; and

r and t are 0 or 1.

Preferred Compounds of Formula (II)

Preferred methods are to use, and preferred compounds are, compounds with the following formula, or pharmaceutically-acceptable salts, solvates, hydrates or prodrugs thereof,



wherein:

L is hydrogen or deuterium;

R₁ and R₅ are attached to any available carbon atom of phenyl ring A and phenyl ring B, respectively, and at each occurrence are independently selected from alkyl, aralkyl, aminoalkyl, halogen, cyano, nitro, hydroxy, alkoxy, alkylthio, NH₂, NH(alkyl), N(alkyl)₂, C(=O)H, acyl, CO₂H, alkoxycarbonyl, carbamyl, sulfonyl, sulfonamide, cycloalkyl, heterocycle, aryl, and heteroaryl, and/or two of R₁ and/or two of R₅ join together to form a fused benzo ring;

R₂, R₃ and R₄ are independently selected from hydrogen and alkyl;

Z is —CO₂—, —SO₂—, or is absent;

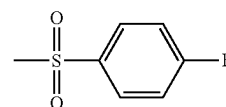
Y, R and T are selected from —CH₂— and —C(=O)—, R₆ is selected from:

C₁₋₄alkyl or C₁₋₄alkenyl optionally substituted with up to three of halogen, aryl and CO₂C₁₋₆alkyl;

phenyl optionally substituted with up to three R₁₂ and/or having fused thereto a benzo-ring or a five to six membered heteroaryl;

heteroaryl selected from thiophenyl, imidazolyl, pyrazolyl, and isoxazolyl wherein said heteroaryl is optionally substituted with up to two R₁₂,

provided that where R₂ is hydrogen, Z—R₆ together are not —SO₂-Me or



R₇ is selected from hydrogen, alkyl, substituted alkyl, alkenyl, substituted alkenyl, aminoalkyl, halogen, cyano, nitro, keto (=O), hydroxy, alkoxy, alkylthio, C(=O)H, acyl, CO₂H, alkoxycarbonyl, carbamyl, sulfonyl, sulfonamide, cycloalkyl, heterocycle, aryl, and heteroaryl;

45

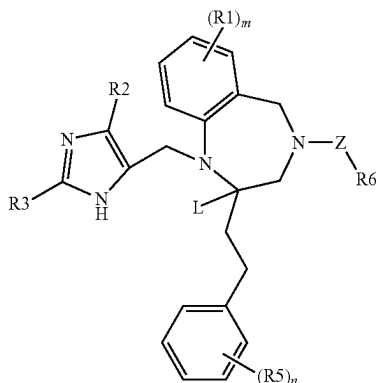
R_{12} at each occurrence is independently selected from each other R_{12} from the group consisting of C_{1-6} alkyl, halogen, nitro, cyano, hydroxy, alkoxy, $NHC(=O)$ alkyl, $-CO_2$ alkyl, $-SO_2$ phenyl, five to six membered monocyclic heteroaryl, and phenyloxy or benzyloxy in turn optionally substituted with halogen, C_{1-4} alkyl, and/or $O(C_{1-4}$ alkyl);

m and n are independently selected from 0, 1, 2 or 3; and

q is 0, 1 or 2; and

r and t are 0 or 1.

More preferred are compounds having the following formula, or pharmaceutically-acceptable salts, solvates, hydrates or prodrugs thereof,



wherein

R_1 and R_5 are attached to any available carbon atom of phenyl ring A and phenyl ring B, respectively, and at each occurrence are independently selected from alkyl, halogen, cyano, hydroxy, alkoxy, NH_2 , $NH(alkyl)$, $N(alkyl)_2$, $C(=O)H$, acyl, CO_2H , alkoxycarbonyl, and/or two of R_1 and/or two of R_5 join together to form a fused benzo ring;

R_2 , R_3 and R_4 are independently selected from hydrogen and lower alkyl;

Z is $-CO_2-$, $-SO_2-$, or is absent;

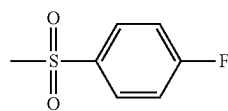
R_6 is selected from:

C_{1-4} alkyl or C_{1-4} alkenyl optionally substituted with up to three of halogen, aryl and CO_2C_{1-6} alkyl;

phenyl optionally substituted with up to three R_{12} and/or having fused thereto a benzo ring or a five to six membered heteroaryl;

heteroaryl selected from thiophenyl, imidazolyl, pyrazolyl, and isoxazolyl, wherein said heteroaryl is optionally substituted with up to two R_{12} ,

provided that where R_2 is hydrogen, Z- R_6 together are not $-SO_2-Me$ or

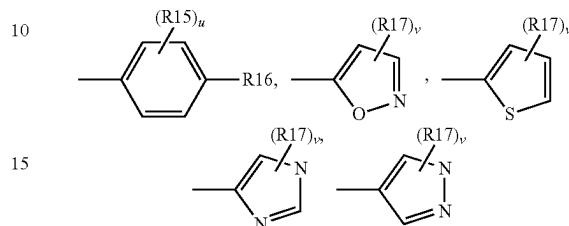


R_{12} at each occurrence is independently selected from each other R_{12} from the group consisting of C_{1-6} alkyl, halogen, nitro, cyano, hydroxy, alkoxy, $NHC(=O)$ alkyl, $-CO_2$ alkyl, $-SO_2$ phenyl, five to six membered monocyclic heteroaryl, and phenyloxy or benzyloxy in turn optionally substituted with halogen, C_{1-4} alkyl, and/or $O(C_{1-4}$ alkyl); and

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m and n are independently selected from 0, 1, or 2.

Even more preferred are compounds as immediately defined above wherein R_6 is selected from C_{1-4} alkyl, trifluoromethyl, benzyl, C_{2-3} alkenyl substituted with phenyl,



wherein:

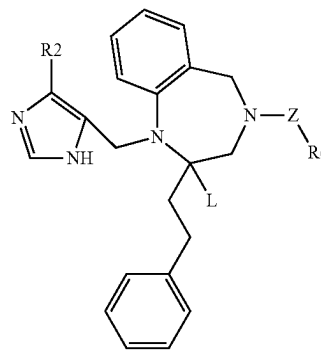
R_{15} is halogen, alkyl, nitro, cyano, hydroxy, alkoxy, $NHC(=O)$ alkyl, and/or two R_{15} groups are taken together to form a fused benzo ring or a five to six membered heteroaryl;

R_{16} is selected from hydrogen, halogen, alkyl, nitro, cyano, hydroxy, alkoxy, $NHC(=O)$ alkyl, and phenyloxy or benzyloxy in turn optionally substituted with 1 to 3 of halogen, cyano, and C_{1-4} alkoxy;

R_{17} is selected from alkyl, alkoxy, CO_2C_{1-6} alkyl, and SO_2 phenyl;

and u and v are independently 0, 1 or 2.

Most preferred compounds of Formula (II) are those having the formula:



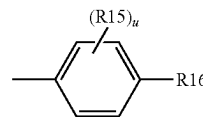
wherein

L is deuterium;

R_2 is hydrogen or CH_3 ;

Z is $-CO_2-$, $-SO_2-$, or is absent; and

R_6 is selected from the groups recited immediately above, most preferably

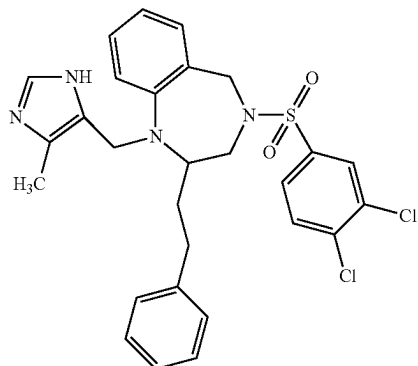


Example Embodiments of Formula (II)

Compounds from [8, 12], selected as specific anti-cancer therapeutics by the invention of this disclosure, selected because they inhibit the reverse, more than the forward, mode of ATP synthase. EC_{50} and IC_{50} used interchangeably.

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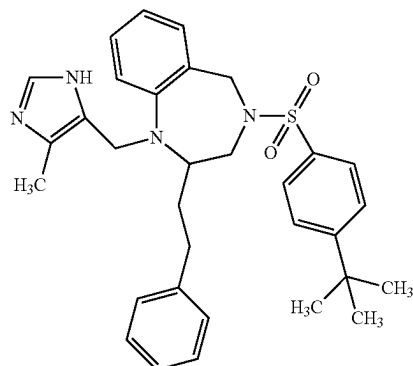
EC₅₀ values for F₁F₀ ATP hydrolysis, and F₁F₀ ATP synthesis, in NADH-linked and NADPH-linked sub-mitochondrial (SMP) assays respectively, sourced from [8], are presented. [8] refer to these EC₅₀ values as IC₅₀ values for inhibiting F₁F₀ ATP hydrolase (reverse mode) and F₁F₀ ATP synthase (forward mode). However, this is incorrect. Because, as identified by the invention of this disclosure, explained herein, although these molecules inhibit F₁F₀ ATP hydrolase, their reducing of F₁F₀ ATP synthesis is not (predominantly) because of inhibiting F₁F₀ ATP synthase, but by uncoupling.



EC₅₀ F₁F₀ ATP hydrolase = 0.022 (μM)
 EC₅₀ F₁F₀ ATP synthesis > 30 (μM)
 EC₅₀ Ratio > 1,364

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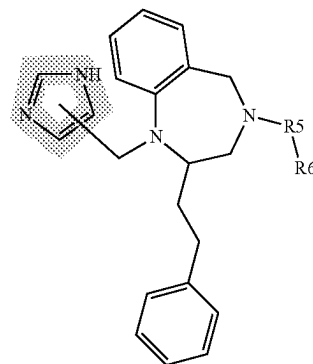
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EC₅₀ F₁F₀ ATP hydrolase = 0.077 (μM)
 EC₅₀ F₁F₀ ATP synthesis > 30 (μM)
 EC₅₀ Ratio > 390

For all compounds;

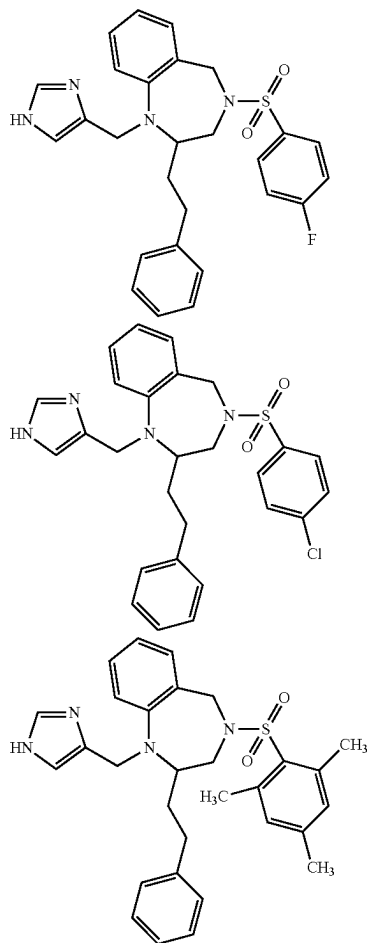
EC ₅₀ F ₁ F ₀ ATP synthesis > 30 μM			
R6	R5	Imidazole	EC ₅₀ F ₁ F ₀ ATP hydrolase (μM)
4-F—Ph	SO ₂	5-yl	0.221



Ph	SO ₂	5-yl	0.282
4-OH—Ph	SO ₂	5-yl	0.667
4-OMe—Ph	SO ₂	5-yl	0.077
2,5-di-Cl—Ph	SO ₂	5-yl	0.158
4-(AcNH)—Ph	SO ₂	5-yl	2.981
4-CN—Ph	SO ₂	5-yl	0.255
2-Cl-4-CN—Ph	SO ₂	5-yl	0.939
3-NO ₂ —Ph	SO ₂	5-yl	0.423
Naphth-1-yl	SO ₂	5-yl	0.338
Thiophen-2-yl	SO ₂	5-yl	0.636
Benzofurazan-7-yl	SO ₂	5-yl	1.777
Quinolin-8-yl	SO ₂	5-yl	2.935
Bn	SO ₂	5-yl	2.405
CF ₃	SO ₂	5-yl	0.077
4-t-Bu—Ph	SO ₂	5-yl	0.008
4-t-Bu—Ph	CH ₂	5-yl	2.138
4-t-Bu—Ph	CH ₂	4-Me-5-yl	2.352

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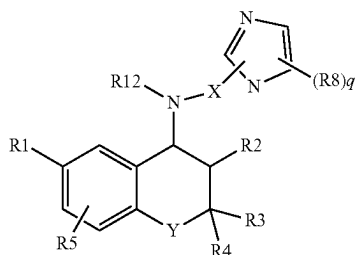
EC ₅₀ F ₁ F ₀ ATP synthesis > 30 μM			
R6	R5	Imidazole	EC ₅₀ F ₁ F ₀ ATP hydrolase (μM)
4-t-Bu—Ph	SO ₂	2-yl	>10
4-F—Ph	SO ₂	2-Me-5-yl	9.623
4-F—Ph	SO ₂	4-Me-5-yl	0.151



Example (III)

Summary of Formula (III)

This invention embodiment relates to compounds having the following formula:



Formula (III)

or their enantiomers, diastereomers, pharmaceutically-acceptable salts, solvates, hydrates or prodrugs thereof, wherein:

50 Optionally, one or more places upon the structure have deuterium in place of hydrogen;

R2 is hydrogen, hydroxy, or —OC(O)R14;

R14 is hydrogen, alkyl, haloalkyl, aryl, arylalkyl, cycloalkyl or (cycloalkyl)alkyl;

55 R3 and R4 are each independently hydrogen, alkyl or arylalkyl;

or R3 and R4 taken together with the carbon atom to which they are attached form a 3- to 7-membered carbocyclic ring;

60 R5 is hydrogen, alkyl, halogen, heterocyclo, nitrile, haloalkyl or aryl;

R12 is aryl or heterocyclo;

X is alkyl;

65 Y is a single bond, —CH₂—, —C(O)—, —O—, —S— or —N(R14)—;

A is nitrogen (N), or N⁺, or carbon;

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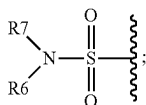
E is absent, or alkyl, or substituted alkyl, or any atom or isotope permitted by valence, for example hydrogen or deuterium;

R8 is hydrogen, alkyl, halogen, carbamyl or carbamylC₁₋₄alkyl, or two R8 groups join to form an optionally substituted fused phenyl ring;

q is 0, 1, 2 or 3.

R1 is any chemical group smaller than 300 Daltons, R9, cyano, hydrogen, halogen, alkyl, substituted alkyl, alkenyl, alkylene, alkoxy, thioalkyl, aminoalkyl, carbamyl, sulfonyl, sulfonamide, cycloalkyl, haloalkyl, haloalkoxy, aryl, heterocyclo, heteroaryl;

R9 is



R6 and R7 are independently hydrogen, alkyl, cycloalkyl, aryl, arylalkyl, haloalkyl, hydroxyalkyl, hydroxyalkyl substituted with a carboxylic ester or carboxylic acid, alkoxyalkyl, thioalkyl, (cycloalkyl)alkyl, morpholinylalkyl, heterocyclo or (heterocyclo)alkyl; or R6 and R7 taken together with the nitrogen atom to which they are attached form a 5- to 7-membered mono or bicyclic ring including fused rings such as 1-pyrrolidinyl, 1-piperidinyl, 1-azepinyl, 4-morpholinyl, 4-thiamorpholinyl, 4-thiamorpholine dioxide, 1-piperaZinyl, 4-alkyl-1-piperaZinyl, 4-arylalkyl-1-piperaZinyl, 4-diarylalkyl-1-piperazinyl; or 1-piperaZinyl, 1-pyrrolidinyl, 1-piperidinyl or 1-azepinyl substituted with one or more alkyl, alkoxy, alkylthio, halo, trifluoromethyl, hydroxy, aryl, arylalkyl, —COOR₁₄ or —CO-substituted amino;

or R5 and R6 taken together with the atoms to which they are attached form a 5- to 7-membered ring optionally substituted with aryl;

Preferred Compounds of Formula (III)

Preferred methods are to use, and preferred compounds are, compounds of Formula (III), their enantiomers, diastereomers, pharmaceutically-acceptable salts, solvates, hydrates or prodrugs thereof, in which:

Y is oxygen;

R2 is hydroxyl;

R3 and R4 are methyl;

R6 and R7 are alkyl; or R6 and R7 taken together with the nitrogen atom to which they are attached form a 6-membered ring;

X is alkyl;

R12 is aryl or heterocyclo;

A is N;

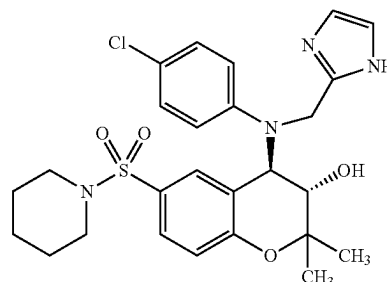
E is absent or hydrogen;

R5 and R8 are hydrogen;

Example embodiments of Formula (III) Compounds from [7], selected as specific anti-cancer therapeutics by the invention of this disclosure. EC₅₀ values for F₁F₀ ATP hydrolysis, and F₁F₀ ATP synthesis, in NADH-linked and NADPH-linked sub-mitochondrial (SMP) assays respectively. [7] refers to these EC₅₀ values as IC₅₀ values for inhibiting F₁F₀ ATP hydrolase (reverse mode) and F₁F₀ ATP synthase (forward mode). However, this is incorrect. Because, as identified by the invention of this disclosure, explained herein, although these molecules inhibit F₁F₀ ATP hydrolase, their reducing of F₁F₀ ATP synthesis is not

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(predominantly) because of inhibiting F₁F₀ ATP synthase, but by uncoupling. The structure on the left is BMS-199264. It does not harm ex vivo rat heart at a concentration (10 μM [11]) that it exerts anti-cancer activity (discovery of this disclosure).

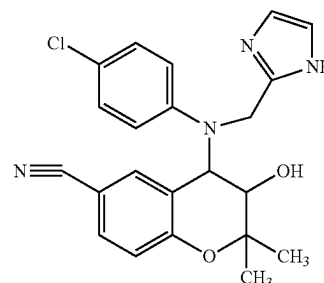


EC₅₀ F₁F₀ ATP hydrolase = 0.48 ± 0.23 (μM)

EC₅₀ F₁F₀ ATP synthesis = 18 ± 9.5 (μM)

EC₅₀ Ratio = 38

10 μM doesn't harm ex vivo rat heart



3S,4R

EC₅₀ F₁F₀ ATP hydrolase = 0.24 ± 0.13 (μM)

EC₅₀ F₁F₀ ATP synthesis = 3.8 ± 2.1 (μM)

3R,4S

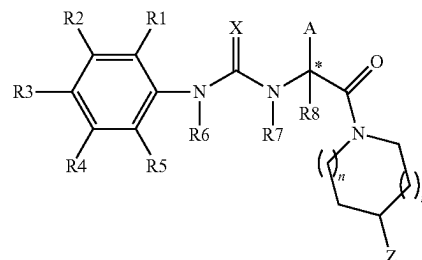
EC₅₀ F₁F₀ ATP hydrolase = 0.48 ± 0.26 (μM)

EC₅₀ F₁F₀ ATP synthesis = 4 ± 0.45 (μM)

Example (IV)

Summary of Formula (IV)

This invention embodiment relates to compounds having the following formula:



Formula (IV)

or their enantiomers, diastereomers, pharmaceutically-acceptable salts, solvates, hydrates or prodrugs thereof, wherein:

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X is selected from O or S;

A is selected from hydrogen, deuterium, alkyl, substituted alkyl, cycloalkyl, heterocycle, aryl, and heteroaryl;

n and m are 0, 1, or 2

R₁ through R₅ are independently selected from hydrogen, halogen, NO₂, CN, C₁₋₈alkyl, substituted C₁₋₈alkyl, C₃₋₈cycloalkyl, aryl, heterocyclo, heteroaryl, OR₉, SR₉, COR₁₁, CO₂R₁₁, CONR₉R₁₀ or NR₉R₁₀;

R₆ and R₇ are independently hydrogen, alkyl or substituted alkyl;

R₈ is hydrogen, deuterium, C₁₋₈alkyl, substituted C₁₋₈alkyl, aryl, heterocyclo or heteroaryl;

Z is hydrogen, alkyl, substituted alkyl, cycloalkyl, aryl, heterocyclo, heteroaryl, COR₁₁, CO₂R₁₁, SO₂R₁₁, S(O)R₁₁ or CONR₉R₁₀;

R₉ and R₁₀ are independently hydrogen, C₁₋₈alkyl, substituted C₁₋₈alkyl, C₃₋₁₀cycloalkyl, aryl, heterocyclo, heteroaryl, COR₁₃, SO₂R₁₃ or S(O)R₁₃; and

R₁₁, R₁₂ and R₁₃ are independently hydrogen, C₁₋₈alkyl, substituted C₁₋₈alkyl, C₃₋₁₀cycloalkyl, aryl, heterocyclo or heteroaryl;

wherein each occurrence of R₉-R₁₃ is chosen independently.

Preferred Compounds of Formula (IV)

Preferred methods are to use, and preferred compounds are, compounds of Formula (IV), their enantiomers, diastereomers, pharmaceutically-acceptable salts, solvates, hydrates or prodrugs thereof, in which:

R₂, R₃ and R₄ are all hydrogen; and/or

R₆ and R₇ are both hydrogen; and/or

n and m are both 1; and/or

R₁ and R₅ are both C₁₋₈ alkyl, preferably both R₁ and R₅ are isopropyl groups.

Other preferred methods use, and preferred compounds are, compounds of Formula (IV), their enantiomers, diastereomers, pharmaceutically-acceptable salts, solvates, hydrates or prodrugs thereof, in which:

Z is C₁₋₈alkyl, C₂₋₈alkenyl, C₁₋₈haloalkyl, cycloalkyl, aryl, arylalkyl, heteroaryl, heteroarylalkyl —COR₁₁, —CO₂R₁₁, —SO₂R₁₁, —S(O)R₁₁ or —CONR₉R₁₀; especially preferable is benzyl, —C(O)₂H or —C(O)₂C₁₋₈alkyl;

R₉ is hydrogen;

R₁₀ is C₁₋₈alkyl or C₃₋₁₀cycloalkyl; aryl or arylalkyl; and

R₁₁ is hydrogen, C₁₋₈alkyl, C₃₋₁₀cycloalkyl, C₃₋₁₀heterocycloalkyl, C₃₋₁₀aryl or C₃₋₁₀ arylalkyl.

Other preferred methods use, and preferred compounds are, compounds of Formula (IV), their enantiomers, diastereomers, pharmaceutically-acceptable salts, solvates, hydrates or prodrugs thereof, in which:

A is hydrogen, deuterium, C₁₋₈alkyl, heteroaryl, aryl, or alkyl substituted with heterocyclo, aryl, OH, SH, ST¹, —C(O), H, T³-NT⁵T⁶, —T⁸-C(O)_tT⁹-NT⁵T⁶ or T³-N(T²)T⁴NT⁵T⁶, T¹ is alkyl, (hydroxy)alkyl, (alkoxy)alkyl, alkenyl, alkynyl, cycloalkyl, (cycloalkyl)alkyl, cycloalkenyl, (cycloalkenyl)alkyl, aryl, (aryl)alkyl, heterocyclo, (heterocyclo)alkyl, heteroaryl or (heteroaryl)alkyl;

T² and T³ are each independently a single bond, —T⁸-S(O)_t-T⁹, —T⁸-C(O)_t-T⁹, —T¹⁸-C(S)-T⁹, —T⁸-S-T⁹, —T⁸-O—C(O)-T⁹, —T⁸-C(O)_tT⁹, —T⁸-C(=NT¹⁰)-T⁹ or —T⁸-C(O)—C(O)-T⁹;

T⁵, T⁶, T⁷, T⁸ and T⁹ are independently hydrogen, alkyl, (hydroxy)alkyl, (alkoxy)alkyl, alkenyl, alkynyl, cycloalkyl, (cycloalkyl)alkyl, cycloalkenyl, (cycloalkenyl)alkyl, aryl, (aryl)alkyl, heterocyclo, (heterocyclo)alkyl, heteroaryl or (heteroaryl)alkyl, each group optionally substituted where valence allows by one to three groups selected from halo, cyano, nitro, OH, oxo, —SH, alkyl, (hydroxy)alkyl,

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(alkoxy)alkyl, alkenyl, alkynyl, cycloalkyl, (cycloalkyl)alkyl, cycloalkenyl, (cycloalkenyl)alkyl, aryl, (aryl)alkyl, heterocyclo, (heterocyclo)alkyl, heteroaryl or (heteroaryl)alkyl, —OT¹¹, —ST¹¹, —C(O)_tH, —C(O)_tT¹¹, —O—C(O)T¹¹, T⁸C(O)_tN(T¹²)T¹¹, —SO₃H, —S(O)_tT¹¹, S(O)_tN(T¹²)T¹¹, —T¹³-NT¹¹T¹², —T¹³-N(T¹²)-T⁴-NT¹¹T²², —T¹³-N(T¹¹)-T¹²-T¹¹ and —T¹³-N(T¹⁸)-T¹⁴-H; or

T⁸ and T⁹ are each independently a single bond, alkylene, alkenylene or alkynylene;

T¹¹ is alkyl, (hydroxy)alkyl, (alkoxy)alkyl, alkenyl, alkynyl, cycloalkyl, (cycloalkyl)alkyl, cycloalkenyl, (cycloalkenyl)alkyl, aryl, (aryl)alkyl, heterocyclo, (heterocyclo)alkyl, heteroaryl or (heteroaryl)alkyl;

T¹² is halo, cyano, nitro, OH, oxo, —SH, alkyl, (hydroxy)alkyl, (alkoxy)alkyl, alkenyl, alkynyl, cycloalkyl, (cycloalkyl)alkyl, cycloalkenyl, (cycloalkenyl)alkyl, aryl, (aryl)alkyl, heterocyclo, (heterocyclo)alkyl, heteroaryl or (heteroaryl)alkyl, —C(O)_tH or —SO₃H;

T¹³ and T¹⁴ are each independently a single bond, —S(O)_t—, —C(O)—, —C(S)—, —O—, —S—, —O—C(O)—, —C(O)_t—, —C(=NT¹³)— or —C(O)—C(O)—;

wherein each occurrence of T¹-T¹⁴ is chosen independently; and

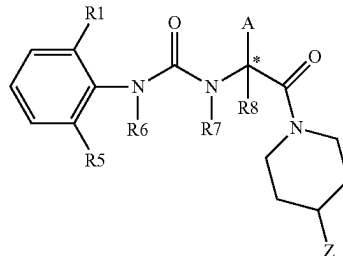
t is 1 or 2.

Preferred compounds of the foregoing section are those in which A is hydrogen, deuterium, C₁₋₈alkyl, hydroxyalkyl, heterocycloalkyl, heteroaryl alkyl, aryl, arylalkyl, or alkyl substituted with a group selected from SH, ST⁴, —C(O)_tH, T⁶-NT⁸T⁹, —T¹¹-C(O)_tT¹²-NT⁸T⁹ and T⁶-N(T⁵)T⁷NT⁸T⁹.

More preferred are those compounds in which A is hydrogen, deuterium, methyl, —CH₂(CH₃)₂, —(CH₂)₂(CH₃)₂, —CH(CH₃)CH₂(CH₃), —(CH₂)OH, hydroxyethyl, —(CH₂)₂SCH₃, —CH₂SH, phenyl, —CH₂(phenyl), —CH₂(p-hydroxyphenyl), —CH₂(indole), —(CH₂)C(O)NH₂, —(CH₂)₂C(O)NH₂, —(CH₂)₂C(O)OH, —CH₂C(O)OH, —(CH₂)₄NH₂, —(CH₂)₃(=NH)CNH₂, or —CH₂(imidazole). Especially preferred A groups are —CH(CH₃)CH₂(CH₃), phenyl, phenyl alkyl or —CH₂(2-indole).

Alternatively preferred methods use, and preferred compounds are, compounds of Formula (IVb), their enantiomers, diastereomers, pharmaceutically-acceptable salts, solvates, hydrates or prodrugs thereof, in which:

Formula (IVb)



wherein:

A is hydrogen, deuterium, C₁₋₈alkyl, heteroaryl, aryl, or alkyl substituted with heterocyclo, aryl, OH, SH, ST¹, —C(O)_tH, T³-NT⁵T⁶, —T⁸-C(O)_tT⁹-NT⁵T⁶ or T³-N(T²)T⁴NT⁵T⁶;

R¹ and R⁵ are independently C₁₋₈alkyl optionally substituted where valence allows;

R⁶ and R⁷ are independently hydrogen or C₁₋₈alkyl;

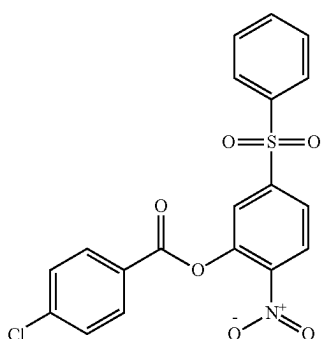
R⁸ is hydrogen, deuterium, C₁₋₈alkyl or substituted C₁₋₈alkyl;

Molecular permutations of BTB06584. Enumerations of this Markush structure, and their pharmaceutically-acceptable salts, solvates, hydrates and prodrugs thereof, are disclosed as anti-cancer molecules: the process/method of their use as anti-cancer molecules is disclosed by this invention. As valence permits: R1 is selected from the options of R1 (independently in each case of R1), X is selected from the options of X (independently in each case

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of X), R₂ is selected from the options of R₂ (independently in each case of R₂), R₃ is selected from the options of R₃ (independently in each case of R₃), R₄ is selected from the options of R₄ (independently in each case of R₄). In other embodiments one or more phenyl groups has one or more of its double bonds replaced with a single bond. In other embodiments, one or more phenyl groups is replaced with cyclohexane, each with the same possible substitutions as the phenyl it replaces. Hydrogen atoms aren't shown in this figure, but in further embodiments one or more hydrogen atoms is replaced with deuterium. In further embodiments: any possible isotopic substitution at one or more places.

Example Embodiment of Formula (V)



BTB06584

Definitions Used to Specify Formulas (I), (II), (III) and (IV)

The initial definition provided for a group or term herein applies to that group or term throughout the present specification, individually or as part of another group, unless otherwise indicated.

The term "alkyl" refers to straight or branched chain hydrocarbon groups having 1 to 21 carbon atoms, preferably 1 to 8 carbon atoms. Lower alkyl groups, that is, alkyl groups of 1 to 4 carbon atoms, are most preferred.

The term "substituted alkyl" refers to an alkyl group as defined above having one, two, three, or four substituents selected from the group consisting of halogen, trifluoromethyl, alkenyl, alkynyl, nitro, cyano, keto ($=O$), OR_a , SR_a , NR_aR_b , NR_aSO_2 , $NR_aSO_2R_c$, SO_2R_c , $SO_2NR_aR_b$, CO_2R_a , $C(=O)R_a$, $C(=O)NR_aR_b$, $OC(=O)R_a$, $-OC(=O)NR_aR_b$, $NR_aC(=O)R_b$, $NR_aCO_2R_b$, $=N-OH$, $=N-O$ -alkyl, aryl, heteroaryl, heterocyclo and cycloalkyl, wherein R_a and R_b are selected from hydrogen, alkyl, alkenyl, cycloalkyl, heterocyclo, aryl, and heteroaryl, and R_c is selected from hydrogen, alkyl, cycloalkyl, heterocyclo aryl and heteroaryl. When a substituted alkyl includes an aryl, heterocyclo, heteroaryl, or cycloalkyl substituent, said ringed systems are as defined below and thus may in turn have zero to four substituents (preferably 0-2 substituents), also as defined below. When either R_a , R_b or R_c is an alkyl, said alkyl may optionally be substituted with 1-2 of halogen, trifluoromethyl, alkenyl, alkynyl, nitro, cyano, keto ($=O$), OH , O (alkyl), phenyloxy, benzyloxy, SH , S (alkyl), NH_2 , NH (alkyl), N (alkyl)₂, $NHSO_2$, $NHSO_2$ (alkyl), SO_2 (alkyl), SO_2NH_2 , SO_2NH (alkyl), CO_2H , CO_2 (alkyl), $C(=O)H$, $C(=O)$ alkyl, $C(=O)NH_2$, $C(=O)NH$ (alkyl), $C(=O)N$ (alkyl)₂, $OC(=O)$ alkyl, $-OC(=O)NH_2$, $-OC(=O)NH$ (alkyl), $NHC(=O)$ alkyl, and/or $NHCO_2$ (alkyl).

"Alkyl" when used in conjunction with another group such as in arylalkyl refers to a substituted alkyl in which at

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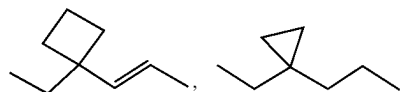
least one of the substituents is the specifically named group. For example, the term arylalkyl includes benzyl, or any other straight or branched chain alkyl having at least one aryl group attached at any point of the alkyl chain. As a further example, the term carbamylalkyl includes the group $-(CH_2)_n-NH-C(=O)alkyl$, Wherein n is 1 to 12.

The term "alkenyl" refers to straight or branched chain hydrocarbon groups having 2 to 21 carbon atoms and at least one double bond. Alkenyl groups of 2 to 6 carbon atoms and having one double bond are most preferred.

The term "alkynyl" refers to straight or branched chain hydrocarbon groups having 2 to 21 carbon atoms and at least one triple bond. Alkynyl groups of 2 to 6 carbon atoms and having one triple bond are most preferred.

The term "alkylene" refers to bivalent straight or branched chain hydrocarbon groups having 1 to 21 carbon atoms, preferably 1 to 8 carbon atoms, e.g., $\{-CH_2-\}_n$, Wherein n is 1 to 12, preferably 1-8. Lower alkylene groups, that is, alkylene groups of 1 to 4 carbon atoms, are most preferred. The terms "alkenylene" and "alkynylene" refer to bivalent radicals of alkenyl and alkynyl groups, respectively, as defined above.

When reference is made to a substituted alkylene, alkenylene, or alkynylene group, these groups are substituted with one to four substituents as defined above for alkyl groups. A substituted alkylene, alkenylene, or alkynylene may have a ringed substituent attached in a spiro fashion as in



and so forth.

The term "alkoxy" refers to an alkyl or substituted alkyl group as defined above having one, two or three oxygen atoms ($-O-$) in the alkyl chain. For example, the term "alkoxy" includes the groups $-O-C_{1-12}alkyl$, $-C_{1-6}alkylene-O-C_{1-6}alkyl$, $-C_{1-4}alkylene-O$ -phenyl, and so forth.

The term "thioalkyl" or "alkylthio" refers to an alkyl or substituted alkyl group as defined above having one or more sulphur ($-S-$) atoms in the alkyl chain. For example, the term "thioalkyl" or "alkylthio" includes the groups $-(CH_2)_n-S-CH_2aryl$, $-(CH_2)_n-S-aryl$, etc. etc.

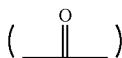
The term "aminoalkyl" or "alkylamino" refers to an alkyl or substituted alkyl group as defined above having one or more nitrogen ($-NR'-$) atoms in the alkyl chain. For example, the term "aminoalkyl" includes the groups $-NR'-C_{1-12}alkyl$ and $-CH_2-NR'-aryl$, etc. (where R' is hydrogen, alkyl or substituted alkyl as defined above.) "Amino" refers to the group $-NH_2$.

When a subscript is used as in $C_{1-8}alkyl$, the subscript refers to the number of carbon atoms the group may contain. Zero when used in a subscript denotes a bond, e.g., $C_{0-4}alkyl$ refers to a bond or an alkyl of 1 to 4 carbon atoms. When used with alkoxy, thioalkyl or aminoalkyl, a subscript refers to the number of carbon atoms that the group may contain in addition to heteroatoms. Thus, for example, monovalent $C_{1-2}aminoalkyl$ includes the groups $-CH_2-NH_2$, $-NH-CH_3$, $-(CH_2)_2-NH_2$, $-NH-CH_2-CH_3$, $-CH_2-NH_2-CH_3$, and $-N-(CH_3)_2$. A lower aminoalkyl comprises an aminoalkyl having one to four carbon atoms.

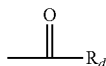
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The alkoxy, thioalkyl, or aminoalkyl groups may be monovalent or bivalent. By "monovalent" it is meant that the group has a valency (i.e., power to combine with another group), of one, and by "bivalent" it is meant that the group has a valency of two. For example, a monovalent alkoxy includes groups such as $\text{—O—C}_{1-12}\text{alkyl}$, $\text{—C}_{1-6}\text{alkylene—O—C}_{1-6}\text{alkyl}$, etc., whereas a bivalent alkoxy includes groups such as $\text{—O—C}_{1-2}\text{alkylene—}$, $\text{—C}_{1-6}\text{alkylene—O—C}_{1-6}\text{alkylene—}$, etc.

The term "acyl" refers to a carbonyl

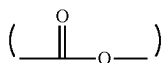


linked to an organic group i.e.

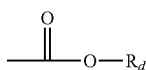


wherein R_d may be selected from alkyl, alkenyl, substituted alkyl, substituted alkenyl, aryl, heterocyclo, cycloalkyl, or heteroaryl, as defined herein.

The term "alkoxycarbonyl" refers to a group having a carboxy or ester group



linked to an organic radical, i.e.,



Wherein R_d is as defined above for acyl.

The term "carbonyl" refers to a functional group in which a nitrogen atom is directly bonded to a carbonyl, i.e., as in $\text{—NR}_e\text{C(=O)R}_f$ or $\text{—C(=O)NR}_e\text{R}_f$, wherein R_e and R_f can be hydrogen, alkyl, substituted alkyl, alkenyl, substituted alkenyl, alkoxy, cycloalkyl, aryl, heterocyclo, or heteroaryl, or they may join to form a ring.

The term "sulfonyl" refers to a sulphoxide group (i.e., —S(O)_{1-2}) linked to an organic radical R_c , as defined above.

The term "sulfonamide" or "sulfonamido" refers to the group $\text{—S(O)}_2\text{NR}_e\text{R}_f$ wherein R_e and R_f are as defined above. Preferably when one of R_e and R_f is optionally substituted heteroaryl or heterocycle (as defined below), the other of R_e and R_f is hydrogen or alkyl.

The term "cycloalkyl" refers to fully saturated and partially unsaturated hydrocarbon rings of 3 to 9, preferably 3 to 7 carbon atoms. The term "cycloalkyl" includes such rings having zero to four substituents (preferably 0-2 substituents), selected from the group consisting of halogen, alkyl, substituted alkyl (e.g., trifluoromethyl), alkenyl, substituted alkenyl, alkynyl, nitro, cyano, keto, OR_d , SR_d , NR_dR_e , NR_dSO_2 , $\text{NR}_d\text{SO}_2\text{R}_e$, C(=O)H , acyl, $\text{—CO}_2\text{H}$, alkoxycarbonyl, carbamyl, sulfonyl, sulfonamide, —OC(=O)R_d , —N—OH , —N—O—alkyl , aryl, heteroaryl, heterocyclo, a 4 to 7 membered carbocyclic ring, and a five or six membered ketal, e.g., 1,3-dioxolane or 1,3-dioxane,

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wherein R_c , R_d and R_e are defined as above. The term "cycloalkyl" also includes such rings having a phenyl ring fused thereto or having a carbon-carbon bridge of 3 to 4 carbon atoms. Additionally, when a cycloalkyl is substituted with a further ring, i.e., aryl, arylalkyl, heteroaryl, heteroarylalkyl, heterocyclo, heterocycloalkyl, cycloalkylalkyl, or a further cycloalkyl ring, such ring in turn may be substituted with one to two of $\text{C}_{0-4}\text{alkyl}$ optionally substituted with halogen, trifluoromethyl, alkenyl, alkynyl, nitro, cyano, keto (=O), OH, O(alkyl), phenyloxy, benzyloxy, SH, S(alkyl), NH_2 , NH(alkyl), N(alkyl) $_2$, NHSO_2 , $\text{NHSO}_2\text{(alkyl)}$, $\text{SO}_2\text{(alkyl)}$, SO_2NH_2 , $\text{SO}_2\text{NH(alkyl)}$, CO_2H , $\text{CO}_2\text{(alkyl)}$, C(=O)H , C(=O)alkyl , C(=O)NH_2 , C(=O)NH(alkyl) , C(=O)N(alkyl)_2 , OC(=O)alkyl , —OC(=O)NH_2 , —OC(=O)NH(alkyl) , NHC(=O)alkyl , and $\text{NHCO}_2\text{(alkyl)}$.

The term "halo" or "halogen" refers to chloro, bromo, fluoro and iodo.

The term "haloalkyl" means a substituted alkyl having one or more halo substituents. For example, "haloalkyl" includes mono, bi, and trifluoromethyl.

The term "haloalkoxy" means an alkoxy group having one or more halo substituents. For example, "haloalkoxy" includes OCF_3 .

The term "aryl" refers to phenyl, biphenyl, 1-naphthyl, 2-naphthyl, and anthracenyl, with phenyl being preferred. The term "aryl" includes such rings having zero to four substituents (preferably 0-2 substituents), selected from the group consisting of halo, alkyl, substituted alkyl (e.g., trifluoromethyl), alkenyl, substituted alkenyl, alkynyl, nitro, cyano, OR_d , SR_d , NR_dR_e , NR_dSO_2 , $\text{NR}_d\text{SO}_2\text{R}_e$, C(=O)H , acyl, $\text{—CO}_2\text{H}$, alkoxycarbonyl, carbamyl, sulfonyl, sulfonamide, —OC(=O)R_d , heteroaryl, heterocyclo, cycloalkyl, phenyl, benzyl, naphthyl, including phenylethyl, phenyloxy, and phenylthio, wherein R_c , R_d and R_e are defined as above. Additionally, two substituents attached to an aryl, particularly a phenyl group, may join to form a further ring such as a fused or spiro-ring, e.g., cyclopentyl or cyclohexyl or fused heterocycle or heteroaryl. When an aryl is substituted with a further ring, such ring in turn may be substituted with one to two of $\text{C}_{0-4}\text{alkyl}$ optionally substituted with halogen, trifluoromethyl, alkenyl, alkynyl, nitro, cyano, keto (=O), OH, O(alkyl), phenyloxy, benzyloxy, SH, S(alkyl), NH_2 , NH(alkyl), N(alkyl) $_2$, NHSO_2 , $\text{NHSO}_2\text{(alkyl)}$, $\text{SO}_2\text{(alkyl)}$, SO_2NH_2 , $\text{SO}_2\text{NH(alkyl)}$, CO_2H , $\text{CO}_2\text{(alkyl)}$, C(=O)H , C(=O)alkyl , C(=O)NH_2 , C(=O)NH(alkyl) , C(=O)N(alkyl)_2 , OC(=O)alkyl , —OC(=O)NH_2 , —OC(=O)NH(alkyl) , NHC(=O)alkyl , and $\text{NHCO}_2\text{(alkyl)}$.

The term "heterocyclo" refers to substituted and unsubstituted non-aromatic 3 to 7 membered monocyclic groups, 7 to 11 membered bicyclic groups, and 10 to 15 membered tricyclic groups, in which at least one of the rings has at least one heteroatom selected from O, S and N. Each ring of the heterocyclo group containing a heteroatom can contain one or two oxygen or sulfur atoms and/or from one to four nitrogen atoms provided that the total number of heteroatoms in each ring is four or less, and further provided that the ring contains at least one carbon atom. The fused rings completing bicyclic and tricyclic groups may contain only carbon atoms and may be saturated, partially saturated, or unsaturated. The nitrogen and sulfur atoms may optionally be oxidized and the nitrogen atoms may optionally be quaternized. The heterocyclo group may be attached at any available nitrogen or carbon atom. The heterocyclo ring may contain zero to four substituents (preferably 0-2 substituents), selected from the group consisting of halo, alkyl, substituted alkyl (e.g., trifluoromethyl), alkenyl, substituted alkenyl, alkynyl, nitro, cyano, keto, OR_d , SR_d , NR_dR_e ,

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NR_dSO_2 , $\text{NR}_d\text{SO}_2\text{R}_e$, SO_2R_d , $\text{C}(=\text{O})\text{H}$, acyl, $-\text{CO}_2\text{H}$, alkoxycarbonyl, carbamyl, sulfonyl, sulfonamide, $-\text{OC}(=\text{O})\text{R}_d$, $=\text{N}-\text{OH}$, $=\text{N}-\text{O}-\text{alkyl}$, aryl, heteroaryl, cycloalkyl, a five or six membered ketal, e.g., 1,3-dioxolane or 1,3-dioxane, or a monocyclic 4 to 7 membered non aromatic ring having one to four heteroatoms, wherein R_c , R_d and R_e are defined as above. The term "heterocyclo" also includes such rings having a phenyl ring fused thereto or having a carbon-carbon bridge of 3 to 4 carbon atoms. Additionally, when a heterocyclo is substituted with a further ring, i.e., aryl, arylalkyl, heteroaryl, heteroarylalkyl, cycloalkyl, cycloalkylalkyl, heterocycloalkyl, or a further heterocyclo ring, such ring in turn may be substituted with one to two of $\text{C}_{0-4}\text{alkyl}$ optionally substituted with halogen, trifluoromethyl, alkenyl, alkynyl, nitro, cyano, keto ($=\text{O}$), OH, O(alkyl), phenyloxy, benzyloxy, SH, S(alkyl), NH_2 , NH(alkyl), N(alkyl) $_2$, NHSO_2 , $\text{NHSO}_2(\text{alkyl})$, $\text{SO}_2(\text{alkyl})$, SO_2NH_2 , $\text{SO}_2\text{NH}(\text{alkyl})$, CO_2H , $\text{CO}_2(\text{alkyl})$, $\text{C}(=\text{O})\text{H}$, $\text{C}(=\text{O})\text{alkyl}$, $\text{C}(=\text{O})\text{NH}_2$, $\text{C}(=\text{O})\text{NH}(\text{alkyl})$, $\text{C}(=\text{O})\text{N}(\text{alkyl})_2$, $\text{OC}(=\text{O})\text{alkyl}$, $-\text{OC}(=\text{O})\text{NH}_2$, $-\text{OC}(=\text{O})\text{NH}(\text{alkyl})$, $\text{NHC}(=\text{O})\text{alkyl}$, and $\text{NHCO}_2(\text{alkyl})$.

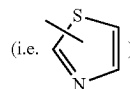
Exemplary monocyclic groups include azetidiny, pyrrolidiny, oxetanyl, imidazolidiny, oxazolidiny, isoxazolidiny, thiazolidiny, isothiazolidiny, tetrahydrofuranyl, piperidiny, piperaziny, 2-oxopiperaziny, 2-oxopiperidiny, 2-oxopyrrolodiny, 2-oxoazepiny, azepiny, 4-piperidonyl, tetrahydropyranyl, morpholiny, thiamorpholiny, thiamorpholiny sulfone, 1,3-dioxolane and tetrahydro-1,1-dioxothienyl and the like. Exemplary bicyclic heterocyclo groups include quinuclidiny.

The term "heteroaryl" refers to substituted and unsubstituted aromatic 5 to 7 membered monocyclic groups, 9 or 10 membered bicyclic groups, and 11 to 14 membered tricyclic groups which have at least one heteroatom selected from O, S and N in at least one of the rings. Each ring of the heteroaryl group containing a heteroatom can contain one or two oxygen or sulfur atoms and/or from one to four nitrogen atoms provided that the total number of heteroatoms in each ring is four or less and each ring has at least one carbon atom. The fused rings completing the bicyclic and tricyclic groups may contain only carbon atoms and may be saturated, partially saturated, or unsaturated. The nitrogen and sulfur atoms may optionally be oxidized and the nitrogen atoms may optionally be quaternized. Heteroaryl groups which are bicyclic or tricyclic must include at least one fully aromatic ring but the other fused ring or rings may be aromatic or non-aromatic. The heteroaryl group may be attached at any available nitrogen or carbon atom of any ring. The heteroaryl ring system may contain zero to four substituents (preferably 0-2 substituents), selected from the group consisting of halo, alkyl, substituted alkyl (e.g., trifluoromethyl), alkenyl, substituted alkenyl, alkynyl, nitro, cyano, OR_d , SR_d , NR_dR_e , NR_dSO_2 , $\text{NR}_d\text{SO}_2\text{R}_e$, SO_2R_d , $\text{C}(=\text{O})\text{H}$, acyl, $-\text{CO}_2\text{H}$, alkoxycarbonyl, carbamyl, sulfonyl, sulfonamide, $-\text{OC}(=\text{O})\text{R}_d$, heterocyclo, cycloalkyl, aryl, or a monocyclic 4 to 7 membered aromatic ring having one to four heteroatoms, including phenylethyl, phenyloxy, and phenylthio, wherein R_c , R_d and R_e are defined as above. Additionally, when a heteroaryl is substituted with a further ring, i.e., aryl, arylalkyl, heterocyclo, heterocycloalkyl, cycloalkyl, cycloalkylalkyl, heteroarylalkyl, or a further heteroaryl ring, such ring in turn may be substituted with one to two of $\text{C}_{0-4}\text{alkyl}$ optionally substituted with halogen, trifluoromethyl, alkenyl, alkynyl, nitro, cyano, keto ($=\text{O}$), OH, O(alkyl), phenyloxy, benzyloxy, SH, S(alkyl), NH_2 , NH(alkyl), N(alkyl) $_2$, NHSO_2 , $\text{NHSO}_2(\text{alkyl})_m$, $\text{SO}_2(\text{alkyl})$, SO_2NH_2 , $\text{SO}_2\text{NH}(\text{alkyl})$, CO_2H , $\text{CO}_2(\text{alkyl})$, $\text{C}(=\text{O})\text{H}$,

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$\text{C}(=\text{O})\text{alkyl}$, $\text{C}(=\text{O})\text{NH}_2$, $\text{C}(=\text{O})\text{NH}(\text{alkyl})$, $\text{C}(=\text{O})\text{N}(\text{alkyl})_2$, $\text{OC}(=\text{O})\text{alkyl}$, $-\text{OC}(=\text{O})\text{NH}_2$, $-\text{OC}(=\text{O})\text{NH}(\text{alkyl})$, $\text{NHC}(=\text{O})\text{alkyl}$, and $\text{NHCO}_2(\text{alkyl})$.

Exemplary monocyclic heteroaryl groups include pyrrolyl, pyrazolyl, pyrazolinyl, imidazolyl, oxazolyl, isoxazolyl, thiazolyl



thiadiazolyl, isothiazolyl, furanyl, thienyl, oxadiazolyl, pyridyl, pyrazinyl, pyrimidinyl, pyridazinyl, triazinyl and the like.

Exemplary bicyclic heteroaryl groups include indolyl, benzothiazolyl, benzodioxolyl, benzoxazolyl, benzothienyl, quinoliny, tetrahydroisoquinoliny, isoquinoliny, benzimidazolyl, benzopyranyl, indoliziny, benzofuranyl, chromonyl, coumariny, benzopyranyl, cinnoliny, quinoxaliny, indazolyl, pyrrolopyridyl, furopyridiny, dihydroisoindolyl, tetrahydroquinoliny and the like.

Exemplary tricyclic heteroaryl groups include carbazolyl, benzidolyl, phenanthrolliny, acridiny, phenanthridiny, xanthenyl and the like.

When the term "unsaturated" is used herein to refer to a ring or group, the ring or group may be fully unsaturated or partially unsaturated.

The phrase "optionally substituted" is intended to include substituted or unsubstituted possibilities. Accordingly, the phrase "each group of which may be optionally substituted" means that each group includes both substituted and unsubstituted groups.

The use of the phrase "Where valency allows" means that the groups may be substituted only to the degree and nature allowed by valency of the group. This is commonly understood by those of skill in the art. For example, a hydrogen substituent cannot be further substituted nor can a phenyl group be directly substituted by an oxo group due to limits on valency.

Stereoisomers

All stereoisomers of Formula [X], such as those, for example, which may exist due to asymmetric carbons, including enantiomeric forms (which may exist even in the absence of asymmetric carbons) and diastereomeric forms, are contemplated and within the scope of this invention. Individual stereoisomers of the compounds of this invention may, for example, be substantially free of other isomers, or may be admixed, for example, as racemates or with all other or other selected, stereoisomers. The chiral centers of the present invention can have the S or R configuration as defined by the IUPAC 1974 Recommendations.

For the molecules presented in this invention's Description and Drawings: the present invention contemplates all geometric/conformational isomers, rotamers, atropisomers, stereoisomers, optically active forms, tautomers, keto-enol tautomers, cis- and trans-isomers, E and Z isomers, R- and S-enantiomers, diastereomers, (D)-isomers, (L)-isomers, the racemic mixtures thereof, other mixtures thereof and isotopic variants (e.g. deuterium in place of hydrogen in some or all places upon the molecule{s}) as falling within the scope of the invention. All such isomers, as well as mixtures thereof, are intended to be included in this invention. As well as analogues and pharmaceutically/physiologically acceptable salts/solvates/hydrates/chelates/metal complexes/mixtures/prodrugs/radionuclides/polymorphs/esters/derivatives/

carriers/crystalline forms/liposomes thereof. Unless indicated otherwise, chemical structures and graphical representations of compounds herein encompass all stereoisomers. Substituents around a carbon-carbon double bond are designated as being in the "Z" or "E" configuration wherein the terms "Z" and "E" are used in accordance with IUPAC standards. Unless otherwise specified, structures depicting double bonds encompass both the "E" and "Z" isomers.

The invention also embraces isotopically labelled compounds of the invention which are identical to those recited herein, except that one or more atoms are replaced by an atom having an atomic mass or mass number different from the atomic mass or mass number usually found in nature. Examples of isotopes that can be incorporated into compounds of the invention include isotopes of hydrogen, carbon, nitrogen, oxygen, phosphorus, sulfur, fluorine and chlorine, such as ^2H , ^3H , ^{13}C , ^{14}C , ^{15}N , ^{18}O , ^{17}O , ^{31}P , ^{32}P , ^{35}S , ^{18}F , and ^{36}Cl , respectively.

Salts, Solvates, Prodrugs

Throughout the specification, groups and substituents thereof may be chosen by one skilled in the field to provide stable moieties and compounds.

The compounds of Formula [X] form salts which are also within the scope of this invention. Reference to a compound of the Formula [X] herein is understood to include reference to salts thereof, unless otherwise indicated.

As used herein, the term "pharmaceutically acceptable salt" refers to any pharmaceutically acceptable salt (e.g., acid or base) of a compound of the present invention which, upon administration to a subject, is capable of providing a compound of this invention or an active metabolite or residue thereof. As is known to those of ordinary skill in the art, "salts" of the compounds of the present invention may be derived from inorganic or organic acids and bases. For therapeutic use, salts of the compounds of the present invention are contemplated as being pharmaceutically acceptable (i.e., non-toxic, physiologically acceptable). However, salts of acids and bases that are non-pharmaceutically acceptable may also find use, for example, in the preparation, isolation or purification of a pharmaceutically acceptable compound.

The term "salt(s)", as employed herein, denotes acidic and/or basic salts formed with inorganic and/or organic acids and bases. In addition, when a compound of Formula [X] contains both a basic moiety, such as, but not limited to an amine or a pyridine or imidazole ring, and an acidic moiety, such as, but not limited to a carboxylic acid, zwitterions ("inner salts") may be formed and are included within the term "salt(s)" as used herein.

Salts of the compounds of the Formula [X] may be formed, for example, by reacting a compound of the Formula [X] with an amount of acid or base, such as an equivalent amount, in a medium such as one in which the salt precipitates or in an aqueous medium followed by lyophilization.

The compounds of Formula [X] which contain a basic moiety, such as, but not limited to an amine or a pyridine or imidazole ring, may form salts with a variety of organic and inorganic acids. Exemplary acid addition salts include acetates (such as those formed with acetic acid or trihaloacetic acid, for example, trifluoroacetic acid), adipates, alginates, ascorbates, aspartates, benzoates, benzenesulfonates, bisulfates, borates, butyrates, citrates, camphorates, camphorsulfonates, cyclopentanepropionates, digluconates, dodecylsulfates, ethanesulfonates, fumarates, glucoheptanoates, glycerophosphates, hemisulfates, heptanoates, hexanoates, hydrochlorides (formed with hydro-

chloric acid), hydrobromides (formed with hydrogen bromide), hydroiodides, 2-hydroxyethanesulfonates, lactates, maleates (formed with maleic acid), methane-sulfonates (formed with methanesulfonic acid), 2-naphthalenesulfonates, nicotines, nitrates, oxalates, pectinates, persulfates, 3-phenylpropionates, phosphates, picrates, pivalates, propionates, salicylates, succinates, sulfates (such as those formed with sulfuric acid), sulfonates (such as those mentioned herein), tartrates, thiocyanates, toluenesulfonates such as tosylates, undecanoates, and the like.

The compounds of Formula [X] which contain an acidic moiety, such as, but not limited to a carboxylic acid, may form salts with a variety of organic and inorganic bases. Exemplary basic salts include ammonium salts, alkali metal salts such as sodium, lithium, and potassium salts, alkaline earth metal salts such as calcium and magnesium salts, salts with organic bases (for example, organic amines) such as benzathines, dicyclohexylamines, hydrabamines [formed with N,N-bis(dehydro-abietyl)ethylenediamine], N-methyl D-glucamines, N-methyl-D-glucamides, t-butyl amines, and salts with amino acids such as arginine, lysine and the like. Basic nitrogen-containing groups may be quaternized with agents such as lower alkyl halides (e.g., methyl, ethyl, propyl, and butyl chlorides, bromides and iodides), dialkyl sulfates (e.g., dimethyl, diethyl, dibutyl, and diamyl sulfates), long chain halides (e.g., decyl, lauryl, myristyl and stearyl chlorides, bromides and iodides), aralkyl halides (e.g., benzyl and phenethyl bromides), and others.

Compounds of the Formula [X], and salts thereof, may exist in their tautomeric form (for example, as an amide or imino ether). All such tautomeric forms are contemplated herein as part of the present invention.

In addition, compounds of the Formulas [X] may have prodrug forms. Any compound that will be converted in vivo to provide the bioactive agent (i.e., a compound of Formula [X]) is a prodrug within the scope and spirit of the invention.

For example, pro-drug compounds of the Formulas [X] may be carboxylate ester moieties. A carboxylate ester may be conveniently formed by esterifying any of the carboxylic acid functionalities found on the disclosed ring structure(s).

Various forms of prodrugs are well known in the art. For examples of such prodrug derivatives, see:

a) *Design of Prodrugs*, edited by H. Bundgaard, (Elsevier, 1985), and *Methods in Enzymology*, Vol. 42, p. 309-396, edited by K. Widder, et. al. (Academic Press, 1985);

b) *A Textbook of Drug Design and Development*, edited by Krosgaard-Larsen and H. Bundgaard, Chapter 5, "Design and Application of Prodrugs," by H. Bundgaard, p. 113-191 (1991);

c) H. Bundgaard, *Advanced Drug Delivery Reviews*, Vol. 8, p. 1-38 (1992);

d) H. Bundgaard, et al., *Journal of Pharmaceutical Sciences*, Vol. 77, p. 285 (1988); and

e) N. Kakeya, et. al., *Chem Phar Bull*, Vol. 32, p. 692 (1984).

It should further be understood that solvates (e.g., hydrates) of the compounds of Formula [X] are also within the scope of the present invention. Methods of solvation are generally known in the art.

Chelates, metal complexes, mixtures, radio-nuclides and liposomes of Formula [X] are within the scope of this invention.

Dosage

As used herein, the term "effective amount" refers to the amount of a compound sufficient to effect beneficial or desired results. An effective amount can be administered in

one or more administrations, applications or dosages and is not intended to be limited to a particular formulation or administration route.

The effective amount of a compound of the present invention may be determined by one of ordinary skill in the art. The specific dose level and frequency of dosage for any particular subject may vary and will depend upon a variety of factors, including the activity of the specific compound employed, the metabolic stability and length of action of that compound, the species, age, body weight, general health, sex and diet of the subject, the mode and time of administration, rate of excretion, drug combination, and severity of the particular condition.

An exemplary effective amount of compounds of Formula [X] may be within the dosage range of about 0.001 to about 300 mg/kg, preferably about 0.2 to about 50 mg/kg and more preferably about 0.5 to about 25 mg/kg (or from about 1 to about 2500 mg, preferably from about 5 to about 2000 mg) on a regimen in single or 2 to 4 divided daily doses. But more exactly it depends upon the compound used, the condition and its advancement/severity, the route of administration, type of dosing (e.g. pulse or consistent etc.), what other treatments are undertaken alongside or previously (e.g. chemotherapeutics, surgery, radiotherapy etc.), the age, sex, condition, previous/other diseases of the patient, pharmacokinetics of compound in that patient, response to treatment and exceptions to this dosage range may be contemplated by the present invention, and they might be changed during treatment to find the optimum. Optimal dosages to be administered to a subject may be determined by those skilled in the art. When the compounds described herein are co-administered with another agent, the effective amount may be less than when the agent is used alone.

Pharmaceutical Composition

As used herein, the term "pharmaceutical composition" refers to the combination of an active agent with a carrier, inert or active, making the composition especially suitable for diagnostic or therapeutic use in vivo. Disclosed is a pharmaceutical composition of a therapeutically effective amount of a compound(s) of Formula [X] or a pharmaceutically acceptable salt thereof, and one or more pharmaceutically acceptable carriers, additives and/or diluents.

As used herein, the term "pharmaceutically acceptable carrier" refers to any of the standard pharmaceutical carriers, such as a phosphate buffered saline solution, water, emulsions (e.g., such as an oil/water or water/oil emulsions), and various types of wetting agents. The compositions also can include stabilizers and preservatives. For examples of carriers, stabilizers and adjuvants, see e.g., Martin, Remington's Pharmaceutical Sciences, 15th Ed., Mack Publ. Co., Easton, Pa. [1975].

Administration

The compounds of Formula [X] may be administered by any means suitable for the condition to be treated. For example: oral, parenteral, enteral, infusion, injection, sublingual, topical, rectal, transdermal, intramuscular and inhalation. The compound may be delivered orally, such as in the form of tablets, capsules, granules, microgranules, pellets, soft-gels, powders, or liquid formulations including syrups, liquids, solutions, elixirs, suspensions, emulsions or magmas; sublingually; buccally; transdermally; parenterally, such as by subcutaneous, intravenous, intramuscular or intrasternal injection or infusion (e.g., as sterile injectable aqueous or non-aqueous solutions or suspensions); nasally such as by inhalation spray; rectally such as in the form of suppositories; or liposomally. Dosage unit formulations containing non-toxic, pharmaceutically acceptable vehicles or diluents

may be administered. The compounds may be administered in a form suitable for immediate release or extended release. Immediate release or extended release may be achieved with suitable pharmaceutical compositions or, particularly in the case of extended release, with devices such as subcutaneous implants or osmotic pumps.

Exemplary compositions for oral administration include suspensions which may contain, for example, microcrystalline cellulose for imparting bulk, alginic acid or sodium alginate as a suspending agent, methylcellulose as a viscosity enhancer, and sweeteners or flavouring agents such as those known in the art; and immediate release tablets which may contain, for example, microcrystalline cellulose, dicalcium phosphate, starch, magnesium stearate and/or lactose and/or other excipients, binders, extenders, disintegrants, diluents and lubricants such as those known in the art. The inventive compounds may be orally delivered by sublingual and/or buccal administration, e.g., with molded, compressed, or freeze-dried tablets. Exemplary compositions may include fast-dissolving diluents such as mannitol, lactose, sucrose, and/or cyclodextrins. Also included in such formulations may be high molecular weight excipients such as celluloses (AVICEL®) or polyethylene glycols (PEG); an excipient to aid mucosal adhesion such as hydroxypropyl cellulose (HPC), hydroxypropyl methyl cellulose (HPMC), sodium carboxymethyl cellulose (SCMC), and/or maleic anhydride copolymer (e.g., GANTREZ®); and agents to control release such as polyacrylic copolymer (e.g., CARBOPOL 934®). Lubricants, glidants, flavours, colouring agents and stabilizers may also be added for ease of fabrication and use.

Exemplary compositions for nasal aerosol or inhalation administration include solutions which may contain, for example, benzyl alcohol or other suitable preservatives, absorption promoters to enhance absorption and/or bioavailability, and/or other solubilizing or dispersing agents such as those known in the art.

Exemplary compositions for parenteral administration include injectable solutions or suspensions which may contain, for example, suitable non-toxic, parenterally acceptable diluents or solvents, such as mannitol, 1,3-butanediol, water, Ringer's solution, an isotonic sodium chloride solution, or other suitable dispersing or wetting and suspending agents, including synthetic mono- or diglycerides, and fatty acids, including oleic acid.

Exemplary compositions for rectal administration include suppositories which may contain, for example, suitable non-irritating excipients, such as cocoa butter, synthetic glyceride esters or polyethylene glycols, which are solid at ordinary temperatures but liquefy and/or dissolve in the rectal cavity to release the drug.

Co-Administration

As used herein, the term "co-administration" refers to the administration of at least two agent(s) (e.g., a compound of the present invention) or therapies to a subject. In some embodiments, the co-administration of two or more agents/therapies is concurrent. In other embodiments, a first agent/therapy is administered prior to a second agent/therapy. Those of skill in the art understand that the formulations and/or routes of administration of the various agents/therapies used may vary. The appropriate dosage for co-administration can be readily determined by one skilled in the art. In some embodiments, when agents/therapies are co-administered, the respective agents/therapies are administered at lower dosages than appropriate for their administration alone. Thus, co-administration is especially desirable in

embodiments where the co-administration of the agents/therapies lowers the requisite dosage of a known potentially harmful (e.g., toxic) agent(s).

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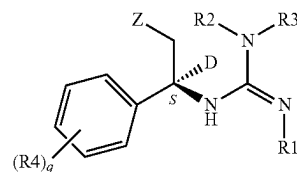
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The invention claimed is:

1. A compound, or a pharmaceutical composition comprising at least one compound (and one or more of a pharmaceutically-acceptable carrier [s], additive [s], diluent [s]), having the formula,



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or a pharmaceutically-acceptable salt, solvate, hydrate or prodrug thereof, wherein:

there is deuterium (D) enrichment at the chiral centre, and optionally at other locations;

wherein there is an enantiomeric excess (ee) of the S stereoisomer;

R₁ is cyano, —SO₂R₈, —C(=O)R₉, or heteroaryl;

R₂ is (i) independently hydrogen, alkyl, benzyl, or substituted alkyl, or (ii) taken together with R₃ forms a heterocyclo;

R₃ is (i) independently alkyl, substituted alkyl, alkylthio, aminoalkyl, carbamyl, B_B-aryl, B_B-heterocyclo, B_B-heteroaryl, or B_B-cycloalkyl, or (ii) phenyl optionally substituted with one or more groups selected from C₁₋₄alkyl, halogen, trifluoromethyl, OCF₃, cyano, nitro, amino, hydroxy, or methoxy, or (iii) independently selected from C₁₋₄alkyl, alkylthio, aminoalkyl, -B_B-aryl, -B_B-heterocyclo, B_B-cycloalkyl, and -B_B-heteroaryl, optionally having one to three substituents selected from R_{3a}; and/or having fused thereto a five or six membered carbocyclic ring, or (iv) taken together with R₂ forms a heterocyclo optionally substituted with alkyl or substituted alkyl;

B_B is a bond, C₁₋₄alkylene, C₂₋₄alkenylene, substituted C₁₋₄alkylene, substituted C₂₋₄alkenylene, substituted C₁₋₄alkylene-C(=O)NH—, —C(=O)NH—, —C₁₋₄alkylene-C(=O)NH—, —C(=O)NR₁₉—, —C₁₋₄alkylene-C(=O)NR₁₉—, or substituted C₁₋₄alkylene-C(=O)NR₁₉—, —(CHR₁₄)_m—(CR₁₅R₁₆)_n— or —(CHR₁₄)_p—C(=O)NH—;

R_{3a} at each occurrence is selected independently from alkyl, substituted alkyl, halogen, haloalkoxy, cyano, nitro, keto, trifluoromethyl, —NR₁₇R₁₈, —SR₁₇, —OR₁₇, —SO₂R_{17a}, —SO₂NR₁₇R₁₈, —NR₁₇C(=O)R₁₈, —CO₂R₁₇, —C(=O)R₁₇, cycloalkyl, aryl, heterocyclo, and heteroaryl, wherein when R_{3a} is cycloalkyl, aryl, heterocyclo or heteroaryl, said cycloalkyl, aryl, heterocyclo and heteroaryl in turn is optionally substituted with alkyl or substituted alkyl;

Z is a heteroaryl, provided that when R₁ is cyano, Z is not 2-pyridinyl; and/or

Z is selected from 5-membered heteroaryl groups containing 1, 2 or 3 heteroatoms each independently selected from N, O and S; and/or

Z is triazolyl optionally substituted with one to two R₇ substituents or imidazolyl optionally substituted with one to two R₇ substituents and/or having fused thereto a benzene ring in turn optionally substituted with one to two R₇ substituents; and

R₇ is alkyl, carbamyl, or substituted alkyl;

R₄ at each occurrence is selected independently of each other R₄ from the group consisting of halogen, trifluoromethyl, OCF₃, alkyl, substituted alkyl, haloalkyl, nitro, cyano, haloalkoxy, OR₂₅, SR₂₅, NR₂₅R₂₆, NR₂₅SO₂R₂₇, SO₂R₂₇, SO₂NR₂₅R₂₆, CO₂R₂₆, C(=O)R₂₆, C(=O)NR₂₅R₂₆, OC(=O)R₂₅, —OC(=O)NR₂₅R₂₆, NR₂₅C(=O)R₂₆, NR₂₅CO₂R₂₆, aryl, heteroaryl, heterocyclo and cycloalkyl;

R₈ is C₁₋₄alkyl or phenyl optionally substituted with alkyl, halogen, haloalkoxy, cyano, nitro, or trifluoromethyl;

R₉ is —NR₁₀R₁₁, alkyl, substituted alkyl, alkoxy, alkylthio, cycloalkyl, aryl, heteroaryl, heterocyclo, or —CO₂R₁₂, alkyl or phenyl optionally substituted with one to four of halogen, cyano, trifluoromethyl, nitro, hydroxy, phenyl, pyridinyl, C₁₋₄alkoxy, haloalkoxy, C₁₋₆alkyl, CO₂alkyl, SO₂alkyl, SO₂NH₂, amino, NH (C₁₋₄alkyl), N (C₁₋₄alkyl)₂, NHC(=O) alkyl, C(=O)

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alkyl, and/or C₁₋₄alkyl optionally substituted with one to three of trifluoromethyl, hydroxy, cyano, phenyl, pyridinyl; and/or a five or six membered heteroaryl or heterocyclo in turn optionally substituted with keto or having a benzene ring fused thereto or

a) C₁₋₄alkyl optionally substituted with one to two of:

i) SR₁₃, OR₁₃, NR_{13a}R_{13b}, halogen, trifluoromethyl, CO₂R_{13a}, and C(=O)NR_{13a}R_{13b};

ii) cycloalkyl optionally substituted with one to two of C(=O)H, C₁₋₄acyl, alkenyl, carbamyl, and/or phenyl in turn optionally substituted with halogen;

iii) phenyl or naphthyl optionally substituted with one to two of halogen, nitro, amino, alkyl, hydroxy, C₁₋₄alkoxy, or having fused thereto a five or six membered heterocyclo;

iv) pyridinyl, thiophenyl, furanyl, tetrahydrofuranyl, or azepinyl, optionally substituted with alkyl or having fused thereto a five to six membered carbocyclic ring optionally substituted with keto or C₁₋₄alkoxy;

b) 3 to 6 membered cycloalkyl optionally having up to four substituents selected from alkyl, halogen, cyano, alkenyl, acyl, alkylthio, carbamyl, phenyl in turn optionally substituted with halogen; or having an aryl fused thereto;

c) phenyl optionally substituted with one to four of halogen, cyano, trifluoromethyl, nitro, hydroxy, C₁₋₄alkoxy, haloalkoxy, C₁₋₄alkyl, CO₂alkyl, SO₂alkyl, SO₂NH₂, amino NH (C₁₋₄alkyl), N (C₁₋₄alkyl)₂, NHC(=O) alkyl, C(=O) alkyl, and/or C₁₋₄alkyl optionally substituted with one to three of trifluoromethyl, hydroxy, cyano, phenyl, pyridinyl; and/or a five or six membered heteroaryl or heterocyclo in turn optionally substituted with keto or having a benzene ring fused thereto;

d) pyridinyl, thiazolyl, furanyl, thiophenyl, and pyrrolyl optionally substituted with one to two of halogen, alkyl, and phenyl in turn optionally substituted with halogen or trifluoromethyl;

R₁₀ and R₁₁ are (i) independently selected from hydrogen, alkyl, substituted alkyl, alkoxy, heterocyclo, cycloalkyl, aryl, heteroaryl or C₁₋₄alkyl optionally substituted with one to two of —CO₂alkyl, —C(=O)NH (aryl), NH (aryl), cycloalkyl, phenyloxy, phenyl in turn optionally substituted with C₁₋₄alkyl, hydroxy, C₁₋₄alkoxy, halogen, amino, nitro, tetrahydrofuranyl, and/or five or six membered heterocyclo, or having a five or six membered heterocyclo fused thereto; pyrrolidinyl optionally substituted with keto; naphthyl, anthracenyl, pyridinyl, thiophenyl, furanyl, imidazolyl, benzimidazolyl, or indolyl in turn optionally substituted with C₁₋₄alkyl or C₁₋₄alkoxy; or (ii) taken together form a heteroaryl or heterocyclo selected from pyrrolidinyl, piperazinyl, piperidiny, morpholinyl, tetrahydropyridinyl, and imidazolidinyl, wherein said heterocyclo formed by R₁₀ and R₁₁ is optionally substituted with one to two of keto, CO₂H, C₁₋₄alkoxy, CO₂alkyl, C₁₋₄carbamyl, benzyl; phenyl in turn optionally substituted with alkyl, halogen, or C₁₋₄alkoxy;

tetrahydropyridinyl in turn optionally substituted with keto and/or phenyl; alkyl optionally substituted with amino or NHR₂₁ wherein R₂₁ is alkyl or phenyl optionally substituted with alkyl; and/or has a benzene ring fused thereto in turn optionally substituted with one to two of alkyl, C₁₋₄alkoxy, CO₂alkyl, and/or C₁₋₄carbamyl;

R₁₂ and R₁₉ are hydrogen or alkyl;

R₁₃ is hydrogen or alkyl;

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R_{13a} and R_{13b} are selected from hydrogen, alkyl, and aryl; R₁₄, R₁₅ and R₁₆ at each occurrence are independently selected from hydrogen, alkyl, hydroxy, hydroxy C₁₋₄alkyl, C₁₋₄alkoxy, and phenyl, and/or one of R₁₅ and one of R₁₆ join together to form a 3 to 6 membered cycloalkyl;

R₁₇ and R₁₈ are independently selected from hydrogen, alkyl, substituted alkyl, aryl, phenyl, or benzyl wherein the phenyl or benzyl is optionally substituted with alkyl, hydroxy, or hydroxyalkyl;

R_{17a} is alkyl or substituted alkyl;

R₂₅ and R₂₆ are independently selected from hydrogen, alkyl, or substituted alkyl, or taken together form a heterocyclo or heteroaryl ring;

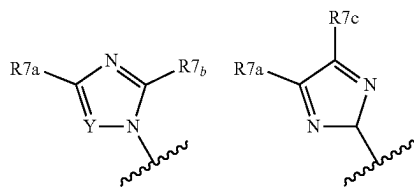
R₂₇ is alkyl or substituted alkyl;

q is 0, 1, 2, or 3;

m and n are 0, 1 or 2; and

p is 0, 1, 2, or 3.

2. A compound according to claim 1, wherein z is



Y is CH, N or CR_{7c};

R₁ is cyano, —SO₂R₈, —C(=O)R₉, or heteroaryl;

R₂ is (i) independently hydrogen, alkyl, or substituted alkyl, or (ii) taken together with R₃ forms a heterocyclo;

R₃ is (i) independently selected from

(a) alkyl optionally substituted with one to two of hydroxy and alkoxy;

(b) alkylthio or aminoalkyl optionally substituted with hydroxy or alkoxy;

(c)-A₁-aryl, wherein the aryl is optionally substituted with up to four substituents selected from alkyl, substituted alkyl, halogen, haloalkoxy, cyano, nitro, —NR₁₇R₁₈, —SR₁₇, —OR₁₇, —SO₂R_{17a}, —SO₂NR₁₇R₁₈, —NR₁₇C(=O)R₁₈, —CO₂R₁₇, —C(=O)R₁₇, cycloalkyl, aryl, heterocyclo, and heteroaryl, and/or has fused thereto a five or six membered cycloalkyl ring;

(d)-A₂-heteroaryl wherein the heteroaryl is a five or six membered monocyclic ring having 1 to 3 heteroatoms selected from N, O, and S, or an eight or nine membered bicyclic ringed system having at least one aromatic ring and 1 to 4 heteroatoms selected from N, O, and S in at least one of the rings, said heteroaryl being optionally substituted with halogen, alkyl, alkoxy, carbonyl, sulfonyl, nitro, cyano, trifluoromethyl, alkylthio, alkoxy, keto, —C(=O)H, acyl, benzyloxy, hydroxy, hydroxyalkyl, or phenyl optionally substituted with alkyl or substituted alkyl;

(e)-A₂-heterocyclo wherein the heterocyclo is optionally substituted with one to two groups selected from alkyl, keto, hydroxy, hydroxyalkyl, —C(=O)H, acyl, CO₂H, alkoxy, carbonyl, phenyl, and/or benzyl, and/or has a bridged carbon-carbon chain or fused benzene ring joined thereto;

(f)-A₂-cycloalkyl wherein the cycloalkyl is optionally substituted with one to two groups selected from alkyl,

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keto, —C(=O)H, acyl, CO₂H, alkoxy, carbonyl, and/or benzyl, and/or has a bridged carbon-carbon chain or fused benzene ring joined thereto; or

(ii) taken together with R₂ forms a heterocyclo;

R₄ at each occurrence is selected independently of each other R₄ from the group consisting of halogen, alkyl, haloalkyl, cyano, and haloalkoxy;

R_{7a}, R_{7b} and R_{7c} are independently selected from hydrogen, alkyl, carbamyl, or carbamylalkyl, or R_{7a} and R_{7c} join to form an aryl or heteroaryl;

R₈ is alkyl, arylalkyl, or aryl;

R₉ is —NR₁₀R₁₁, alkyl, substituted alkyl, alkoxy, alkylthio, cycloalkyl, aryl, heteroaryl, heterocyclo, CO₂R₁₂, or phenyl optionally substituted with one to four of halogen, cyano, trifluoromethyl, nitro, hydroxy, C₁₋₄alkoxy, haloalkoxy, C₁₋₆alkyl, CO₂alkyl, SO₂alkyl, SO₂NH₂, amino, NH (C₁₋₄alkyl), N (C₁₋₄alkyl)₂, NHC(=O)alkyl, C(=O)alkyl, and/or C₁₋₄alkyl optionally substituted with one to three of trifluoromethyl, hydroxy, cyano, phenyl, pyridinyl; and/or a five or six membered heteroaryl or heterocyclo in turn optionally substituted with keto or having a benzene ring fused thereto;

R₁₀ is independently hydrogen, alkyl, or alkoxy; and

R₁₁ is independently hydrogen, alkyl, substituted alkyl, alkoxy heterocyclo cycloalkyl, aryl, or heteroaryl; or

R₁₀ and R₁₁ taken together form a heterocyclo or heteroaryl optionally substituted with alkyl, keto, CO₂H, alkoxy, carbonyl, hydroxy, alkoxy, alkyl, carbamyl, aryl, or substituted alkyl, wherein when the R₁₀ and R₁₁ group comprises a phenyl ring, said phenyl ring is optionally substituted with one to two of alkyl, halogen, and alkoxy;

R₁₂ is hydrogen or alkyl;

A₁ is —(CHR₁₄)_m—V—(CR₁₅R₁₆)_n—or —(CHR₁₄)_p—(C=O)NH—;

A₂ is —(CHR₁₄)_m—V—(CR₁₅R₁₆)_n;

V is a bond, S, or —NR₂₂—;

R₁₄, R₁₅ and R₁₆ at each occurrence are independently selected from hydrogen, alkyl, hydroxy, hydroxy C₁₋₄alkyl, C₁₋₄alkoxy, and phenyl, and/or one of R₁₅ and one of R₁₆ join together to form a three to six membered cycloalkyl;

R₁₇ and R₁₈ are independently selected from hydrogen, alkyl, phenyl, and benzyl, wherein the phenyl and benzyl is optionally substituted with alkyl, hydroxy, or hydroxyalkyl;

R_{17a} is alkyl or substituted alkyl;

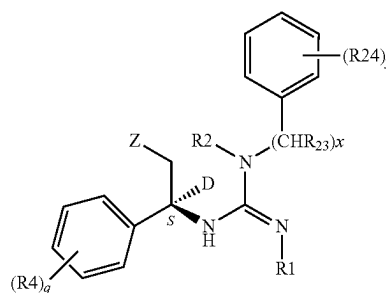
R₂₂ is hydrogen or alkyl;

m and n are 0, 1, 2, or 3;

p is 0, 1, 2, or 3; and

q is 0, 1, 2, or 3.

3. A compound according to claim 2 having the formula,



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or a pharmaceutically-acceptable salt, solvate, hydrate or prodrug thereof, wherein:

R_{7a} , R_{7b} and R_{7c} are independently selected from hydrogen, alkyl, carbamyl or carbamylC₁₋₄alkyl, or R_{7a} and R_{7c} join to form a fused phenyl ring;

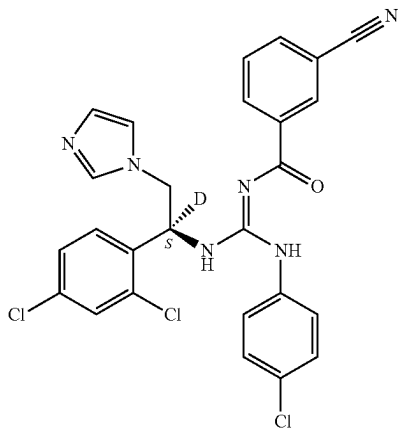
R_{23} is selected from hydrogen, alkyl, hydroxyalkyl, or phenyl;

R_{24} is selected from alkyl, halogen, trifluoromethyl, cyano, halogen, hydroxy, OCF₃, methoxy, phenyloxy, benzyloxy, cyano, acyl, or two R_{24} groups join to form a fused cycloalkyl or benzene ring; and

x is 0, 1, or 2; and

y is 0, 1, 2, or 3.

4. A compound according to claim 1 having the structure



or a pharmaceutically-acceptable salt, solvate, hydrate or prodrug thereof, wherein:

there is deuterium (D) enrichment at the chiral centre, and optionally at other locations; and

wherein there is an enantiomeric excess (ee) of the S stereoisomer.

5. A compound/composition according to claim 1,

wherein R_1 is thiazolyl; and/or

wherein Z is an optionally-substituted bicyclic heteroaryl.

6. A compound/composition according to claim 4,

wherein deuterium (D) enrichment at the chiral centre exceeds 40% deuterium incorporation, and optionally exceeds 90%; and

wherein the enantiomeric excess (ee) of the S stereoisomer exceeds 70%, and optionally exceeds 90%.

7. A method of treating or ameliorating cancer in a subject wherein the cancer is selected from breast cancer, prostate cancer, renal cancer, ovarian cancer, skin cancer, Central Nervous System (CNS) cancer, colon cancer, lung cancer, blood cancer;

wherein

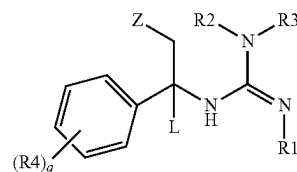
the method comprises administering an effective amount to the subject of at least one compound and/or composition selected from claim 1.

8. A method of treating or ameliorating cancer in a subject wherein the cancer is selected from breast cancer, prostate cancer, renal cancer, ovarian cancer, skin cancer, Central Nervous System (CNS) cancer, colon cancer, lung cancer, blood cancer;

wherein the method comprises administering to the subject an effective amount of at least one compound, or a pharmaceutical composition(s) comprising at least one

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compound (and one or more of a pharmaceutically-acceptable carrier [s], additive [s], diluent [s]), of the following formula:



or a pharmaceutically-acceptable salt, solvate, hydrate or prodrug thereof, or a stereoisomer

or a pharmaceutically-acceptable salt, solvate, hydrate or prodrug thereof, wherein:

L is alkyl, substituted alkyl or any atom or isotope permitted by valence except hydrogen at natural abundance;

R_1 is cyano, $-\text{SO}_2\text{R}_8$, $-\text{C}(=\text{O})\text{R}_9$, or heteroaryl;

R_2 is (i) independently hydrogen, alkyl, benzyl, or substituted alkyl, or (ii) taken together with R_3 forms a heterocyclo;

R_3 is (i) independently alkyl, substituted alkyl, alkylthio, aminoalkyl, carbamyl, B_B -aryl, B_B -heterocyclo, B_B -heteroaryl, or B_B -cycloalkyl, or (ii) phenyl optionally substituted with one or more groups selected from C₁₋₄alkyl, halogen, trifluoromethyl, OCF₃, cyano, nitro, amino, hydroxy, or methoxy, or (iii) independently selected from C₁₋₄alkyl, alkylthio, aminoalkyl, B_B -aryl, B_B -heterocyclo, B_B -cycloalkyl, and B_B -heteroaryl, optionally having one to three substituents selected from R_{3a} ; and/or having fused thereto a five or six membered carbocyclic ring, or (iv) taken together with R_2 forms a heterocyclo optionally substituted with alkyl or substituted alkyl; B_B is a bond, C₁₋₄alkylene, C₂₋₄alkenylene, substituted C₁₋₄alkylene, substituted C₂₋₄alkenylene, substituted C₁₋₄alkylene-C(=O)NH—, $-\text{C}(=\text{O})\text{NH}$ —, $-\text{C}(=\text{O})\text{NH}$ —, $-\text{C}(=\text{O})\text{NR}_{19}$ —, $-\text{C}(=\text{O})\text{NR}_{19}$ —, or substituted C₁₋₄alkylene-C(=O)NR₁₉—, $-(\text{CHR}_{14})_m-(\text{CR}_{15}\text{R}_{16})$ nor $-(\text{CHR}_{14})_p-\text{C}(=\text{O})\text{NH}$ —;

R_{3a} at each occurrence is selected independently from alkyl, substituted alkyl, halogen, haloalkoxy, cyano, nitro, keto, trifluoromethyl, $-\text{NR}_{17}\text{R}_{18}$, $-\text{SR}_{17}$, $-\text{OR}_{17}$, $-\text{SO}_2\text{R}_{17a}$, $-\text{SO}_2\text{NR}_{17}\text{R}_{18}$, $-\text{NR}_{17}\text{C}(=\text{O})\text{R}_{18}$, $-\text{CO}_2\text{R}_{17}$, $-\text{C}(=\text{O})\text{R}_{17}$, cycloalkyl, aryl, heterocyclo, and heteroaryl, wherein when R_{3a} is cycloalkyl, aryl, heterocyclo or heteroaryl, said cycloalkyl, aryl, heterocyclo and heteroaryl in turn is optionally substituted with alkyl or substituted alkyl;

Z is a heteroaryl, provided that when R_1 is cyano, Z is not 2-pyridinyl; and/or

Z is selected from 5-membered heteroaryl groups containing 1, 2 or 3 heteroatoms each independently selected from N, O and S; and/or

Z is triazolyl optionally substituted with one to two R_7 substituents or imidazolyl optionally substituted with one to two R_7 substituents and/or having fused thereto a benzene ring in turn optionally substituted with one to two R_7 substituents; and

R_7 is alkyl, carbamyl, or substituted alkyl;

R_4 at each occurrence is selected independently of each other R_4 from the group consisting of halogen, trifluoromethyl, OCF₃, alkyl, substituted alkyl, haloalkyl,

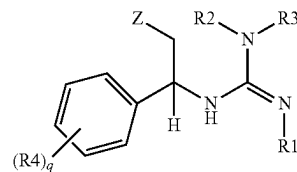
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- nitro, cyano, haloalkoxy, OR₂₅, SR₂₅, NR₂₅R₂₆, NR₂₅SO₂R₂₇, SO₂R₂₇, SO₂NR₂₅R₂₆, CO₂R₂₆, C(=O)R₂₆, C(=O)NR₂₅R₂₆, OC(=O)R₂₅, —OC(O)NR₂₅R₂₆, NR₂₅C(=O)R₂₆, NR₂₅CO₂R₂₆, aryl, heteroaryl, heterocyclo and cycloalkyl;
- R₈ is C₁₋₄alkyl or phenyl optionally substituted with alkyl, halogen, haloalkoxy, cyano, nitro, or trifluoromethyl;
- R₉ is —NR₁₀R₁₁, alkyl, substituted alkyl, alkoxy, alkylthio, cycloalkyl, aryl, heteroaryl, heterocyclo, or —CO₂R₁₂, alkyl or phenyl optionally substituted with one to four of halogen, cyano, trifluoromethyl, nitro, hydroxy, C₁₋₄alkoxy, haloalkoxy, C₁₋₆alkyl, CO₂alkyl, SO₂alkyl, SO₂NH₂, amino, NH (C₁₋₄alkyl), N (C₁₋₄alkyl)₂, NHC(=O) alkyl, C(=O) alkyl, and/or C₁₋₄alkyl optionally substituted with one to three of trifluoromethyl, hydroxy, cyano, phenyl, pyridinyl; and/or a five or six membered heteroaryl or heterocyclo in turn optionally substituted with keto or having a benzene ring fused thereto or
- a) C₁₋₄alkyl optionally substituted with one to two of:
- i) SR₁₃, OR₁₃, NR_{13a}R_{13b}, halogen, trifluoromethyl, CO₂R_{13a}, and C(=O)NR_{13a}R_{13b};
- ii) cycloalkyl optionally substituted with one to two of C(=O)H, C₁₋₄acyl, alkenyl, carbamyl, and/or phenyl in turn optionally substituted with halogen;
- iii) phenyl or naphthyl optionally substituted with one to two of halogen, nitro, amino, alkyl, hydroxy, C₁₋₄alkoxy, or having fused thereto a five or six membered heterocyclo;
- iv) pyridinyl, thiophenyl, furanyl, tetrahydrofuranyl, or azepinyl, optionally substituted with alkyl or having fused thereto a five to six membered carbocyclic ring optionally substituted with keto or C₁₋₄alkoxy;
- b) 3 to 6 membered cycloalkyl optionally having up to four substituents selected from alkyl, halogen, cyano, alkenyl, acyl, alkylthio, carbamyl, phenyl in turn optionally substituted with halogen; or having an aryl fused thereto;
- c) phenyl optionally substituted with one to four of halogen, cyano, trifluoromethyl, nitro, hydroxy, C₁₋₄alkoxy, haloalkoxy, C₁₋₄alkyl, CO₂alkyl, SO₂alkyl, SO₂NH₂, amino NH (C₁₋₄alkyl), N (C₁₋₄alkyl)₂, NHC(=O) alkyl, C(=O) alkyl, and/or C₁₋₄alkyl optionally substituted with one to three of trifluoromethyl, hydroxy, cyano, phenyl, pyridinyl; and/or a five or six membered heteroaryl or heterocyclo in turn optionally substituted with keto or having a benzene ring fused thereto;
- d) pyridinyl, thiazolyl, furanyl, thiophenyl, and pyrrolyl optionally substituted with one to two of halogen, alkyl, and phenyl in turn optionally substituted with halogen or trifluoromethyl;
- R₁₀ and R₁₁ are (i) independently selected from hydrogen, alkyl, substituted alkyl, alkoxy, heterocyclo, cycloalkyl, aryl, heteroaryl or C₁₋₄alkyl optionally substituted with one to two of —CO₂alkyl, —C(=O)NH (aryl), NH (aryl), cycloalkyl, phenyloxy, phenyl in turn optionally substituted with C₁₋₄alkyl, hydroxy, C₁₋₄alkoxy, halogen, amino, nitro, tetrahydrofuranyl, and/or five or six membered heterocyclo, or having a five or six membered heterocyclo fused thereto; pyrrolidinyl optionally substituted with keto; naphthyl, anthracenyl, pyridinyl, thiophenyl, furanyl, imidazolyl, benzimidazolyl, or indolyl in turn optionally substituted with C₁₋₄alkyl or C₁₋₄alkoxy; or
- (ii) taken together form a heteroaryl or heterocyclo selected from pyrrolidinyl, piperazinyl, piperidinyl,

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- morpholinyl, tetrahydropyridinyl, and imidazolidinyl, wherein said heterocyclo formed by R₁₀ and R₁₁ is optionally substituted with one to two of keto, CO₂H, C₁₋₄alkoxy, CO₂alkyl, C₁₋₄carbamyl, benzyl; phenyl in turn optionally substituted with alkyl, halogen, or C₁₋₄alkoxy; tetrahydropyridinyl in turn optionally substituted with keto and/or phenyl; alkyl optionally substituted with amino or NHR₂₁ wherein R₂₁ is alkyl or phenyl optionally substituted with alkyl; and/or has a benzene ring fused thereto in turn optionally substituted with one to two of alkyl, C₁₋₄alkoxy, CO₂alkyl, and/or C₁₋₄carbamyl;
- R₁₂ and R₁₉ are hydrogen or alkyl;
- R₁₃ is hydrogen or alkyl;
- R_{13a} and R_{13b} are selected from hydrogen, alkyl, and aryl;
- R₁₄, R₁₅ and R₁₆ at each occurrence are independently selected from hydrogen, alkyl, hydroxy, hydroxy C₁₋₄alkyl, C₁₋₄alkoxy, and phenyl, and/or one of R₁₅ and one of R₁₆ join together to form a 3 to 6 membered cycloalkyl;
- R₁₇ and R₁₈ are independently selected from hydrogen, alkyl, substituted alkyl, aryl, phenyl, or benzyl wherein the phenyl or benzyl is optionally substituted with alkyl, hydroxy, or hydroxyalkyl;
- R_{17a} is alkyl or substituted alkyl;
- R₂₅ and R₂₆ are independently selected from hydrogen, alkyl, or substituted alkyl, or taken together form a heterocyclo or heteroaryl ring;
- R₂₇ is alkyl or substituted alkyl;
- q is 0, 1, 2, or 3;
- m and n are 0, 1 or 2; and
- p is 0, 1, 2, or 3.
9. A method of treating or ameliorating cancer in a subject wherein the cancer is selected from breast cancer, prostate cancer, renal cancer, ovarian cancer, skin cancer, Central Nervous System (CNS) cancer, colon cancer, lung cancer, blood cancer;

wherein the method comprises administering to the subject an effective amount of at least one compound, or a pharmaceutical composition(s) comprising at least one compound (and one or more of a pharmaceutically-acceptable carrier [s], additive [s], diluent [s]), of the following formula:



- or a pharmaceutically-acceptable salt, solvate, hydrate or prodrug thereof, or a stereoisomer or a pharmaceutically-acceptable salt, solvate, hydrate or prodrug thereof, wherein:
- R₁ is cyano, —SO₂R₈, —C(=O)R₉, or heteroaryl;
- R₂ is (i) independently hydrogen, alkyl, benzyl, or substituted alkyl, or (ii) taken together with R₃ forms a heterocyclo;
- R₃ is (i) independently alkyl, substituted alkyl, alkylthio, aminoalkyl, carbamyl, B_B-aryl, B_B-heterocyclo, B_B-heteroaryl, or B_B-cycloalkyl, or (ii) phenyl optionally substituted with one or more groups selected from C₁₋₄alkyl, halogen, trifluoromethyl, OCF₃, cyano, nitro, amino, hydroxy, or methoxy, or (iii) independently selected from C₁₋₄alkyl, alkylthio, aminoalkyl,

-B_B-aryl, -B_B-heterocyclo, B_B-cycloalkyl, and -B_B-heteroaryl, optionally having one to three substituents selected from R_{3a}; and/or having fused thereto a five or six membered carbocyclic ring, or (iv) taken together with R₂ forms a heterocyclo optionally substituted with alkyl or substituted alkyl;

B_B is a bond, C₁₋₄alkylene, C₂₋₄alkenylene, substituted C₁₋₄alkylene, substituted C₂₋₄alkenylene, substituted C₁₋₄alkylene-C(=O)NH—, —C(=O)NH—, —C₁₋₄alkylene-C(=O)NH—, —C(=O)NR₁₉—, —C₁₋₄alkylene-C(=O)NR₁₉—, or substituted C₁₋₄alkylene-C(=O)NR₁₉—, —(CHR₁₄)_m—(CR₁₅R₁₆) nor —(CHR₁₄)_p—C(=O)NH—;

R_{3a} at each occurrence is selected independently from alkyl, substituted alkyl, halogen, haloalkoxy, cyano, nitro, keto, trifluoromethyl, —NR₁₇R₁₈, —SR₁₇, —OR₁₇, —SO₂R_{17a}, —SO₂NR₁₇R₁₈, —NR₁₇C(=O)R₁₈, —CO₂R₁₇, —C(=O)R₁₇, cycloalkyl, aryl, heterocyclo, and heteroaryl, wherein when R_{3a} is cycloalkyl, aryl, heterocyclo or heteroaryl, said cycloalkyl, aryl, heterocyclo and heteroaryl in turn is optionally substituted with alkyl or substituted alkyl;

Z is a heteroaryl, provided that when R₁ is cyano, Z is not 2-pyridinyl; and/or

Z is selected from 5-membered heteroaryl groups containing 1, 2 or 3 heteroatoms each independently selected from N, O and S; and/or

Z is triazolyl optionally substituted with one to two R₇ substituents or imidazolyl optionally substituted with one to two R₇ substituents and/or having fused thereto a benzene ring in turn optionally substituted with one to two R₇ substituents; and

R₇ is alkyl, carbamyl, or substituted alkyl;

R₄ at each occurrence is selected independently of each other R₄ from the group consisting of halogen, trifluoromethyl, OCF₃, alkyl, substituted alkyl, haloalkyl, nitro, cyano, haloalkoxy, OR₂₅, SR₂₅, NR₂₅R₂₆, NR₂₅SO₂R₂₇, SO₂R₂₇, SO₂NR₂₅R₂₆, CO₂R₂₆, C(=O)R₂₆, C(=O)NR₂₅R₂₆, OC(=O)R₂₅, —OC(=O)NR₂₅R₂₆, NR₂₅C(=O)R₂₆, NR₂₅CO₂R₂₆, aryl, heteroaryl, heterocyclo and cycloalkyl;

R₈ is C₁₋₄alkyl or phenyl optionally substituted with alkyl, halogen, haloalkoxy, cyano, nitro, or trifluoromethyl;

R₉ is —NR₁₀R₁₁, alkyl, substituted alkyl, alkoxy, alkylthio, cycloalkyl, aryl, heteroaryl, heterocyclo, or —CO₂R₁₂, alkyl or phenyl optionally substituted with one to four of halogen, cyano, trifluoromethyl, nitro, hydroxy, C₁₋₄alkoxy, haloalkoxy, C₁₋₆alkyl, CO₂alkyl, SO₂alkyl, SO₂NH₂, amino, NH (C₁₋₄alkyl), N (C₁₋₄alkyl)₂, NHC(=O) alkyl, C(=O) alkyl, and/or C₁₋₄alkyl optionally substituted with one to three of trifluoromethyl, hydroxy, cyano, phenyl, pyridinyl; and/or a five or six membered heteroaryl or heterocyclo in turn optionally substituted with keto or having a benzene ring fused thereto or

a) C₁₋₄alkyl optionally substituted with one to two of:

i) SR₁₃, OR₁₃, NR_{13a}R_{13b}, halogen, trifluoromethyl, CO₂R_{13a}, and C(=O)NR_{13a}R_{13b};

ii) cycloalkyl optionally substituted with one to two of C(=O)H, C₁₋₄acyl, alkenyl, carbamyl, and/or phenyl in turn optionally substituted with halogen;

iii) phenyl or naphthyl optionally substituted with one to two of halogen, nitro, amino, alkyl, hydroxy, C₁₋₄alkoxy, or having fused thereto a five or six membered heterocyclo;

iv) pyridinyl, thiophenyl, furanyl, tetrahydrofuranyl, or azepinyl, optionally substituted with alkyl or having

fused thereto a five to six membered carbocyclic ring optionally substituted with keto or C₁₋₄alkoxy;

b) 3 to 6 membered cycloalkyl optionally having up to four substituents selected from alkyl, halogen, cyano, alkenyl, acyl, alkylthio, carbamyl, phenyl in turn optionally substituted with halogen; or having an aryl fused thereto;

c) phenyl optionally substituted with one to four of halogen, cyano, trifluoromethyl, nitro, hydroxy, C₁₋₄alkoxy, haloalkoxy, C₁₋₄alkyl, CO₂alkyl, SO₂alkyl, SO₂NH₂, amino NH (C₁₋₄alkyl), N (C₁₋₄alkyl)₂, NHC(=O) alkyl, C(=O) alkyl, and/or C₁₋₄alkyl optionally substituted with one to three of trifluoromethyl, hydroxy, cyano, phenyl, pyridinyl; and/or a five or six membered heteroaryl or heterocyclo in turn optionally substituted with keto or having a benzene ring fused thereto;

d) pyridinyl, thiazolyl, furanyl, thiophenyl, and pyrrolyl optionally substituted with one to two of halogen, alkyl, and phenyl in turn optionally substituted with halogen or trifluoromethyl;

R₁₀ and R₁₁ are (i) independently selected from hydrogen, alkyl, substituted alkyl, alkoxy, heterocyclo, cycloalkyl, aryl, heteroaryl or C₁₋₄alkyl optionally substituted with one to two of —CO₂alkyl, —C(=O)NH (aryl), NH (aryl), cycloalkyl, phenyloxy, phenyl in turn optionally substituted with C₁₋₄alkyl, hydroxy, C₁₋₄alkoxy, halogen, amino, nitro, tetrahydrofuranyl, and/or five or six membered heterocyclo, or having a five or six membered heterocyclo fused thereto; pyrrolidinyl optionally substituted with keto; naphthyl, anthracenyl, pyridinyl, thiophenyl, furanyl, imidazolyl, benzimidazolyl, or indolyl in turn optionally substituted with C₁₋₄alkyl or C₁₋₄alkoxy; or

(ii) taken together form a heteroaryl or heterocyclo selected from pyrrolidinyl, piperazinyl, piperidinyl, morpholinyl, tetrahydropyridinyl, and imidazolidinyl, wherein said heterocyclo formed by R₁₀ and Ru is optionally substituted with one to two of keto, CO₂H, C₁₋₄alkoxy, CO₂alkyl, C₁₋₄carbamyl, benzyl; phenyl in turn optionally substituted with alkyl, halogen, or C₁₋₄alkoxy; tetrahydropyridinyl in turn optionally substituted with keto and/or phenyl; alkyl optionally substituted with amino or NHR₂₁ wherein R₂₁ is alkyl or phenyl optionally substituted with alkyl; and/or has a benzene ring fused thereto in turn optionally substituted with one to two of alkyl, C₁₋₄alkoxy, CO₂alkyl, and/or C₁₋₄carbamyl;

R₁₂ and R₁₉ are hydrogen or alkyl;

R₁₃ is hydrogen or alkyl;

R_{13a} and R_{13b} are selected from hydrogen, alkyl, and aryl;

R₁₄, R₁₅ and R₁₆ at each occurrence are independently selected from hydrogen, alkyl, hydroxy, hydroxy C₁₋₄alkyl, C₁₋₄alkoxy, and phenyl, and/or one of R₁₅ and one of R₁₆ join together to form a 3 to 6 membered cycloalkyl;

R₁₇ and R₁₈ are independently selected from hydrogen, alkyl, substituted alkyl, aryl, phenyl, or benzyl wherein the phenyl or benzyl is optionally substituted with alkyl, hydroxy, or hydroxyalkyl;

R_{17a} is alkyl or substituted alkyl;

R₂₅ and R₂₆ are independently selected from hydrogen, alkyl, or substituted alkyl, or taken together form a heterocyclo or heteroaryl ring;

R₂₇ is alkyl or substituted alkyl;

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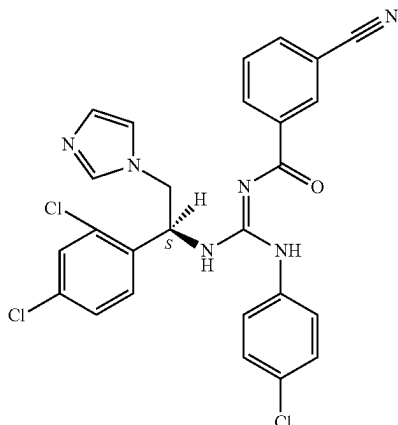
q is 0, 1, 2, or 3;

m and n are 0, 1 or 2; and

p is 0, 1, 2, or 3.

10. A method according to claim 9, wherein there is an enantiomeric excess (ee) of the S stereoisomer.

11. A method according to claim 10, using the compound:



or a pharmaceutically-acceptable salt, solvate, hydrate or prodrug thereof, wherein:

there is an enantiomeric excess (ee) of the S stereoisomer.

12. A method according to claim 8 wherein a compound administered to the subject has the ability/property to preferentially inhibit/and reduce the activity of the ATP-hydrolysing, as compared to the ATP-synthesizing, mode of ATP synthase,

wherein the differential between the compound's EC_{50} F_1F_0 ATP synthesis and a smaller valued EC_{50} F_1F_0 ATP hydrolysis in a Sub-Mitochondrial Particle [SMP] assay of the art is one or more of >10, >100, >1000, >5000 times difference.

13. A method according to claim 9 wherein a compound administered to the subject has the ability to preferentially inhibit and reduce the activity of the ATP-hydrolysing, as compared to the ATP-synthesizing, mode of ATP synthase,

wherein the differential between the compound's EC_{50} F_1F_0 ATP synthesis and a smaller valued EC_{50} F_1F_0 ATP hydrolysis in a Sub-Mitochondrial Particle [SMP] assay of the art is one or more of >10, >100, >1000, >5000 times difference.

14. A method according to claim 10,

wherein the enantiomeric excess (ee) of the S stereoisomer exceeds 70%, and optionally exceeds 90%.

15. A method according to claim 11,

wherein the enantiomeric excess (ee) of the S stereoisomer exceeds 70%, and optionally exceeds 90%.

16. A method of claim 8 wherein the Central Nervous system (CNS) cancer is astrocytoma or Glioblastoma Multiforme (GBM).

17. A method of claim 8 wherein the lung cancer is Non-Small Cell Lung Cancer (NSCLC).

18. A method of claim 8 wherein the skin cancer is melanoma.

19. A method of claim 8 wherein the blood cancer is selected from leukemia, lymphoma, Multiple myeloma.

20. A method of claim 19 wherein the leukemia is selected from Chronic Myeloid Leukemia (CML), Acute Myeloid Leukemia (AML), Acute Lymphoblastic Leukemia (ALL).

21. A method of claim 8 wherein the cancer is carcinoma.

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22. A method of claim 21 wherein the carcinoma is selected from adenocarcinoma, squamous cell carcinoma, large cell carcinoma.

23. A method of claim 8 wherein the cancer is selected from Triple Negative Breast Cancer (TNBC), Androgen Receptor (AR) negative prostate cancer, amelanotic melanoma, clear cell renal carcinoma, mesothelioma.

24. A method of claim 9 wherein the Central Nervous system (CNS) cancer is astrocytoma or Glioblastoma Multiforme (GBM).

25. A method of claim 9 wherein the lung cancer is Non-Small Cell Lung Cancer (NSCLC).

26. A method of claim 9 wherein the skin cancer is melanoma.

27. A method of claim 9 wherein the blood cancer is selected from leukemia, lymphoma, Multiple myeloma.

28. A method of claim 27 wherein the leukemia is selected from Chronic Myeloid Leukemia (CML), Acute Myeloid Leukemia (AML), Acute Lymphoblastic Leukemia (ALL).

29. A method of claim 9 wherein the cancer is carcinoma.

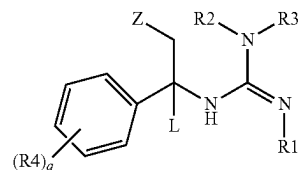
30. A method of claim 29 wherein the carcinoma is selected from adenocarcinoma, squamous cell carcinoma, large cell carcinoma.

31. A method of claim 9 wherein the cancer is selected from Triple Negative Breast Cancer (TNBC), Androgen Receptor (AR) negative prostate cancer, amelanotic melanoma, clear cell renal carcinoma, mesothelioma.

32. A method of claim 8 wherein the cancer exhibits the Warburg effect.

33. A method of claim 9 wherein the cancer exhibits the Warburg effect.

34. A method of treating or ameliorating cancer in a subject wherein the cancer performs F_1F_0 ATP hydrolysis; wherein the method comprises administering to the subject an effective amount of at least one compound, or a pharmaceutical composition(s) comprising at least one compound (and one or more of a pharmaceutically-acceptable carrier [s], additive [s], diluent [s]), of the following formula:



or a pharmaceutically-acceptable salt, solvate, hydrate or prodrug thereof, or a stereoisomer or a pharmaceutically-acceptable salt, solvate, hydrate or prodrug thereof, wherein:

L is alkyl, substituted alkyl or any atom or isotope permitted by valence;

R_1 is cyano, $-SO_2R_8$, $-C(=O)R_9$, or heteroaryl;

R_2 is (i) independently hydrogen, alkyl, benzyl, or substituted alkyl, or (ii) taken together with R_3 forms a heterocycle;

R_3 is (i) independently alkyl, substituted alkyl, alkylthio, aminoalkyl, carbamyl, B_B -aryl, B_B -heterocyclo, B_B -heteroaryl, or B_B -cycloalkyl, or (ii) phenyl optionally substituted with one or more groups selected from C_{1-4} alkyl, halogen, trifluoromethyl, OCF_3 , cyano, nitro, amino, hydroxy, or methoxy, or (iii) independently selected from C_{1-4} alkyl, alkylthio, aminoalkyl, $-B_B$ -aryl, $-B_B$ -heterocyclo, B_B -cycloalkyl, and $-B_B$ -heteroaryl, optionally having one to three substituents

selected from R_{3a} , and/or having fused thereto a five or six membered carbocyclic ring, or (iv) taken together with R_2 forms a heterocyclo optionally substituted with alkyl or substituted alkyl;

B_b is a bond, C_{1-4} alkylene, C_{2-4} alkenylene, substituted C_{1-4} alkylene, substituted C_{2-4} alkenylene, substituted C_{1-4} alkylene- $C(=O)NH-$, $-C(=O)NH-$, $-C_{1-4}$ alkylene- $C(=O)NH-$, $-C(=O)NR_{19}-$, $-C_{1-4}$ alkylene- $C(=O)NR_{19}-$, or substituted C_{1-4} alkylene- $C(=O)NR_{19}-$, $-(CHR_{14})_m-(CR_{15}R_{16})$ nor $-(CHR_{14})_p-C(=O)NH-$;

R_{3a} at each occurrence is selected independently from alkyl, substituted alkyl, halogen, haloalkoxy, cyano, nitro, keto, trifluoromethyl, $-NR_{17}R_{18}$, $-SR_{17}$, $-OR_{17}$, $-SO_2R_{17a}$, $-SO_2NR_{17}R_{18}$, $-NR_{17}C(=O)R_{18}$, $-CO_2R_{17}$, $-C(=O)R_{17}$, cycloalkyl, aryl, heterocyclo, and heteroaryl, wherein when R_{3a} is cycloalkyl, aryl, heterocyclo or heteroaryl, said cycloalkyl, aryl, heterocyclo and heteroaryl in turn is optionally substituted with alkyl or substituted alkyl;

Z is a heteroaryl, provided that when R_1 is cyano, Z is not 2-pyridinyl; and/or

Z is selected from 5-membered heteroaryl groups containing 1, 2 or 3 heteroatoms each independently selected from N, O and S; and/or

Z is triazolyl optionally substituted with one to two R_7 substituents or imidazolyl optionally substituted with one to two R_7 substituents and/or having fused thereto a benzene ring in turn optionally substituted with one to two R_7 substituents; and

R_7 is alkyl, carbamyl, or substituted alkyl;

R_4 at each occurrence is selected independently of each other R_4 from the group consisting of halogen, trifluoromethyl, OCF_3 , alkyl, substituted alkyl, haloalkyl, nitro, cyano, haloalkoxy, OR_{25} , SR_{25} , $NR_{25}R_{26}$, $NR_{25}SO_2R_{27}$, SO_2R_{27} , $SO_2NR_{25}R_{26}$, CO_2R_{26} , $C(=O)R_{26}$, $C(=O)NR_{25}R_{26}$, $OC(=O)R_{25}$, $-OC(=O)NR_{25}R_{26}$, $NR_{25}C(=O)R_{26}$, $NR_{25}CO_2R_{26}$, aryl, heteroaryl, heterocyclo and cycloalkyl;

R_8 is C_{1-4} alkyl or phenyl optionally substituted with alkyl, halogen, haloalkoxy, cyano, nitro, or trifluoromethyl;

R_9 is $-NR_{10}R_{11}$, alkyl, substituted alkyl, alkoxy, alkylthio, cycloalkyl, aryl, heteroaryl, heterocyclo, or $-CO_2R_{12}$, alkyl or phenyl optionally substituted with one to four of halogen, cyano, trifluoromethyl, nitro, hydroxy, C_{1-4} alkoxy, haloalkoxy, C_{1-6} alkyl, CO_2 alkyl, SO_2 alkyl, SO_2NH_2 , amino, $NH(C_{1-4}alkyl)$, $N(C_{1-4}alkyl)_2$, $NHC(=O)alkyl$, $C(=O)alkyl$, and/or C_{1-4} alkyl optionally substituted with one to three of trifluoromethyl, hydroxy, cyano, phenyl, pyridinyl; and/or a five or six membered heteroaryl or heterocyclo in turn optionally substituted with keto or having a benzene ring fused thereto or a) C_{1-4} alkyl optionally substituted with one to two of:

- SR_{13} , OR_{13} , $NR_{13a}R_{13b}$, halogen, trifluoromethyl, CO_2R_{13a} , and $C(=O)NR_{13a}R_{13b}$;
- cycloalkyl optionally substituted with one to two of $C(=O)H$, $C_{1-4}acyl$, alkenyl, carbamyl, and/or phenyl in turn optionally substituted with halogen;
- phenyl or naphthyl optionally substituted with one to two of halogen, nitro, amino, alkyl, hydroxy, C_{1-4} alkoxy, or having fused thereto a five or six membered heterocyclo;
- pyridinyl, thiophenyl, furanyl, tetrahydrofuranyl, or azepinyl, optionally substituted with alkyl or having fused thereto a five to six membered carbocyclic ring optionally substituted with keto or C_{1-4} alkoxy;

- 3 to 6 membered cycloalkyl optionally having up to four substituents selected from alkyl, halogen, cyano, alkenyl, acyl, alkylthio, carbamyl, phenyl in turn optionally substituted with halogen; or having an aryl fused thereto;
- phenyl optionally substituted with one to four of halogen, cyano, trifluoromethyl, nitro, hydroxy, C_{1-4} alkoxy, haloalkoxy, C_{1-4} alkyl, CO_2 alkyl, SO_2 alkyl, SO_2NH_2 , amino $NH(C_{1-4}alkyl)$, $N(C_{1-4}alkyl)_2$, $NHC(=O)alkyl$, $C(=O)alkyl$, and/or C_{1-4} alkyl optionally substituted with one to three of trifluoromethyl, hydroxy, cyano, phenyl, pyridinyl; and/or a five or six membered heteroaryl or heterocyclo in turn optionally substituted with keto or having a benzene ring fused thereto;
- pyridinyl, thiazolyl, furanyl, thiophenyl, and pyrrolyl optionally substituted with one to two of halogen, alkyl, and phenyl in turn optionally substituted with halogen or trifluoromethyl;

R_{10} and R_{11} are (i) independently selected from hydrogen, alkyl, substituted alkyl, alkoxy, heterocyclo, cycloalkyl, aryl, heteroaryl or C_{1-4} alkyl optionally substituted with one to two of $-CO_2alkyl$, $-C(=O)NH(aryl)$, $NH(aryl)$, cycloalkyl, phenyloxy, phenyl in turn optionally substituted with $C_{1-4}alkyl$, hydroxy, $C_{1-4}alkoxy$, halogen, amino, nitro, tetrahydrofuranyl, and/or five or six membered heterocyclo, or having a five or six membered heterocyclo fused thereto; pyrrolidinyl optionally substituted with keto; naphthyl, anthracenyl, pyridinyl, thiophenyl, furanyl, imidazolyl, benzimidazolyl, or indolyl in turn optionally substituted with $C_{1-4}alkyl$ or $C_{1-4}alkoxy$; or

(ii) taken together form a heteroaryl or heterocyclo selected from pyrrolidinyl, piperazinyl, piperidinyl, morpholinyl, tetrahydropyridinyl, and imidazolidinyl, wherein said heterocyclo formed by R_{10} and R_{11} is optionally substituted with one to two of keto, CO_2H , $C_{1-4}alkoxy$, CO_2alkyl , $C_{1-4}carbamyl$, benzyl; phenyl in turn optionally substituted with alkyl, halogen, or $C_{1-4}alkoxy$; tetrahydropyridinyl in turn optionally substituted with keto and/or phenyl; alkyl optionally substituted with amino or NHR_{21} wherein R_{21} is alkyl or phenyl optionally substituted with alkyl; and/or has a benzene ring fused thereto in turn optionally substituted with one to two of alkyl, $C_{1-4}alkoxy$, CO_2alkyl , and/or $C_{1-4}carbamyl$;

R_{12} and R_{19} are hydrogen or alkyl;

R_{13} is hydrogen or alkyl;

R_{13a} and R_{13b} are selected from hydrogen, alkyl, and aryl;

R_{14} , R_{15} and R_{16} at each occurrence are independently selected from hydrogen, alkyl, hydroxy, hydroxy $C_{1-4}alkyl$, $C_{1-4}alkoxy$, and phenyl, and/or one of R_{15} and one of R_{16} join together to form a 3 to 6 membered cycloalkyl;

R_{17} and R_{18} are independently selected from hydrogen, alkyl, substituted alkyl, aryl, phenyl, or benzyl wherein the phenyl or benzyl is optionally substituted with alkyl, hydroxy, or hydroxyalkyl;

R_{17a} is alkyl or substituted alkyl;

R_{25} and R_{26} are independently selected from hydrogen, alkyl, or substituted alkyl, or taken together form a heterocyclo or heteroaryl ring;

R_{27} is alkyl or substituted alkyl;

q is 0, 1, 2, or 3;

m and n are 0, 1 or 2; and

p is 0, 1, 2, or 3.

* * * * *